# The Topology of REM Mentation (Dreams):

## Attractors, Affective Projections, and the Dual-Regime Architecture of Consciousness

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ORCID: 0009-0001-0034-2903 February 2026

**Abstract**

This paper introduces a unified mathematical framework for the formal analysis of REM mentation—the cognitive, spatial, and affective processes that constitute dreaming. The framework comprises eight interlocking components: (i) a dual-regime model of consciousness in which waking and dreaming states are characterized as distinct regions of a continuous cognitive phase space, separated by bifurcation-like transitions governed by measurable parameters including executive control, associative density, censorship level, and

temporal coherence; (ii) a theory of topological attractors positing that recurring spatial motifs in dream reports may correspond to stable or slowly drifting attractors in the phase space of an internal world-model; (iii) an affective projection mapping that formalizes how emotional states may give rise to spatial configurations experienced as dream environments; (iv) a distinction between logical and affective validation operators, capturing the observation that dream cognition appears to rely on internal coherence of felt meaning rather than external factual consistency; (v) a predictive-processing account in which the self-model parameter undergoes systematic attenuation during REM sleep, potentially serving as a form of identity-boundary testing; (vi) a characterization of dreaming as an inverted information filter that may preferentially amplify signals suppressed during waking cognition; (vii) a structural analogy between autonomous dream processes and unsupervised recalibration in artificial neural networks; and (viii) a proposal for statistical psychogeography—the empirical clustering of collective dream-space motifs across populations. A ninth component extends the attractor theory to stochastic dynamics, modeling the pre-dream hypnagogic state as a probabilistic distribution over competing attractors and the onset of coherent dreaming as a crystallization event—a noise-driven selection process that bears a structural (though not ontological) analogy with wavefunction collapse in quantum measurement theory. A tenth component introduces the interrogative collapse model, formalizing the observation that dream content exists in a state of radical indeterminacy that resolves only when internal cognitive acts—implicit questions of the form “where,” “what,” “who”—force ad hoc specification, and that excessive interrogation destabilizes the dream by generating inconsistencies faster than affective validation can absorb them. An eleventh component analyzes the dream interface as an instance of virtual complexity: the dissociation between the compactness of the generative mechanism (low Kolmogorov complexity), the richness of the experiential stream (high Shannon entropy), and the moderate structure of its regularities (intermediate Gell-Mann effective complexity). A twelfth component addresses the non-topological character of dream space, arguing that the drifting relational structure of dream content—where spatial adjacency, identity, and causal relations change discontinuously—requires mathematical tools beyond classical topology, including pretopological spaces, drifting relational graphs, and sheaf-theoretic models of local-but-not-global consistency. A thirteenth component examines dream amnesia—the near-total loss of dream content upon awakening—reviewing neurobiological mechanisms including noradrenergic suppression, active hippocampal inhibition by melanin-concentrating hormone neurons, and encoding-consolidation competition, and proposing four experimental interventions to test the framework's predictions. A fourteenth component develops a theory of dream atemporality, modeling the variable temporality of dream experience through content-dependent pseudo-Riemannian metrics in which cognitive processing demands curve subjective dream spacetime—a structural analogy with general relativistic time dilation. A fifteenth component addresses the paradox of physical law preservation in dreams: the generative prior explains why dreams overwhelmingly obey waking physics despite the absence of external constraints, the ontological reframing mechanism explains why violations feel natural, and the generator-experiencer decoupling formalizes the agency paradox whereby the dreamer is simultaneously sole author and powerless spectator. A sixteenth component extends the regime parameter space beyond dreaming to a terminal regime, modeling the near-death experience as the limit of the framework: complete self-model dissolution, maximal ontological flexibility, extreme time dilation, and zero encoding—a state in which seconds of cortical activity may produce subjective lifetimes of experience. A seventeenth component formalizes the dream-reality indistinguishability problem and nested dreams as recursive confabulation, demonstrating that the distinction between waking and dreaming is epistemically underdetermined from the inside. An eighteenth component addresses the negativity bias of dream content through the inverted filter mechanism, formalizes affective leakage as the channel through which dreams influence waking psychic state, and identifies intervention points including imagery rehearsal therapy, lucid dreaming, and pre-sleep affective priming. A nineteenth component recasts prophetic dreams as a natural consequence of predictive processing operating without executive constraints, and formalizes dream symbolism as a compression phenomenon under culturally modulated suppression—yielding the testable prediction that dream symbolism should be denser in restrictive cultures than in permissive ones. A twentieth component establishes a formal correspondence between dreaming and the idle generation mode of large language models, demonstrating that hallucination, confabulation, local coherence with global inconsistency, and the absence of source monitoring are structural properties shared by any sufficiently complex generative system operating without external input—suggesting that dreaming may be an inevitable computational consequence of possessing a generative model, not a specific biological adaptation. A twenty-first component addresses the dream-memory boundary, formalizing source monitoring

failure as the mechanism by which dream episodes contaminate autobiographical memory: the same generative system produces both dreams and memory retrievals, the internal format is identical, and the source tag distinguishing them can be lost—with implications for false memory formation, legal epistemology, and the hypothesis that every memory that passes through a sleep cycle is, to some degree, a dream. Each component is developed with formal definitions, propositions, and proofs where tractable, accompanied by critical discussion of limitations, alternative explanations, and conditions under which the proposed models might fail. The framework is offered as a set of potentially falsifiable conjectures rather than established theory, with the aim of stimulating interdisciplinary investigation at the intersection of dynamical systems theory, cognitive neuroscience, and the phenomenology of sleep mentation.

**Keywords:** REM mentation, dream topology, cognitive phase space, attractor dynamics, affective projection, dual-regime consciousness, predictive processing, self-model, statistical psychogeography, unsupervised learning, stochastic attractors, crystallization dynamics, interrogative collapse, virtual complexity, Kolmogorov complexity, effective complexity, pretopological space, sheaf cohomology, dream amnesia, MCH neurons, dream atemporality, subjective spacetime metric, generative prior, ontological reframing, agency paradox, near-death experience, terminal regime, reality testing, nested dreams, negativity bias, threat simulation, affective leakage, prophetic dreams, symbolic compression, cultural modulation, large language models, hallucination, idle generation, source monitoring, dream-memory boundary, false memory, epistemic contamination

# Contents

# 1 Introduction

The phenomenon of dreaming—technically, mentation during rapid eye movement (REM) sleep and, to a lesser extent, during non-REM stages [1]—occupies a peculiar position in the sciences of mind. It is universal among mammals, phylogenetically ancient, metabolically costly, and accompanied by a characteristic neurochemical and electrophysiological signature [2]. Yet despite decades of neuroscientific investigation, there remains no consensus on whether dreaming serves a specific adaptive function, constitutes an epiphenomenon of sleep-related neural maintenance, or represents something else entirely [3–5].

The present work does not claim to resolve this question. Instead, it proposes a mathematical framework within which several observable regularities of dream experience—its spatial recurrences, its distinctive logic, its affective saturation, its relationship to waking cognition—can be described with formal precision and subjected to principled investigation. The approach draws on dynamical systems theory [6], predictive processing accounts of brain function [7, 8], quantitative dream content analysis [9], and the phenomenological tradition of attending carefully to the structure of subjective experience [10].

The motivation for this undertaking arises from a persistent observation: dream reports, when collected systematically, reveal striking regularities that resist dismissal as noise. Recurring spatial environments—houses with impossible architectures, staircases leading to unvisited floors, transit stations where the intended vehicle never arrives—appear across individuals, cultures, and historical periods [11, 12]. The internal logic of dream experience, while violating ordinary causal reasoning, appears to maintain its own form of coherence [5]. Emotional tone in dreams often bears a complex, non-trivial relationship to waking affective states [13]. These patterns suggest the possibility that dream cognition, rather than being random neural discharge, may reflect the operation of structured processes amenable to formal characterization.

**Epistemological Note on the Status of the Framework.** The framework developed in this monograph is a mathematical model, not a direct empirical description of neural processes during REM sleep. Its primary object is a formal generative system—a set of interacting mathematical structures—that produces dynamics exhibiting properties recognizable as dreamlike: local coherence with global inconsistency, lazy specification of detail, drifting relational structure, virtual complexity, and near-total loss of content at termination. The relationship between this model and actual dreaming is analogous to the relationship between the Ising model and ferromagnetism, or between cellular automata and biological morphogenesis: the model captures structural regularities without claiming ontological identity with the target system. Where the text refers to “dreams,” “dream content,” or “dream experience,” this should be understood as shorthand for “the output of the formal generative system, which shares specified structural properties with reported dream phenomenology.” The empirical predictions listed in Section 14.5 identify points where the model’s output can be compared against data, but the framework’s value does not depend entirely on these comparisons: a self-consistent formal system that produces dreamlike dynamics from compact generative rules is of independent mathematical and theoretical interest.

The framework developed here is organized as a progressive construction. Section 2 introduces the foundational model: a cognitive phase space in which waking and dreaming correspond to distinct dynamical regimes, with transitions governed by continuously varying parameters. Section 3 develops the theory of topological attractors. Section 4 formalizes the mapping from emotional states to spatial experience. Section 5 introduces the distinction between logical and affective validation. Section 6 addresses the dissolution of self-boundaries during dreaming. Section 7 proposes a characterization of dreaming as an inverted information filter. Section 8 develops the structural analogy with artificial neural networks. Section 9 outlines the program of statistical psychogeography. Section 10 extends the attractor theory to stochastic dynamics, introducing the concept of dream crystallization and its structural analogy with measurement-like selection processes. Section 11 develops the interrogative collapse model, formalizing the ad hoc, demand-driven generation of dream content and its implications for dream stability. Section 12 analyzes the dream interface as virtual complexity, comparing Kolmogorov, Shannon, and Gell-Mann effective complexity measures. Section 13 addresses the non-topological character of dream space and the drifting relational structure of dream content. Section 14 examines the mechanisms of dream amnesia and proposes experimental interventions. Section 15 develops a theory of dream atemporality using content-dependent spacetime metrics. Section 16 addresses the paradox of physical law preservation, ontological reframing, and the agency paradox. Section 17 extends the framework to the terminal regime and near-death experience. Section 18 formalizes the dream-reality indistinguishability problem and nested dreams. Section 19 addresses the negativity bias, psychic impact, and intervention. Section 20 recasts prophetic dreams and dream symbolism. Section 21 establishes the LLM-dreaming correspondence. Section 22 addresses the dream-memory boundary and source confusion. Section 23 provides an integrative discussion, and Section 24 summarizes the principal contributions and their limitations.

Throughout, each formal result is followed by a discussion of its scope, assumptions, conditions under which it might not hold, and alternative interpretations that the formalism cannot exclude. The framework is offered as a set of conjectures and modeling tools, not as established theory.

# 2 The Dual-Regime Model of Consciousness

## 2.1 Phenomenological Motivation

The most fundamental structural observation about human consciousness is that it alternates, with approximate circadian regularity, between two broadly distinguishable modes. In the waking mode, cognition is characterized by executive control over attention, sequential reasoning, adherence to external causal regularities, and suppression of associations deemed irrelevant to current goals. In the dreaming mode—particularly during REM sleep—these constraints are substantially relaxed: associative connections proliferate, temporal sequencing becomes fluid or absent, logical contradictions pass without notice, and the boundary between self and environment becomes unstable [2, 5].

This alternation is accompanied by well-documented neurochemical shifts: the suppression of noradrenergic and serotonergic neuromodulation during REM sleep, the relative preservation of cholinergic and dopaminergic activity, and characteristic changes in prefrontal cortical engagement [2]. These signatures suggest that the transition between waking and dreaming is not a binary switch but a continuous modulation of underlying parameters.

## 2.2 Formal Construction

The central mathematical object is a state space representing all possible instantaneous configurations of a cognitive system. Each point encodes the current pattern of neural activation, the current associative landscape, and the current mode of information processing.

**Definition 2.1** (Cognitive Phase Space)**.** Let $\mathcal{S}$ be a smooth manifold of dimension $n$, called the *cognitive phase space*. A point $\mathbf{s} \in \mathcal{S}$ represents an instantaneous cognitive state. The system evolves according to:

$$\frac{d\mathbf{s}}{dt} = F(\mathbf{s}; \boldsymbol{\lambda}), \tag{1}$$

where $F : \mathcal{S} \times \Lambda \to T\mathcal{S}$ is a smooth vector field, $T\mathcal{S}$ denotes the tangent bundle of $\mathcal{S}$, and $\boldsymbol{\lambda} = (\sigma, \kappa, \alpha, \tau) \in \Lambda \subseteq \mathbb{R}^4$ is a vector of regime parameters.

The tangent bundle $T\mathcal{S}$ collects all possible directions and rates of change at every point of the state space. Assigning a tangent vector to each point $\mathbf{s}$ determines, at each instant, the direction and speed at which the cognitive state evolves. The regime parameters $\boldsymbol{\lambda}$ modulate this evolution. The dimension $n$ of $\mathcal{S}$ is left unspecified; in principle it could be extremely large (reflecting the full neural state), but for the framework to be tractable, it should be understood as the effective dimensionality of a low-dimensional manifold embedded in the full neural state space—consistent with recent evidence that neural population dynamics during both waking and sleep evolve on low-dimensional manifolds [14]. A toy instantiation at $n = 3$ is developed in Appendix B.

**Definition 2.2** (Regime Parameters)**.** The regime parameter vector $\boldsymbol{\lambda} = (\sigma, \kappa, \alpha, \tau)$ consists of:

(i) *Executive control* $\sigma \in [0, 1]$: the degree to which top-down, goal-directed processes constrain the trajectory of cognitive states.

(ii) *Associative censorship* $\kappa \in [0, 1]$: the threshold below which associative connections are suppressed.

(iii) *Associative density* $\alpha \in [0, \infty)$: the number of active associative connections per unit state-space volume.

(iv) *Temporal coherence* $\tau \in [0, 1]$: the degree to which successive cognitive states preserve causal and temporal ordering.

These parameters are selected because each corresponds to a documented dimension along which waking and dreaming cognition differ [2, 5]. Additional parameters could extend the model without altering its structure.

**Definition 2.3** (Waking and Dreaming Regimes)**.** The *waking regime* $\mathcal{W} \subset \Lambda$ is:

$$\mathcal{W} = \{\boldsymbol{\lambda} \in \Lambda \mid \sigma > \sigma^*,\ \kappa > \kappa^*,\ \alpha < \alpha^*,\ \tau > \tau^*\}, \tag{2}$$

where $\sigma^*, \kappa^*, \alpha^*, \tau^*$ are threshold values. The *dreaming regime* $\mathcal{D} \subset \Lambda$ is:

$$\mathcal{D} = \{\boldsymbol{\lambda} \in \Lambda \mid \sigma < \sigma^{**},\ \kappa < \kappa^{**},\ \alpha > \alpha^{**},\ \tau < \tau^{**}\}, \tag{3}$$

with $\sigma^{**} \leq \sigma^*$, etc. The region $\Lambda \setminus (\mathcal{W} \cup \mathcal{D})$ constitutes the *transitional zone*, encompassing hypnagogic, hypnopompic, and other intermediate states.

The transitional zone accommodates lucid dreaming, sleep-onset mentation, and various dissociative states [15, 16].

## 2.3 Phase Transition Structure

Neurophysiological evidence suggests that the onset of REM sleep involves relatively abrupt shifts in neuromodulatory tone [2]. This motivates the hypothesis that the transition may exhibit bifurcation-like properties.

**Definition 2.4** (Effective Regime Parameter)**.** Let $\lambda_{\text{eff}} : \Lambda \to \mathbb{R}$ be:

$$\lambda_{\text{eff}}(\boldsymbol{\lambda}) = w_1(1-\sigma) + w_2(1-\kappa) + w_3\frac{\alpha}{\alpha_{\max}} + w_4(1-\tau), \tag{4}$$

where $w_i > 0$, $\sum_i w_i = 1$, and $\alpha_{\max}$ is a normalization constant. The parameter $\lambda_{\text{eff}}$ takes values near 0 in waking and near 1 in REM dreaming.

**Proposition 2.5** (Existence of Critical Transitions)**.** *Suppose $F(\mathbf{s};\boldsymbol{\lambda})$ satisfies standard regularity conditions. If the Jacobian*

$$J(\mathbf{s}^*, \boldsymbol{\lambda}) = \left.\frac{\partial F}{\partial \mathbf{s}}\right|_{\mathbf{s}=\mathbf{s}^*} \tag{5}$$

*at a fixed point $\mathbf{s}^*$ (where $F(\mathbf{s}^*;\boldsymbol{\lambda}) = \mathbf{0}$) has an eigenvalue crossing the imaginary axis as $\lambda_{\text{eff}}$ passes through $\lambda_c$, then a qualitative change in the local attractor structure occurs at $\lambda_c$.*

The Jacobian matrix $J$ encodes how sensitive the flow is to small displacements from the fixed point. Its eigenvalues determine stability: when all eigenvalues have negative real parts, nearby trajectories converge to $\mathbf{s}^*$; when an eigenvalue crosses to the positive side, the fixed point loses stability.

*Proof.* By the implicit function theorem for parameterized dynamical systems [17], fixed points persist under smooth parameter variation so long as $J$ is nonsingular.

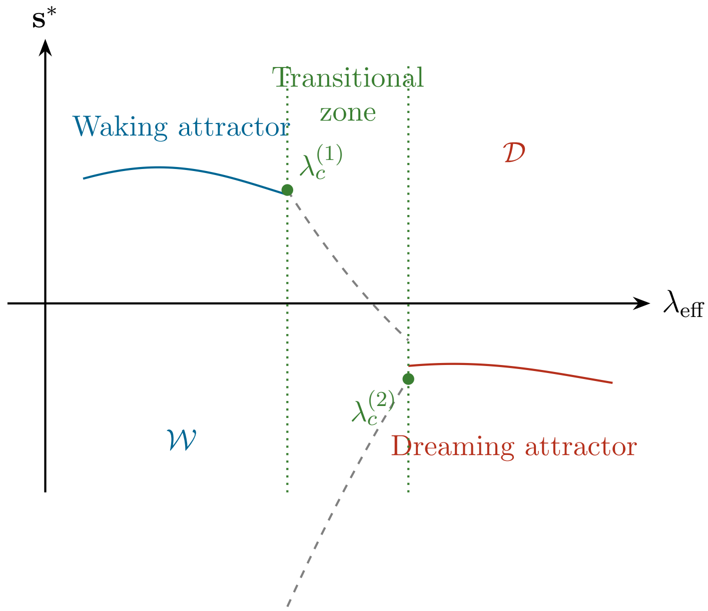


Figure 1: Schematic bifurcation diagram. As $\lambda_{\text{eff}}$ increases, the stable waking attractor (blue) loses stability near $\lambda_c^{(1)}$, and a stable dreaming attractor (red) emerges near $\lambda_c^{(2)}$. Dashed lines: unstable equilibria. Green region: transitional zone. The diagram is schematic; actual bifurcation structure may be more complex.

When an eigenvalue reaches the imaginary axis, nonsingularity fails, and center manifold reduction guarantees the existence of a local bifurcation [17]. The specific type is determined by the normal form coefficients of the reduced system. □ □

*Caveat* 2.6. This application of bifurcation theory requires several strong assumptions. The cognitive phase space $\mathcal{S}$ is assumed to be a smooth manifold, whereas the actual neural state space may have fractal or discrete structure. The flow $F$ is assumed smooth and autonomous, whereas neural dynamics are noisy and non-autonomous. The reduction to a scalar $\lambda_{\text{eff}}$ may obscure multi-parameter effects. The identification of neural quantities with $\sigma, \kappa, \alpha, \tau$ remains an open empirical challenge. Without such identification, the analysis, while mathematically valid, lacks direct empirical content.

## 2.4 The Transitional Zone and Lucid Dreaming

**Definition 2.7** (Lucid Dreaming Region)**.** The *lucid dreaming region* $\mathcal{L} \subset \Lambda$ is provisionally defined as:

$$\mathcal{L} = \{\boldsymbol{\lambda} \in \Lambda \mid \sigma \geq \sigma^{**} + \delta_\sigma,\ \kappa < \kappa^*,\ \alpha > \alpha^*,\ \tau < \tau^*\}\,, \tag{6}$$

where $\delta_\sigma > 0$ represents the minimal increment in executive control required for metacognitive awareness.

Within this framework, lucid dreaming occupies a region where executive control is partially restored while associative freedom, relaxed censorship, and temporal fluidity persist. This characterization is consistent with neuroimaging findings showing increased frontal gamma activity during lucid episodes [15], though it should be noted that lucid dreaming exists on a spectrum of lucidity and likely involves additional neural mechanisms not captured by $\sigma$ alone [16].

# 3 Topological Attractors of Dream Space

## 3.1 Phenomenological Motivation

Among the most widely reported features of dream experience is the recurrence of spatial environments. Individuals frequently report returning, across months or years, to the same houses, staircases, corridors, or urban landscapes—places that do not correspond to any waking location, yet are recognized with familiarity stronger than that accorded to actual places [9, 12].

These environments are not static. Reports indicate slow evolution: new rooms appear, configurations shift subtly, spatial relationships alter [9]. The rate of change appears to operate on a timescale substantially longer than individual dream episodes.

## 3.2 Formal Construction

**Definition 3.1** (Endogenous Dream Dynamics)**.** Let $\boldsymbol{\lambda}_D \in \mathcal{D}$. The *endogenous dream dynamics* is:

$$\frac{d\mathbf{s}}{dt} = F_D(\mathbf{s}) \coloneqq F(\mathbf{s}; \boldsymbol{\lambda}_D). \tag{7}$$

An *attractor* of $F_D$ is a compact invariant set $A \subset \mathcal{S}$ that is Lyapunov stable and attracts all trajectories originating in some open neighborhood $U \supset A$.

A compact invariant set is a bounded, closed subset preserved by the flow: any trajectory starting in $A$ remains in $A$. Lyapunov stability requires that trajectories starting near $A$ remain near $A$. Together, these conditions ensure that $A$ represents a configuration the system reliably revisits.

**Definition 3.2** (Basin of Attraction)**.** The *basin of attraction* of an attractor $A$ is:

$$\mathfrak{B}(A) = \left\{ \mathbf{s}_0 \in \mathcal{S} \;\middle|\; \lim_{t\to\infty} d(\phi_t(\mathbf{s}_0), A) = 0 \right\}, \tag{8}$$

where $\phi_t$ is the time-$t$ flow map and $d(\cdot, A)$ is the distance to the set $A$.

A large basin implies that many different pre-sleep cognitive configurations lead to the same attractor—and thus potentially to the same recurring dream environment.

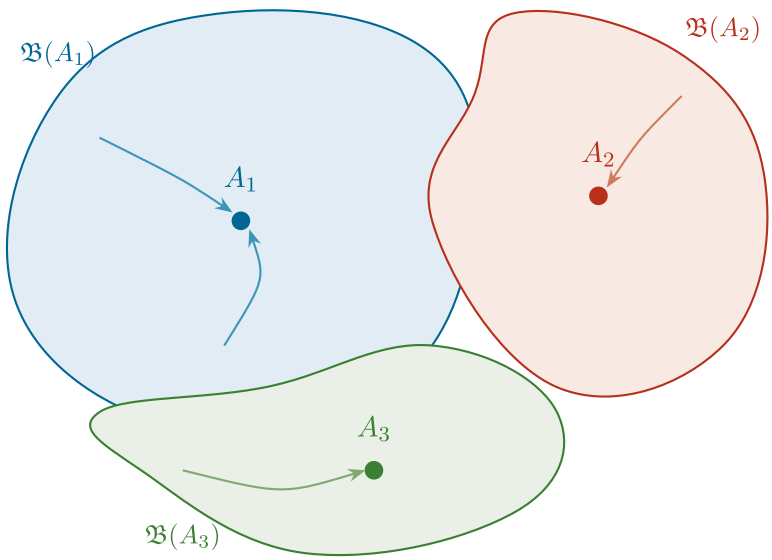


Figure 2: Three attractors $(A_1, A_2, A_3)$ in the endogenous dream dynamics, with basins of attraction. Trajectories from diverse initial conditions converge to the nearest attractor, corresponding to distinct recurring dream environments.

**Definition 3.3** (Slowly Drifting Attractor)**.** Let $A(t_s)$ be a family of attractors parameterized by slow time $t_s = \epsilon t$, $0 < \epsilon \ll 1$. The attractor is *slowly drifting* if:

(i) For each fixed $t_s$, $A(t_s)$ is a stable attractor of $F_D$.

(ii) The Hausdorff distance satisfies $d_H(A(t_s), A(t_s + \Delta t_s)) = \mathcal{O}(\epsilon\, \Delta t_s)$.

The Hausdorff distance measures the maximum extent to which two sets differ: the largest distance from any point in one set to the nearest point in the other. Condition (ii) states that the attractor changes at rate $\epsilon$, much slower than individual trajectories converge. Within a single dream, the attractor appears fixed; over months and years, it drifts.

**Proposition 3.4** (Persistence Under Perturbation)**.** *Let $A_0$ be a hyperbolic attractor of $F_D$. Then for sufficiently small perturbations $\delta F$, the perturbed system possesses an attractor $A_\delta$ with:*

$$d_H(A_0, A_\delta) \leq C\|\delta F\|, \tag{9}$$

*where $C > 0$ depends on the spectral gap of the Jacobian.*

Hyperbolicity means that all eigenvalues of the Jacobian at the attractor have strictly negative real parts, ensuring rapid convergence of nearby trajectories. The proposition states that such attractors survive moderate perturbations—formalizing the robustness of recurring dream spaces to changes in daily experience.

*Proof.* The hyperbolicity condition ensures a uniform spectral gap $\mu > 0$. By the persistence theorem for normally hyperbolic invariant manifolds [17], any $C^1$-small perturbation preserves the attractor, with displacement bounded linearly by perturbation magnitude. The constant $C$ depends inversely on $\mu$. □ □

*Caveat* 3.5. The assumption of hyperbolicity is strong; neural attractors may be non-hyperbolic or chaotic [18]. The identification of mathematical attractors with experiential dream spaces is interpretive, not proven. The separation of timescales has not been directly measured. Finally, recurring dream spaces may arise from simpler mechanisms such as hippocampal replay [19, 20] without requiring dynamical systems formalism.

# 4 Affective Projection Mapping

## 4.1 Phenomenological Motivation

Dream reports suggest that spatial environments are not neutral backdrops but are constituted by accompanying emotions [9, 21]. Confinement manifests as a narrow corridor; loss gives rise to an empty room; anxiety produces a labyrinthine building with no exit.

## 4.2 Formal Construction

**Definition 4.1** (Affect Space)**.** The *affect space* $\mathcal{E}$ is a finite-dimensional smooth manifold with metric $g_\mathcal{E}$. A point $\mathbf{e} \in \mathcal{E}$ represents an instantaneous affective state.

**Definition 4.2** (Dream Topography Space)**.** A dream topography $\mathbf{t} \in \mathcal{T} \cong \mathbb{R}^m$ is described by:

$$\mathbf{t} = (t_{\text{enc}},\ t_{\text{vert}},\ t_{\text{conn}},\ t_{\text{bnd}},\ t_{\text{fam}},\ \ldots), \tag{10}$$

where $t_{\text{enc}} \in [0,1]$ (enclosure), $t_{\text{vert}} \in [-1,1]$ (verticality), $t_{\text{conn}} \in [0,1]$ (connectivity), $t_{\text{bnd}} \in [0,1]$ (boundedness), and $t_{\text{fam}} \in [0,1]$ (felt familiarity).

**Definition 4.3** (Affective Projection)**.** The *affective projection* is a smooth map $\Pi : \mathcal{E} \to \mathcal{T}$ satisfying:

(i) *Continuity*: nearby emotional states produce similar spatial configurations.

(ii) *Non-injectivity*: distinct emotions may produce the same spatial experience.

(iii) *Context-dependence*: $\Pi$ may depend on biographical, cultural, and memory parameters.

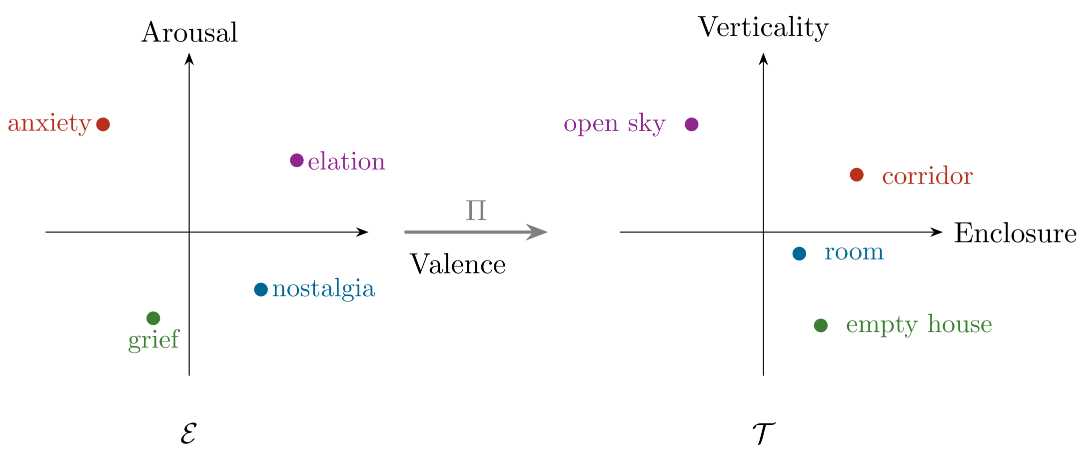


Figure 3: Schematic of the affective projection $\Pi : \mathcal{E} \to \mathcal{T}$. Affect-space points (left) map to dream topography points (right). Labels are illustrative hypotheses.

**Proposition 4.4** (Affective Projection Induces Dream Topography)**.** *Let* $\mathbf{e}(t)$ *be the affective trajectory during a dream episode. The experienced topography is* $\mathbf{t}(t) = \Pi(\mathbf{e}(t))$*, and if* $\Pi$ *is smooth and* $\mathbf{e}(t)$ *piecewise smooth:*

$$\frac{d\mathbf{t}}{dt} = D\Pi\big|_{\mathbf{e}(t)} \cdot \frac{d\mathbf{e}}{dt}, \tag{11}$$

*where* $D\Pi$ *is the Jacobian of* $\Pi$*.*

The rate of spatial change in the dream is determined by the product of the emotional rate of change and the local sensitivity of the projection. Rapid emotional shifts produce rapid spatial transformations; stable emotions yield stable environments. In regions of $\mathcal{E}$ where $\|D\Pi\|$ is large, even small emotional fluctuations produce notable spatial changes.

*Proof.* Direct application of the chain rule for smooth maps between manifolds. □

□

*Caveat* 4.5. The existence of a well-defined map from emotions to spatial configurations is an assumption. Dream spaces may be determined by hippocampal replay independently of affect [19]. Causation may be reversed or bidirectional. The functional form of $\Pi$ is unknown and may be stochastic.

# 5 Logical and Affective Validation

## 5.1 Phenomenological Motivation

Dream cognition accepts logical impossibilities without surprise. A figure bearing no resemblance to one's grandmother is recognized as her with complete certainty [5]. This suggests that dreaming employs a mode of internal validation distinct from the logical validation of waking thought.

## 5.2 Formal Construction

**Definition 5.1** (Validation Operators)**.** A *validation operator* is $V : \mathcal{P} \times \mathcal{C} \to [0, 1]$, where $\mathcal{P}$ is the space of cognitive propositions and $\mathcal{C}$ a context space. Two operators are distinguished:

(i) *Logical validation* $V_L$ evaluates against perceptual consistency, causal coherence, and factual knowledge:

$$V_L(p, c) = f\big(\text{percept}(p, c),\ \text{causal}(p, c),\ \text{factual}(p, c)\big). \tag{12}$$

(ii) *Affective validation* $V_A$ evaluates against felt coherence, emotional resonance, and identity-level recognition:

$$V_A(p, c) = g\big(\text{felt}(p, c),\ \text{resonance}(p, c),\ \text{identity}(p, c)\big). \tag{13}$$

In waking, objects are categorized primarily by perceptual features (the aggregation function $f$ dominates). In dreaming, categorization operates via what might be termed an "affective signature"—a pattern of felt meaning that identifies an entity independently of its appearance ($g$ dominates).

**Definition 5.2** (Regime-Dependent Validation)**.** The effective validation in regime $\boldsymbol{\lambda}$ is:

$$V_{\boldsymbol{\lambda}}(p,c) = \big(1-\omega(\boldsymbol{\lambda})\big)\, V_L(p,c) + \omega(\boldsymbol{\lambda})\, V_A(p,c), \tag{14}$$

where $\omega : \Lambda \to [0,1]$ satisfies $\omega \approx 0$ in $\mathcal{W}$ and $\omega \approx 1$ in $\mathcal{D}$.

**Proposition 5.3** (Affective Override)**.** *In the dreaming regime,* $V_{\boldsymbol{\lambda}_D}(p,c) \approx V_A(p,c)$*. A proposition may receive high validation even when* $V_L(p,c) \approx 0$*, provided* $V_A(p,c) \approx 1$*.*

*Proof.* With $\omega \approx 1$ in $\mathcal{D}$, the contribution of $V_L$ is negligible. □ □

*Counterpoint* 5.4. An alternative account holds that dreaming involves the same validation operator $V_L$ with dramatically lowered thresholds, not a qualitatively distinct operator. Distinguishing these hypotheses would require probing the basis of acceptance judgments during dreaming—a significant methodological challenge, though lucid dreaming protocols offer limited avenues [22].

# 6 Self-Model Dissolution

## 6.1 Phenomenological Motivation

During dreaming, the sense of bounded, continuous selfhood becomes unstable. Dreamers report being simultaneously themselves and another, observing themselves externally, or losing identity entirely [23, 24]. Within predictive processing, the self-model is a set of high-level prior beliefs about the agent [7, 25].

## 6.2 Formal Construction

**Definition 6.1** (Self-Model)**.** The *self-model* is a distribution $\mathcal{S}_\xi$ over agent configurations $\mathcal{A}$, with precision $\xi \in [0,1]$:

$$\mathcal{S}_\xi(\mathbf{a}) = \frac{1}{Z(\xi)} \exp\left(-\frac{\xi}{2}\|\mathbf{a}-\mathbf{a}_0\|^2\right), \tag{15}$$

where $\mathbf{a}_0$ is the expected configuration and $Z(\xi)$ normalizes.

Large $\xi$ produces a sharply peaked distribution: high certainty about identity. Small $\xi$ produces a broad distribution: uncertain, fluid identity. In the limit $\xi \to 0$, the system has no preferential self-model.

**Definition 6.2** (Self-Model Attenuation)**.** In the dreaming regime:

$$\xi_D = \xi_W \cdot h(\boldsymbol{\lambda}), \tag{16}$$

where $h(\boldsymbol{\lambda}) = 1$ in $\mathcal{W}$ and $h(\boldsymbol{\lambda}) = \eta \ll 1$ in $\mathcal{D}$.

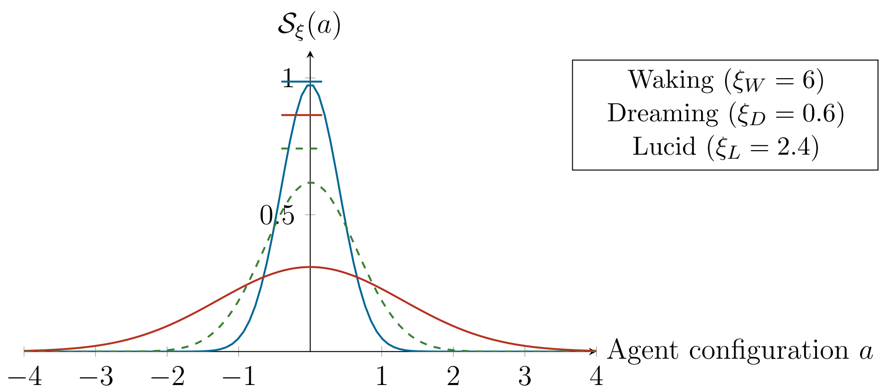


Figure 4: Self-model distributions. Waking (blue): sharply peaked. Dreaming (red): broad, uncertain identity. Lucid (dashed green): partially restored precision.

**Proposition 6.3** (Entropy Increase During Dreaming)**.** *For the Gaussian self-model with $\mathcal{A} \cong \mathbb{R}^d$:*

$$H(\mathcal{S}_\xi) = \frac{d}{2}\left(1 + \ln\frac{2\pi}{\xi}\right). \tag{17}$$

*Since $\xi_D < \xi_W$, it follows that $H(\mathcal{S}_{\xi_D}) > H(\mathcal{S}_{\xi_W})$.*

The differential entropy $H$ quantifies the spread of the self-model. The proposition states that dreaming is associated with increased self-uncertainty.

*Proof.* The differential entropy of a $d$-dimensional Gaussian with precision $\xi I$ is $H = \frac{d}{2}(1 + \ln(2\pi/\xi))$. Since ln is monotonically increasing and $\xi_D < \xi_W$, the result follows. □ □

*Caveat* 6.4. The Gaussian parameterization is adopted for tractability, not biological realism. The self-model likely involves hierarchical, multi-modal structures [24]. The claim that attenuation serves a "testing" function is speculative: it may be an epiphenomenal consequence of REM neurochemistry. The entropy calculation quantifies a formal consequence, not evidence for functional significance.

# 7 Dreaming as an Inverted Information Filter

## 7.1 Phenomenological Motivation

Waking cognition selects a small subset of available information, suppressing the rest [26]. Dream content frequently features elements that were peripheral during waking [9, 13]—suggesting that dreaming may preferentially amplify suppressed signals.

## 7.2 Formal Construction

**Definition 7.1** (Inverted Filter Operator)**.** Let $\mathcal{I}$ be the space of information signals with waking salience $s_W(i) \in [0, 1]$ and suppression index $\bar{s}_W(i) = 1 - s_W(i)$. The *inverted filter*:

$$\Phi(i) = \frac{\bar{s}_W(i)^\beta}{\sum_{j \in \mathcal{I}} \bar{s}_W(j)^\beta}, \tag{18}$$

where $\beta > 0$ controls inversion sharpness. At $\beta = 1$, dream activation is proportional to waking suppression. At $\beta > 1$, the most suppressed signals are disproportionately amplified.

**Proposition 7.2** (Entropy Maximization Under Inversion)**.** *If waking salience $S_W$ is concentrated (low entropy), then the dream activation distribution $S_D = \{\Phi(i)\}$ at $\beta = 1$ has $H(S_D) \geq H(S_W)$, with equality iff $S_W$ is uniform.*

The inverted filter redistributes activation toward suppressed signals, increasing entropy and broadening the range of processed content. This may serve as a corrective against over-specialization of the waking model.

*Proof.* For $\beta = 1$, $q_i = (1 - s_W(i))/\sum_j(1 - s_W(j))$. If $p_i = s_W(i)/\sum_j s_W(j)$ is non-uniform, the inversion redistributes weight away from dominant signals. By strict concavity of entropy, $H(q) \geq H(p)$ with equality iff $p$ is uniform, in which case $q = p =$ uniform. □ □

*Caveat* 7.3. The continuity hypothesis [27] holds that dream content reflects waking concerns (often high-salience) rather than suppressed percepts, complicating a simple inversion model. The evolutionary rationale is speculative. The model should be understood as formalizing a tendency, not a complete account.

# 8 Structural Analogy with Artificial Neural Networks

## 8.1 Motivation and Scope

The wake-sleep algorithm [28], in which a generative model alternates between externally driven and internally driven phases, bears structural resemblance to the waking-dreaming cycle. The present section develops this analogy while emphasizing fundamental disanalogies.

## 8.2 Formal Construction

**Definition 8.1** (Wake-Sleep Architecture)**.** A *wake-sleep cognitive architecture* ( $\mathcal{N}, \theta, \mathcal{D}_{\text{ext}}, \mathcal{D}_{\text{int}}$) alternates between:

(i) **Wake phase:** $\theta \leftarrow \theta - \eta_W \nabla_\theta \mathcal{L}_W(\theta; \mathcal{D}_{\text{ext}})$.
(ii) **Sleep phase:** $\theta \leftarrow \theta - \eta_S \nabla_\theta \mathcal{L}_S(\theta; \mathcal{D}_{\text{int}}(\theta))$.

The sleep-phase update is autonomous: it depends only on internally generated representations. This autonomy is the structural parallel with dreaming.

**Proposition 8.2** (Recalibration Through Autonomous Dynamics)**.** *If the sleep-phase loss is $\mathcal{L}_S(\theta) = D_{KL}(\mathcal{D}_{int}(\theta) \| p_{target})$, then gradient descent on $\mathcal{L}_S$ moves the generative model toward $p_{target}$, independent of current sensory input.*

The Kullback-Leibler divergence $D_{\text{KL}}(p\|q)$ measures the information lost when $q$ is used to approximate $p$. It is non-negative and zero iff $p = q$. Minimizing $\mathcal{L}_S$ thus drives the internal model toward an optimal prior or consolidated structure.

*Proof.* Standard convergence of gradient descent on smooth, lower-bounded objectives ensures $\mathcal{L}_S$ decreases at each step for sufficiently small $\eta_S$. □ □

*Caveat* 8.3. The analogy is structural, not ontological. The wake-sleep algorithm has an explicit loss function; biological dreaming does not. The phenomenological richness of dreaming has no analogue in Boltzmann machines [29]. Whether the parallel reflects a deep computational principle or a superficial resemblance remains open.

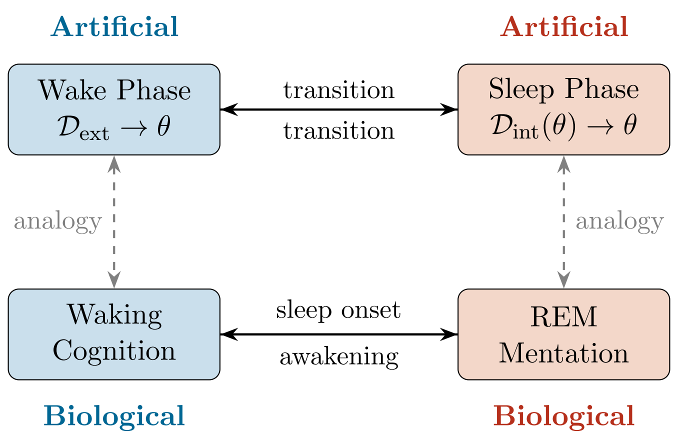


Figure 5: Structural analogy between the wake-sleep algorithm (top) and biological waking-dreaming cycle (bottom). Dashed lines indicate analogical correspondence; solid arrows indicate transitions. The analogy is structural, not ontological.

# 9 Statistical Psychogeography of Collective Dream Motifs

## 9.1 Phenomenological Motivation

Dream content analysis has documented spatial motifs recurring across individuals, cultures, and periods: impossible houses, endless staircases, missed transit connections, inescapable schools [9, 11, 12]. This section proposes a quantitative approach.

## 9.2 Formal Construction

**Definition 9.1** (Dream Report Corpus)**.** A corpus $\mathcal{R} = \{r_1, \ldots, r_N\}$ of textual dream reports, each annotated with demographic metadata.

**Definition 9.2** (Spatial Motif Extraction)**.** A function $\mu : \mathcal{R} \to \mathcal{P}(\mathcal{T})$ maps each report to a set of spatial motif vectors in $\mathcal{T}$.

**Definition 9.3** (Collective Motif Distribution)**.** The empirical distribution over $\mathcal{T}$:

$$\hat{P}(\mathbf{t}) = \frac{1}{|\mathcal{M}|} \sum_{r \in \mathcal{R}} \sum_{\mathbf{t}_j \in \mu(r)} \delta(\mathbf{t} - \mathbf{t}_j), \tag{19}$$

where $\mathcal{M} = \bigcup_r \mu(r)$ is the total motif set.

**Definition 9.4** (Motif Cluster)**.** A connected region $C \subset \mathcal{T}$ identified by density-based clustering of $\hat{P}$, with prevalence $\hat{P}(C) = \int_C \hat{P}(\mathbf{t})\, d\mathbf{t}$.

**Conjecture 9.5** (Existence of Universal Motif Clusters)**.** *There exist motif clusters $C_1, \ldots, C_K$ with $K$ small (plausibly $K \leq 20$) that account for a substantial fraction ($> 50\%$) of total motif density across culturally diverse corpora. These may correspond to spatial archetypes arising from the interaction of universal neuroanatomical constraints with common developmental experiences [30].*

*Caveat* 9.6. Conjecture 9.5 is stated as a conjecture because its truth is empirical. Obstacles include: inter-rater variability in spatial coding [11]; possible inadequacy of the $\mathcal{T}$ parameterization; insufficient corpus diversity; the possibility that apparent universality is linguistic rather than experiential; and the continuity hypothesis's emphasis on individual differences [27].

**Proposition 9.7** (Demographic Modulation)**.** *Let $\hat{P}_G$ be the motif distribution restricted to subgroup $G$. If $\Pi$ depends on lived experience:*

$$D_{KL}(\hat{P}_G \| \hat{P}) > 0 \quad \textit{for most non-trivial } G. \tag{20}$$

*Proof.* If the affective projection $\Pi$ depends on context (Definition 4.3), and different groups differ in affective landscapes, the pushforward $\hat{P}_G = \Pi_\# \mathcal{E}_G$ generically differs from $\hat{P} = \Pi_\# \mathcal{E}$. Exact equality of pushforward measures under a smooth map is a non-generic condition. □ □

# 10 Probabilistic Attractors and Collapse Dynamics

## 10.1 Motivation: From Deterministic to Stochastic Selection

The attractor theory developed in Section 3 treats dream-space selection as a deterministic process: given an initial condition $\mathbf{s}_0$, the flow $F_D$ carries the system to a unique attractor $A_k$, determined entirely by the basin $\mathfrak{B}(A_k)$ in which $\mathbf{s}_0$ falls. This account, while mathematically clean, faces a phenomenological difficulty. Dream experience at sleep onset does not typically begin with a fully formed, definite spatial environment. Instead, the hypnagogic period is characterized by fragmented, overlapping, uncommitted imagery—fleeting impressions that dissolve and recombine before a coherent dream scene eventually "crystallizes" [1, 2]. This phenomenology suggests that the initial phase of dreaming involves not a deterministic trajectory toward a single attractor but a probabilistic process in which multiple potential dream configurations coexist, compete, and are eventually resolved into one.

The present section develops this observation formally, introducing stochastic attractor dynamics and drawing a careful structural analogy with the selection problem in quantum measurement theory. The analogy is *structural*, not ontological: no claim is made that quantum mechanical processes are involved in dream-state selection. The parallel concerns the formal pattern of probabilistic resolution among discrete alternatives, a pattern that arises in both quantum measurement and noise-driven dynamical systems for reasons that may be mathematically related but are physically distinct.

## 10.2 Stochastic Dream Dynamics

The deterministic flow of Equation (7) is now generalized to include stochastic perturbations, reflecting the inherent noise in neural dynamics—thermal fluctuations, synaptic variability, and spontaneous firing events.

**Definition 10.1** (Stochastic Dream Dynamics)**.** The *stochastic dream dynamics* is described by the Itô stochastic differential equation:

$$d\mathbf{s} = F_D(\mathbf{s})\, dt + \sqrt{2\varepsilon}\, \Sigma(\mathbf{s})\, d\mathbf{W}_t, \tag{21}$$

where $F_D$ is the deterministic drift (the endogenous dream dynamics of Definition 3.1), $\varepsilon > 0$ is the noise intensity, $\Sigma(\mathbf{s})$ is a state-dependent diffusion matrix encoding the spatial structure of neural noise, and $\mathbf{W}_t$ is a standard $n$-dimensional Wiener process.

The Itô stochastic differential equation extends the deterministic evolution law by adding a random forcing term. The Wiener process $\mathbf{W}_t$ is the mathematical idealization of continuous random fluctuations: at each instant, an unpredictable displacement is added to the trajectory. The parameter $\varepsilon$ controls the magnitude of these fluctuations. When $\varepsilon$ is very small, trajectories closely follow the deterministic flow and settle reliably into the nearest attractor. When $\varepsilon$ is appreciable, the system may be deflected from one basin of attraction into another before settling—introducing genuine probabilistic selection among attractors.

**Definition 10.2** (Occupation Probability)**.** For a stochastic trajectory $\mathbf{s}(t)$ governed by Equation (21) with initial condition $\mathbf{s}(0) = \mathbf{s}_0$, the *occupation probability* of attractor $A_k$ over a time horizon $T$ is:

$$p_k(\mathbf{s}_0, T) = \mathbb{P}\left[\exists\, t^* \in [0, T] : \mathbf{s}(t) \in \mathcal{U}_k \text{ for all } t \in [t^*, T]\right], \tag{22}$$

where $\mathcal{U}_k$ is a neighborhood of $A_k$. This is the probability that the trajectory, starting from $\mathbf{s}_0$, has settled into the vicinity of $A_k$ by time $T$ and remains there.

**Proposition 10.3** (Kramers-Type Attractor Selection)**.** *In the regime of small noise ($\varepsilon \ll 1$), the occupation probability $p_k$ satisfies the asymptotic relation:*

$$p_k(\mathbf{s}_0, T) \;\approx\; \frac{\exp\bigl(-\Delta V_k/\varepsilon\bigr)}{\sum_{j=1}^{K} \exp\bigl(-\Delta V_j/\varepsilon\bigr)} \quad \text{as } \varepsilon \to 0, \tag{23}$$

*where $\Delta V_k$ is the quasi-potential barrier between $\mathbf{s}_0$ and attractor $A_k$, and $K$ is the number of attractors.*

The quasi-potential $V(\mathbf{s})$ generalizes the notion of an energy landscape to systems that are not necessarily gradient flows. It measures the "cost" of reaching a given state against the deterministic drift: states that the drift naturally carries the system toward have low quasi-potential, while states that require the system to move against its natural tendency have high quasi-potential. The barrier $\Delta V_k$ is the minimum quasi-potential that must be traversed to reach attractor $A_k$ from the initial state $\mathbf{s}_0$. Equation (23) states that the probability of settling into a given attractor decreases exponentially with the height of the barrier, modulated by the noise intensity $\varepsilon$. This is a form of the classical Kramers escape rate theory [6].

The expression in Equation (23) has a recognizable mathematical form: it is a Boltzmann-Gibbs distribution, with the noise intensity $\varepsilon$ playing the role of temperature. At low noise (low "temperature"), the system almost certainly settles into the attractor with the lowest barrier—the most accessible configuration. At higher noise, the distribution broadens, and less accessible attractors receive non-negligible probability.

*Proof.* The proof relies on the Freidlin-Wentzell theory of large deviations for diffusion processes. For the stochastic system Equation (21) with $\varepsilon \to 0$, the probability of a trajectory deviating from the deterministic flow by following a path $\varphi(t)$ over the interval $[0, T]$ is governed by the action functional:

$$I_T[\varphi] = \frac{1}{2} \int_0^T \left\| \Sigma(\varphi(t))^{-1} \bigl(\dot{\varphi}(t) - F_D(\varphi(t))\bigr) \right\|^2 dt.$$

The quasi-potential from $\mathbf{s}_0$ to a set $B$ is $V(\mathbf{s}_0, B) = \inf\{I_T[\varphi] : \varphi(0) = \mathbf{s}_0, \varphi(T) \in B, T > 0\}$. The probability of the trajectory reaching $B$ before other attractors satisfies $\mathbb{P} \sim \exp(-V(\mathbf{s}_0, B)/\varepsilon)$ as $\varepsilon \to 0$. Normalizing over all attractors yields the Boltzmann-type expression. The rigorous statement requires uniform hyperbolicity of the attractors and regularity of the quasi-potential; see Strogatz [6] for the framework and Kuznetsov [17] for technical conditions. □ □

## 10.3 The Pre-Crystallization State

Before the stochastic dynamics has settled into a specific attractor, the system occupies a state that is not committed to any single dream configuration. This transient, uncommitted phase can be formalized as a probability distribution over the set of attractors.

**Definition 10.4** (Dream Superposition State)**.** The *dream superposition state* at time $t$ is the probability vector:

$$\mathbf{p}(t) = \big(p_1(t),\ p_2(t),\ \ldots,\ p_K(t)\big), \quad p_k(t) = \mathbb{P}\big[\mathbf{s}(t) \in \mathfrak{B}(A_k)\big], \tag{24}$$

where $K$ is the number of attractors and $\mathfrak{B}(A_k)$ is the basin of $A_k$.

The term “superposition” is used here by analogy with quantum mechanics, where a system may exist in a probabilistic combination of distinct eigenstates before measurement. The analogy is terminological and structural, not physical. In the present context, $\mathbf{p}(t)$ is a classical probability distribution, not a quantum state vector: it does not involve complex amplitudes, interference, or entanglement. The system is in one basin at any given time; the distribution $\mathbf{p}(t)$ reflects epistemic uncertainty about which basin it currently occupies, not a superposition in the quantum-mechanical sense.

**Definition 10.5** (Crystallization Event)**.** A *crystallization event* at time $t_c$ occurs when the superposition state $\mathbf{p}(t)$ transitions from a distributed configuration (multiple $p_k > 0$) to a concentrated one (a single $p_{k^*} \approx 1$, all others $\approx 0$):

$$H(\mathbf{p}(t_c^-)) > H_{\text{thresh}} \quad \text{and} \quad H(\mathbf{p}(t_c^+)) < H_{\text{thresh}}, \tag{25}$$

where $H(\mathbf{p}) = -\sum_k p_k \ln p_k$ is the Shannon entropy and $H_{\text{thresh}}$ is a threshold value below which the distribution is considered effectively concentrated.

The Shannon entropy $H(\mathbf{p})$ quantifies the uncertainty in the distribution. When $H$ is high, multiple attractors have comparable probability and the dream has not yet committed to a specific spatial configuration. When $H$ drops below threshold, the system has effectively selected one attractor: the dream has "crystallized" into a definite environment.

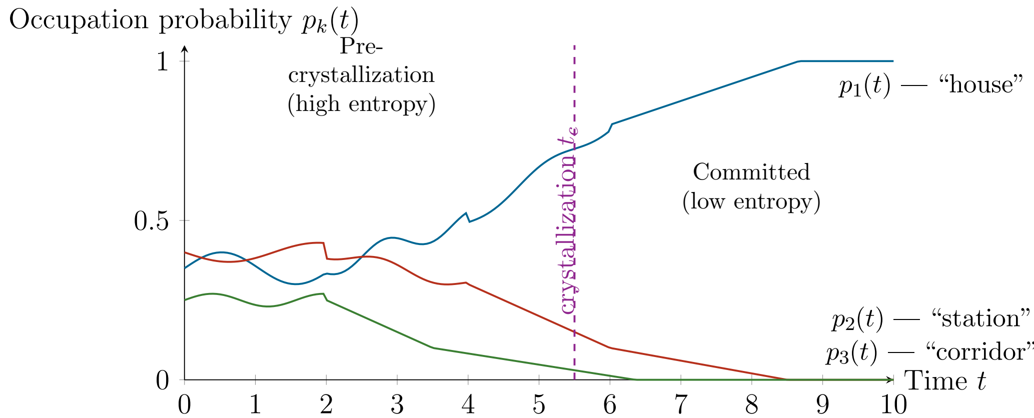


Figure 6: Temporal evolution of occupation probabilities during dream onset. Before crystallization ($t < t_c$), multiple attractors carry substantial probability—the dream is "uncommitted." At $t_c$, one attractor (here $A_1$, the "house") achieves dominance. After crystallization, the dream unfolds within the selected attractor's basin. The fluctuations in the pre-crystallization phase reflect the stochastic dynamics of Equation (21). Labels are illustrative.

## 10.4 Structural Analogy with Measurement-Like Selection

The crystallization dynamics described above bear a formal resemblance to certain features of quantum measurement. In both cases, a system transitions from a state involving multiple coexisting possibilities to a state involving a single, definite outcome. In both cases, the selection is probabilistic: the specific outcome is not determined by the pre-selection state alone but depends on stochastic factors. And in both cases, the transition involves an irreversible loss of information about the alternatives that were not selected.

The purpose of this subsection is to make this analogy precise—and, equally, to delineate its limits.

**Definition 10.6** (Formal Selection Process)**.** A *formal selection process* is a triple $(\mathcal{H}, \mathbf{p}_{\text{pre}}, \mathbf{p}_{\text{post}})$ where:

(i) $\mathcal{H} = \{H_1, H_2, \ldots, H_K\}$ is a finite set of *outcome configurations.*

(ii) $\mathbf{p}_{\text{pre}} = (p_1, \ldots, p_K)$ is a probability distribution over $\mathcal{H}$ representing the pre-selection state, with $H(\mathbf{p}_{\text{pre}}) > 0$ (non-trivial uncertainty).

(iii) $\mathbf{p}_{\text{post}}$ is a distribution concentrated on a single outcome: $p_{k^*} = 1$, $p_j = 0$ for $j \neq k^*$.

The *information loss* of the selection is:

$$\Delta I = H(\mathbf{p}_{\text{pre}}) - H(\mathbf{p}_{\text{post}}) = H(\mathbf{p}_{\text{pre}}) \geq 0. \tag{26}$$

This definition is intentionally abstract: it captures the common formal structure shared by quantum measurement, classical noise-driven attractor selection, and other processes in which a distributed state resolves into a definite one. The following table identifies the correspondences and disanalogies.

Table 1: Structural comparison between quantum measurement and stochastic dream crystallization. Correspondences are formal; no physical identity is claimed.

| **Feature** | **Quantum Measurement** | **Dream Crystallization** |
|---|---|---|
| Pre-selection state | Superposition $\lvert\psi\rangle = \sum_k c_k \lvert k\rangle$ | Probability vector $\mathbf{p}(t)$ over attractors |
| Outcome probabilities | $p_k = \lvert c_k\rvert^2$ (Born rule) | $p_k \approx \exp(-\Delta V_k/\varepsilon)/Z$ (Kramers) |
| Selection mechanism | Interaction with measurement apparatus | Noise-driven basin crossing |
| Irreversibility | Wavefunction collapse | Loss of access to unchosen attractors |
| Interference | Yes (complex amplitudes) | **No** (classical probabilities) |
| Entanglement | Yes | **No** |
| Mathematical space | Hilbert space $\mathcal{H}$ | Phase space $\mathcal{S}$ with measure |
| Nature of uncertainty | Ontological (standard interpretation) | Epistemic (classical noise) |

**Proposition 10.7** (Entropy Reduction at Crystallization)**.** *Let* $\mathbf{p}(t)$ *be the dream superposition state of Definition 10.4. As the system approaches crystallization at* $t_c$*:*

$$\left.\frac{dH(\mathbf{p})}{dt}\right|_{t \to t_c^-} < 0, \tag{27}$$

*and the total entropy reduction satisfies:*

$$\Delta H = H(\mathbf{p}(0)) - H(\mathbf{p}(t_c)) \leq \ln K, \tag{28}$$

*where $K$ is the number of competing attractors. The upper bound $\ln K$ is achieved when the initial distribution is uniform and the final state is fully concentrated.*

The entropy reduction quantifies the information gained (or, equivalently, the uncertainty resolved) by the crystallization event. A dream that crystallizes from among $K = 5$ competing spatial configurations eliminates at most $\ln 5 \approx 1.6$ nats of uncertainty. This is formally identical to the information gain in a measurement that selects one outcome from $K$ equiprobable alternatives.

*Proof.* Since $\mathbf{p}(t_c)$ is concentrated on a single attractor, $H(\mathbf{p}(t_c)) \approx 0$. The maximum entropy of a distribution over $K$ outcomes is $\ln K$ (achieved by the uniform distribution). Therefore $\Delta H = H(\mathbf{p}(0)) - H(\mathbf{p}(t_c)) \leq \ln K - 0 = \ln K$. That $dH/dt < 0$ near $t_c$ follows from the convergence of $p_{k^*} \to 1$: as one component approaches unity, the entropy function, being strictly concave and achieving its minimum at the vertices of the probability simplex, necessarily decreases. □ □

## 10.5 The Quasi-Potential Landscape as a Dream Energy Surface

The quasi-potential $V(\mathbf{s})$ introduced in Proposition 10.3 provides a landscape interpretation of dream-space selection. Each attractor sits at a local minimum of $V$; the barriers between attractors determine the relative accessibility of different dream configurations.

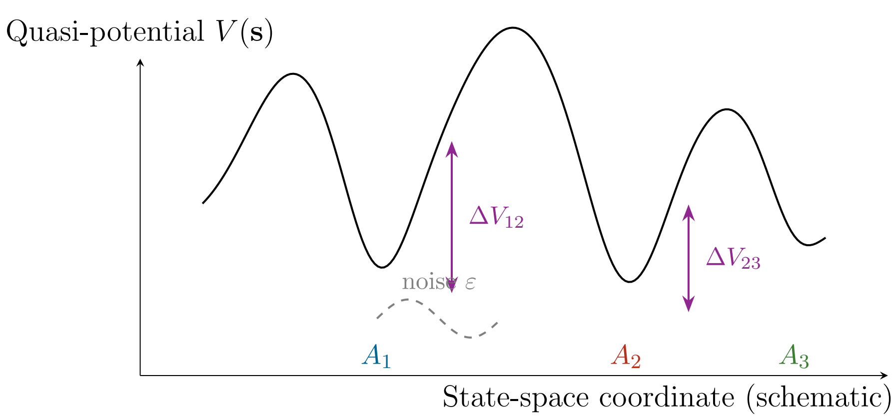


Figure 7: Quasi-potential landscape for stochastic dream dynamics. Attractors $(A_1, A_2, A_3)$ correspond to local minima. Barriers $(\Delta V_{12}, \Delta V_{23})$ determine transition probabilities: lower barriers yield higher crossing probability. At low noise $\varepsilon$, the system is trapped in the nearest well; at higher noise, inter-basin transitions become possible, yielding the probabilistic selection of Equation (23).

**Proposition 10.8** (Noise-Dependent Resolution Time)**.** *The expected time $\langle t_c \rangle$ until crystallization satisfies, for small $\varepsilon$:*

$$\langle t_c \rangle \;\sim\; C_0 \,\exp\!\left(\frac{\Delta V_{\min}}{\varepsilon}\right), \tag{29}$$

*where $\Delta V_{\min} = \min_k \Delta V_k$ is the lowest barrier to any attractor and $C_0$ is a prefactor depending on the local curvature of the quasi-potential at the initial state and at the barrier saddle.*

This result states that the time required for the dream to crystallize depends exponentially on the ratio of the smallest barrier height to the noise intensity. When the initial state is close to a basin boundary (low $\Delta V_{\min}$), crystallization is rapid. When the initial state lies in a region equidistant from several attractors (high $\Delta V_{\min}$), the system may wander for a prolonged period before committing—potentially corresponding to extended hypnagogic imagery.

*Proof.* This is a standard result from Kramers escape theory for multi-well systems. The expected transition time from a metastable state over a barrier of height $\Delta V$ in a noise-driven system with intensity $\varepsilon$ scales as $\exp(\Delta V/\varepsilon)$ in the small-noise limit, with the prefactor determined by the curvature at the well bottom and saddle point [6]. The crystallization time is bounded by the escape time over the lowest barrier. □ □

## 10.6 Implications for Dream Phenomenology

The probabilistic attractor model generates several qualitative predictions about dream experience that are at least partially consistent with reported phenomenology.

First, the model predicts that the hypnagogic period should exhibit characteristics of an uncommitted state: fragmented imagery, rapid shifting between incompatible spatial environments, and a sense of instability or "flickering" between alternatives. These features are indeed characteristic of sleep-onset mentation [1, 2].

Second, the model predicts that the probability of a given dream environment being selected depends on the quasi-potential landscape, which is shaped by long-term memory structure, recent experience, and current affective state. Environments associated with deep, well-established attractors (low quasi-potential minima) should recur more frequently than those associated with shallow attractors. This is consistent with the observation that recurring dream environments tend to involve emotionally significant or biographically important spatial configurations [9].

Third, the model predicts that external perturbations during sleep onset (sounds, temperature changes, bodily sensations) may bias the crystallization toward specific attractors by altering the effective initial condition $\mathbf{s}_0$. This is consistent with the well-documented phenomenon of dream incorporation, in which external stimuli during REM sleep become integrated into dream content [2].

*Caveat* 10.9. The analogy with quantum measurement, while structurally coherent, carries significant risks of overinterpretation and must be handled with care. The following disanalogies are critical:

(i) The pre-crystallization state in dreaming is a *classical* probability distribution reflecting epistemic uncertainty about a determinate (but unknown) system state. In quantum mechanics, the pre-measurement state is typically interpreted as representing ontological indeterminacy. This distinction is fundamental: the mathematics of probability distributions and the mathematics of Hilbert-space amplitudes are structurally different (the latter permits interference, the former does not).

(ii) Quantum measurement involves non-local correlations (entanglement) and violates Bell inequalities. Nothing in the dream crystallization model has this character.

(iii) The quasi-potential landscape is a classical construct derivable from the underlying stochastic differential equation. There is no analogue of the measurement problem—the "collapse" is fully explained by the dynamics of Equation (21) without requiring any additional postulate.

(iv) Some authors have proposed that quantum effects play a functional role in consciousness [10]. The present framework takes no position on this question. The formal analogy developed here holds regardless of whether any quantum processes are involved in neural dynamics; it is an analogy between mathematical structures, not between physical mechanisms.

(v) There exists a risk that the use of quantum-mechanical terminology ("superposition," "collapse") may suggest a deeper physical connection than the mathematics supports. This terminology is adopted solely for its communicative value in highlighting the structural parallel; it should not be interpreted as a commitment to quantum-mind hypotheses.

# 11 Interrogative Collapse and the Ad Hoc Structure of Dream Reality

## 11.1 Phenomenological Motivation

The crystallization model of Section 10 addresses how a definite dream environment is selected from among competing attractors at sleep onset. However, it does not capture a subtler and perhaps more fundamental feature of dream phenomenology: the observation that dream content appears to exist in a state of radical indeterminacy *throughout* the dream, not only at its inception. A dream corridor does not, in any obvious sense, "have" a destination until the dreamer proceeds along it. A dream figure does not "possess" a definite identity until something in the dream narrative requires it. The spatial layout of a dream building is not specified beyond the room currently occupied.

This indeterminacy is not experienced as vagueness or incompleteness. The dream feels complete at each moment. But it achieves this completeness by a striking strategy: it does not pre-specify content that is not currently under interrogation. The dream is, in this sense, *lazy*—it generates detail only on demand, and the detail it generates is confabulated ad hoc from available associative material rather than retrieved from a pre-existing blueprint.

The critical observation is that the act of internal cognitive questioning—"where am I?", "who is this?", "how did I get here?"—functions as a collapse operator, forcing indeterminate content to resolve into something specific. Before the question, the content is multiply compatible with many possible specifications. After the question, it is committed to one. And the commitment is irreversible within the dream: once the corridor has been determined to lead to a kitchen, it leads to a kitchen, and subsequent experience is consistent with this determination (until, perhaps, a new interrogation destabilizes it).

This section formalizes this process as a model of interrogative collapse operating on a space of indeterminate dream content.

## 11.2 The Indeterminacy Structure of Dream Content

The central proposal is that the dream state, at any given moment, specifies only a subset of its potentially specifiable features, and that the unspecified features exist in a state of superposed compatibility.

**Definition 11.1** (Dream Content State)**.** A *dream content state* is a pair $(\mathbf{d}, \mathcal{U})$ where:

(i) $\mathbf{d} \in \mathbb{R}^N$ is the *determinate content vector*, specifying the values of $N$ content features (spatial layout, character identities, temporal context, narrative situation, etc.).

(ii) $\mathcal{U} \subseteq \{1, 2, \ldots, N\}$ is the *indeterminacy set*: the set of feature indices that are currently unresolved. For $i \in \mathcal{U}$, the value $d_i$ is undefined (or, more precisely, is drawn from a distribution rather than taking a definite value).

The *resolved content* is the restriction of $\mathbf{d}$ to the complement $\mathcal{U}^c = \{1, \ldots, N\} \setminus \mathcal{U}$. The *degree of indeterminacy* is $|\mathcal{U}|/N \in [0, 1]$.

A fully determinate state has $\mathcal{U} = \varnothing$: every feature is specified. A maximally indeterminate state has $\mathcal{U} = \{1, \ldots, N\}$: nothing is specified. The claim is that ordinary (non-lucid) dreaming operates at high degrees of indeterminacy, with only the features currently relevant to the experiential foreground in a resolved state.

**Definition 11.2** (Compatibility Ensemble)**.** For a dream content state $(\mathbf{d}, \mathcal{U})$, the *compatibility ensemble* is the set of all complete content vectors consistent with the resolved content:

$$\mathcal{C}(\mathbf{d}, \mathcal{U}) = \left\{ \mathbf{d}' \in \mathbb{R}^N \;\middle|\; d'_i = d_i \text{ for all } i \in \mathcal{U}^c \right\}. \tag{30}$$

The ensemble may be equipped with a probability measure $\nu$ reflecting the associative landscape: the likelihood that a given completion would be generated if the corresponding features were to be resolved.

The compatibility ensemble $\mathcal{C}$ is the set of all "possible dreams" consistent with what has been determined so far. When few features are resolved (large $\mathcal{U}$), the ensemble is vast: the dream is compatible with many different continuations. As features are resolved, the ensemble contracts. This contraction is the formal content of what is experienced as the dream "filling in" its details.

## 11.3 The Interrogative Collapse Operator

The mechanism by which indeterminate features become resolved is modeled as an interrogative operator: a cognitive event, implicit or explicit, that demands the specification of a currently unresolved feature.

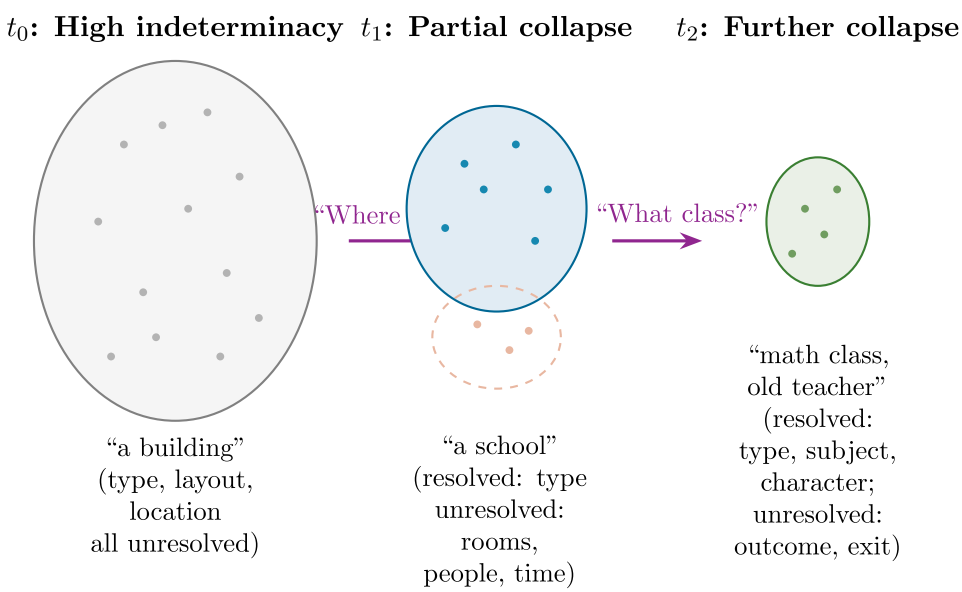


Figure 8: Progressive collapse of the compatibility ensemble through successive interrogative acts. Each internal question ("Where am I?", "What class?") resolves additional features, contracting the set of compatible dream continuations. The resolved content is generated ad hoc at the moment of interrogation. Eliminated alternatives (dashed, faded) become inaccessible.

**Definition 11.3** (Interrogative Operator)**.** An *interrogative operator* $Q_i$ targets the $i$-th content feature ($i \in \mathcal{U}$) and resolves it by sampling from a conditional distribution:

$$Q_i : (\mathbf{d}, \mathcal{U}) \ \mapsto \ (\mathbf{d}', \mathcal{U} \setminus \{i\}), \tag{31}$$

where $d'_j = d_j$ for $j \neq i$, and $d'_i$ is drawn from the *confabulation distribution*:

$$d'_i \sim \nu_i\big(\cdot \mid \{d_j : j \in \mathcal{U}^c\}\big), \tag{32}$$

a distribution conditioned on the already-resolved features. The indeterminacy set shrinks by one element: $|\mathcal{U}'| = |\mathcal{U}| - 1$.

The confabulation distribution $\nu_i$ encodes the associative, affective, and mnemonic resources available to the dreaming mind at the moment of interrogation. It is conditioned on the already-resolved features to ensure consistency: if the building has already been determined to be a school, then the question "what room is this?" will sample from school-compatible rooms, not kitchens or forests. However, the specific room selected is not pre-determined; it is generated at the moment the question is posed.

This is the key distinction from the crystallization model of Section 10: there, the system selects among pre-existing attractors; here, the specific content is *constructed* at the moment of resolution. The dream does not choose among ready-made worlds; it assembles its world piecewise, on demand.

**Definition 11.4** (Confabulation Distribution)**.** The confabulation distribution for feature $i$, given the resolved content $\mathbf{d}_{\mathcal{U}^c}$, is:

$$\nu_i(x \mid \mathbf{d}_{\mathcal{U}^c}) = \frac{1}{Z_i} \exp\left( -\frac{1}{\gamma}\, E_i(x, \mathbf{d}_{\mathcal{U}^c}) \right), \tag{33}$$

where $E_i(x, \mathbf{d}_{\mathcal{U}^c})$ is an *associative energy* measuring the incompatibility of the value $x$ for feature $i$ with the already-resolved content, $\gamma > 0$ is a *confabulation temperature* controlling the randomness of the resolution, and $Z_i$ is the normalizing constant.

The associative energy $E_i$ encodes the long-term associative structure of the dreamer's memory: values that are strongly associated with the current resolved context have low energy and high probability; values that are weakly associated have high energy and low probability. The temperature $\gamma$ modulates the sharpness of this selection. At low $\gamma$ ("cold" confabulation), the system selects the most strongly associated value with near-certainty, producing stereotypical, predictable dream content. At high $\gamma$ ("hot" confabulation), the system samples broadly, producing bizarre, loosely connected content.

**Proposition 11.5** (Monotonic Entropy Reduction Under Sequential Interrogation)**.** *Let $H_k$ denote the entropy of the compatibility ensemble after $k$ interrogative operations have been applied. Then:*

$$H_0 \geq H_1 \geq H_2 \geq \cdots \geq H_N = 0, \tag{34}$$

*where $H_0$ is the entropy of the initial (maximally indeterminate) ensemble and $H_N = 0$ corresponds to full determination. The expected entropy reduction per interrogation is:*

$$\mathbb{E}[H_k - H_{k+1}] = H(\nu_{i_{k+1}} \mid \mathbf{d}_{\mathcal{U}_k^c}), \tag{35}$$

*the conditional entropy of the confabulation distribution for the $(k+1)$-th feature given the current resolved state.*

Each interrogation reduces the overall indeterminacy of the dream by resolving one feature. The amount of reduction equals the entropy of the distribution from which the resolved value is drawn. If the confabulation distribution is sharply peaked (the "answer" is nearly determined by context), little entropy is lost; if it is broad (the answer is genuinely uncertain), much entropy is lost.

*Proof.* The entropy of the compatibility ensemble is $H_k = \sum_{i \in \mathcal{U}_k} H(\nu_i \mid \mathbf{d}_{\mathcal{U}_k^c})$, where the sum is over the remaining unresolved features. Each interrogation removes one term from this sum and may alter the conditioning of the remaining terms. Since conditioning cannot increase entropy (by the data processing inequality), $H_{k+1} \leq H_k$. After $N$ interrogations, $\mathcal{U}_N = \varnothing$, and $H_N = 0$. The expected per-step reduction is the entropy of the removed feature's confabulation distribution. □ □

## 11.4 The Ad Hoc Consistency Principle

A central feature of dream confabulation is that it typically produces content that is *locally consistent*—consistent with the immediately surrounding resolved content—but may be *globally inconsistent*—inconsistent with content resolved at earlier or later points in the dream. The same building may contain a room on the second floor that was earlier experienced on the ground floor. The same person may have been named differently in an earlier dream scene. These inconsistencies are not perceived as contradictions during the dream (cf. the affective validation of Section 5), but they are detectable upon waking reflection.

**Definition 11.6** (Local Consistency Radius)**.** The *local consistency radius* $r > 0$ defines the scope within which the confabulation distribution enforces consistency. Feature $i$ is resolved consistently with feature $j$ if $|i - j|_{\text{narrative}} \leq r$, where $|\cdot|_{\text{narrative}}$ measures the narrative distance between features (temporal separation in the dream narrative, spatial proximity in the dream environment, or associative distance in the content graph). Features separated by more than $r$ are resolved independently, permitting global inconsistency.

**Proposition 11.7** (Emergence of Global Inconsistency)**.** *If the local consistency radius $r$ is finite and the total number of resolved features $|\mathcal{U}^c|$ exceeds a critical value $N_c \sim r \cdot d$ (where $d$ is the effective dimensionality of the content space), then the probability of at least one global inconsistency approaches 1:*

$$\mathbb{P}\big[\textit{at least one inconsistency}\big] \;\geq\; 1 - \exp\!\left(-c\,\frac{|\mathcal{U}^c|^2}{r \cdot N}\right), \tag{36}$$

*where $c > 0$ is a constant depending on the structure of the confabulation distributions.*

This result formalizes the intuition that dreams become more internally contradictory as they become more detailed. A short, sparse dream (few resolved features) is likely to be self-consistent simply because there are few features to conflict. A long, detailed dream (many resolved features) will almost certainly contain inconsistencies, because the local consistency constraint cannot enforce global coherence.

*Proof.* The proof proceeds by a coupon-collector-type argument. Consider resolved features as nodes in a graph, with edges connecting features within narrative distance $r$. Consistency is enforced along edges. For features separated by more than $r$, their values are conditionally independent given the resolved content within distance $r$ of each. As the number of resolved features grows, the number of pairs at distance $> r$ grows quadratically ($\sim |\mathcal{U}^c|^2/2$), while the consistency constraint only covers pairs at distance $\leq r$ ($\sim |\mathcal{U}^c| \cdot r$). Each unconstrained pair has a probability $p_{\text{incon}} > 0$ of being mutually inconsistent. By the union bound complement, the probability of zero inconsistencies is at most $(1 - p_{\text{incon}})^{\binom{m}{2}/rN}$ where $m = |\mathcal{U}^c|$, which decreases exponentially in $m^2/(rN)$. □ □

## 11.5 Interrogation Rate and Dream Stability

The rate at which interrogative operators are applied determines the phenomenological character of the dream. Slow interrogation (few features resolved per unit time) preserves high indeterminacy and yields the fluid, underdetermined quality characteristic of ordinary dreaming. Rapid interrogation (many features resolved per unit time) forces the dream into high determination but, by Proposition 11.7, also increases the probability of detectable inconsistency.

**Definition 11.8** (Interrogation Rate)**.** The *interrogation rate* $\eta(t) \geq 0$ is the expected number of interrogative operators applied per unit dream-time. The cumulative number of resolved features at time $t$ is:

$$R(t) = |\mathcal{U}^c(t)| = R_0 + \int_0^t \eta(s)\, ds, \tag{37}$$

where $R_0$ is the number of features already resolved at dream onset (e.g., through the crystallization process of Section 10).

**Proposition 11.9** (Instability Under High Interrogation Rate)**.** *There exists a critical interrogation rate $\eta^*$ such that for $\eta > \eta^*$, the expected number of detectable* *inconsistencies per unit time exceeds a threshold at which the dream's internal coherence is compromised:*

$$\eta^* \approx \sqrt{\frac{r \cdot N \cdot \ln(1/\delta)}{c \cdot t}}, \tag{38}$$

*where $\delta$ is the maximum tolerable inconsistency probability and $t$ is the elapsed dream time. For $\eta \gg \eta^*$, the rate of inconsistency production may exceed the capacity of affective validation (Section 5) to absorb contradictions, potentially triggering either dream dissolution (awakening) or the onset of lucidity.*

This result connects directly to the phenomenology of lucid dreaming. The lucid dreamer, by definition, applies waking-type cognitive interrogation to dream content—asking "is this real?", examining objects closely, reading text, checking clocks. Each such act is an interrogative operator that forces resolution of previously indeterminate features. If the rate of such interrogation exceeds $\eta^*$, the resulting cascade of inconsistencies may destabilize the dream. This is consistent with the well-documented difficulty of maintaining lucid dreams: experienced lucid dreamers report that excessive "testing" of the dream environment tends to cause the dream to dissolve or the dreamer to awaken [16, 22].

*Proof.* From Proposition 11.7, the probability of at least one inconsistency is $\geq 1 - \exp(-cR(t)^2/(rN))$. Setting this equal to $1 - \delta$ and solving for $R(t)$: $R(t) = \sqrt{rN \ln(1/\delta)/c}$. Since $R(t) \approx \eta \cdot t$ (for constant $\eta$), the critical rate is $\eta^* = R(t)/(t) = \sqrt{rN \ln(1/\delta)/(c\,t^2)}$. For $\eta > \eta^*$, the inconsistency probability exceeds $1 - \delta$, and the rate of new inconsistencies grows as $d\mathbb{P}/dt \sim 2c\eta^2 t/(rN)$, which can outpace the smoothing capacity of affective validation. □ □

## 11.6 Comparison with the Crystallization Model

The interrogative collapse model of the present section and the crystallization model of Section 10 operate at different scales and address different aspects of dream indeterminacy. Table 2 summarizes the distinctions.

The computational analogy in the final row is suggestive. Crystallization resembles a phase transition in which the system commits to a macroscopic configuration. Interrogative collapse resembles lazy evaluation in programming: values are not computed until they are needed, and once computed, they are cached for local reuse but may become stale if the context shifts. Alternatively, it resembles just-in-time compilation: the dream "compiles" its content from high-level associative specifications only at the moment of experiential access.

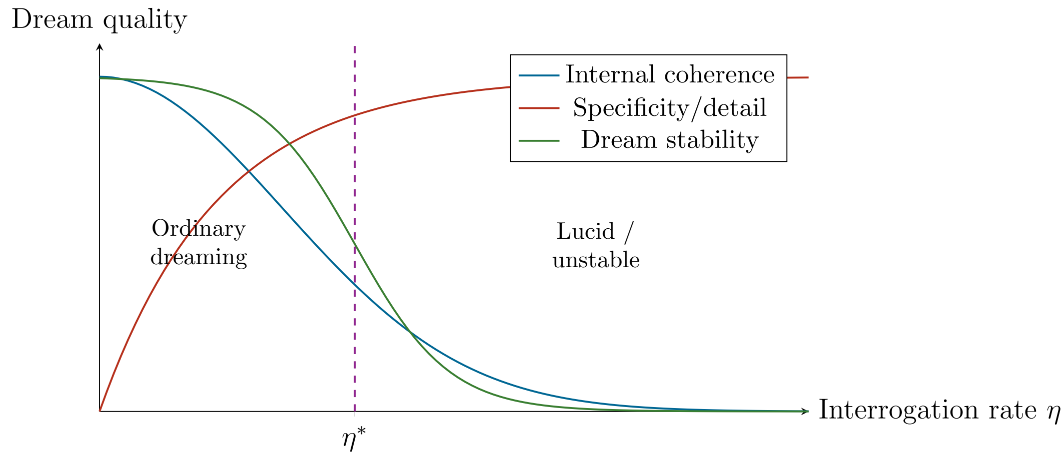


Figure 9: Trade-off between interrogation rate and dream qualities. At low interrogation rates (ordinary dreaming), coherence is high but detail is low—the dream is self-consistent because it is largely unspecified. At high rates (lucid or analytically active dreaming), detail increases but coherence and stability decrease as inconsistencies accumulate faster than affective validation can absorb them. The critical rate $\eta^*$ marks the transition to instability.

Table 2: Comparison of the two collapse models.

| **Feature** | **Crystallization (Section 10)** | **Interrogative Collapse** |
|---|---|---|
| Scope | Selection of global dream environment | Resolution of local content features |
| Timing | Sleep onset (single event) | Throughout the dream (ongoing) |
| Mechanism | Noise-driven basin selection | Cognitive interrogation triggers confabulation |
| Outcome space | Pre-existing attractors | Ad hoc constructions from associative material |
| Reversibility | Effectively irreversible | Locally irreversible, globally mutable |
| Analogue | Phase transition / measurement | Lazy evaluation / just-in-time compilation |

*Caveat* 11.10. The interrogative collapse model faces several significant objections.

(i) The model assumes that dream content is genuinely indeterminate rather than merely inaccessible. An alternative account holds that the dream is fully specified at each moment but that only a fraction of this specification is accessible to the experiential subject. On this account, there is no indeterminacy—only limited access. Distinguishing these hypotheses empirically appears extremely difficult, as both predict the same phenomenological reports.

(ii) The notion of an "interrogative operator" presupposes that dream cognition involves something like questioning, even if implicit. It is unclear whether the dreaming mind engages in anything that can meaningfully be described as interrogation, or whether the apparent demand-driven generation of content is better explained by associative spreading activation without any interrogative component.

(iii) The confabulation distribution $\nu_i$ is defined abstractly; its actual form, if it exists, would depend on the detailed structure of long-term memory, current neurochemical state, and recent experience in ways that the present formalism does not specify.

(iv) The local consistency radius $r$ is assumed to be finite and fixed. In practice, it may vary with dream state, emotional intensity, and the nature of the content being resolved.

(v) The model generates qualitative predictions (e.g., that lucid interrogation destabilizes dreams) that are consistent with existing reports but are not uniquely predicted by this model—other accounts of lucid dream instability (e.g., arousal-based models) exist and may be equally or more parsimonious.

# 12 The Dream Interface as Virtual Complexity

## 12.1 Motivation: The Complexity Gap

The preceding sections have modeled dream content as the output of a generative system—an endogenous dynamics operating under specific parameter regimes, projecting affective states into spatial configurations, confabulating content on demand. A natural question arises: how complex is this system, and how complex is its output? These are not the same question. A simple rule can produce apparently complex behavior (as in cellular automata), and a complex system can produce simple output (as in an equilibrium thermodynamic system). The relationship between the complexity of the generating mechanism and the complexity of the generated experience is central to understanding what dreaming *is* as an information-processing phenomenon.

Three distinct notions of complexity bear on this question, each capturing a different aspect of the problem. Kolmogorov complexity [31] measures the length of the shortest program that generates a given string—it captures *generative compactness*. Shannon entropy [32] measures the average unpredictability of a signal—it captures *informational richness*. Gell-Mann effective complexity [33, 34] captures the length of a concise description of the *regularities* in a string, as opposed to its random components—it captures *structured content*. The proposal of this section is that dreaming is characterized by a distinctive dissociation among these three quantities: low Kolmogorov complexity, high Shannon entropy, and moderate effective complexity. This dissociation is what is meant by *virtual complexity*—the production of experientially rich, apparently complex content from a generatively compact mechanism.

## 12.2 Formal Definitions

**Definition 12.1** (Dream Output Stream)**.** A *dream output stream* $\omega = (\omega_1, \omega_2, \ldots, \omega_T)$ is a sequence of experiential states sampled at discrete time steps from a dream episode. Each $\omega_t \in \Omega$ belongs to a finite alphabet $\Omega$ encoding the discretized content of the dream at time $t$ (spatial configuration, character presence, emotional tone, narrative event, etc.).

**Definition 12.2** (Kolmogorov Complexity of the Dream Generator)**.** The *Kolmogorov complexity* of the dream output stream $\omega$ is the length of the shortest program $\pi^*$ on a universal Turing machine $U$ that produces $\omega$:

$$K(\omega) = \min\{|\pi| : U(\pi) = \omega\}, \tag{39}$$

where $|\pi|$ denotes the length of program $\pi$ in bits.

Kolmogorov complexity is uncomputable in general, but its conceptual content is clear: it measures the minimum amount of information required to *specify* the dream, not the amount of information the dream appears to *contain*. A dream that can be generated by a compact rule ("traverse the attractor landscape under confabulation distribution $\nu$ with temperature $\gamma$") has low Kolmogorov complexity regardless of how long, varied, or surprising its content appears.

**Definition 12.3** (Shannon Entropy Rate of the Dream Stream)**.** The *Shannon entropy rate* of the dream output stream is:

$$h(\omega) = \lim_{T\to\infty} \frac{1}{T} H(\omega_1, \omega_2, \ldots, \omega_T) = \lim_{T\to\infty} \frac{1}{T} \left[ - \sum_{\omega\in\Omega^T} P(\omega) \ln P(\omega) \right], \tag{40}$$

where $H$ is the joint Shannon entropy and $P$ is the probability distribution over length-$T$ sequences generated by the dream process.

The Shannon entropy rate measures the average unpredictability per time step. A dream with high entropy rate produces content that is difficult to predict from one moment to the next—the next scene, character, or spatial configuration carries substantial new information. A dream with low entropy rate is repetitive and predictable.

**Definition 12.4** (Effective Complexity)**.** Following the framework of Gell-Mann and Lloyd, the *effective complexity* $C_{\text{eff}}(\omega)$ of the dream output stream is the Kolmogorov complexity of the ensemble of regularities—the shortest description of the statistical patterns, recurring motifs, and structural rules that characterize $\omega$, as distinguished from its random or idiosyncratic components:

$$C_{\text{eff}}(\omega) = K(\mathcal{R}_\omega), \tag{41}$$

where $\mathcal{R}_\omega$ is a model (in the minimum description length sense) that captures the regularities of $\omega$ without encoding its random fluctuations.

Effective complexity is high when the signal has rich, non-trivial structure (many regularities that require substantial description). It is low both for perfectly ordered signals (whose regularities are trivially simple) and for perfectly random signals (which have no regularities to describe). It achieves its maximum for signals that are neither fully ordered nor fully random—precisely the character of dream experience.

## 12.3 The Virtual Complexity Hypothesis

**Proposition 12.5** (Complexity Dissociation in Dream Mentation)**.** *Under the framework developed in the preceding sections, the dream output stream $\omega$ is expected to satisfy the following inequalities:*

$$K(\omega) \ll T \cdot h(\omega) \gg C_{\mathit{eff}}(\omega), \tag{42}$$

*where $T$ is the length of the stream. That is, the Kolmogorov complexity of the generator is much smaller than the total Shannon information in the output, which in turn is much larger than the effective complexity of the regularities.*

The first inequality ($K(\omega) \ll T \cdot h(\omega)$) states that the dream is highly compressible: despite its apparent informational richness, it can be generated by a compact program. This program consists of the attractor landscape (Section 3), the affective projection (Section 4), the confabulation distribution (Section 11), and a noise source. The total description length of these components is finite and does not grow with the length of the dream, whereas the Shannon information content grows linearly.

The second inequality ($T \cdot h(\omega) \gg C_{\text{eff}}(\omega)$) states that most of the information in the dream stream is random variation rather than structured regularity. The regularities—recurring spaces, emotional logic, narrative patterns—constitute only a small fraction of the total information content. The majority of the stream consists of the specific, unrepeatable details that fill in the framework: the particular face of the stranger, the exact shade of light in the corridor, the specific words spoken. These details are generated by the stochastic components of the model (the noise term in Equation (21), the sampling in Equation (32)) and carry Shannon entropy but not effective complexity.

*Proof.* The proof is structural rather than quantitative, as the quantities involved are uncomputable or require empirical estimation.

For the first inequality: the dream generator, as formalized in this paper, consists of (a) the vector field $F_D$, parameterized by a finite-dimensional parameter vector $\boldsymbol{\lambda}$; (b) the attractor set $\{A_1, \ldots, A_K\}$, each describable as a region of $\mathcal{S}$; (c) the affective projection $\Pi$; (d) the confabulation distribution $\nu_i$ with its energy function $E_i$ and temperature $\gamma$; and (e) a random seed for the noise process. The total description length of (a)–(d) is bounded by a constant $L_{\text{gen}}$ independent of the dream length $T$. The noise process adds $\mathcal{O}(T)$ bits of randomness, but this randomness is drawn from a simple distribution (Wiener process) and thus requires only a seed of fixed length to specify. Hence $K(\omega) \leq L_{\text{gen}} + \mathcal{O}(\log T)$, while $T \cdot h(\omega) = \mathcal{O}(T)$, establishing $K(\omega) \ll T \cdot h(\omega)$ for large $T$.

For the second inequality: the regularities $\mathcal{R}_\omega$ consist of the attractor structure, the affective projection, and the confabulation rules—the same components as the generator but without the specific random realizations. Their description length $C_{\text{eff}} \approx L_{\text{gen}}$ is again bounded by a constant, while $T \cdot h(\omega) = \mathcal{O}(T)$, establishing

$C_{\text{eff}} \ll T \cdot h(\omega)$. □ □

## 12.4 Productivity: Finite Generativity, Infinite Output

The virtual complexity of dreaming is closely related to the linguistic notion of *productivity*—the capacity of a finite system of rules to generate an unbounded variety of outputs. A grammar with a finite number of rules can generate infinitely many sentences; a dream generator with a finite description can produce an indefinite variety of dream episodes.

**Definition 12.6** (Dream Generativity)**.** The *generativity* of a dream generator $G = (F_D, \{A_k\}, \Pi, \nu, \gamma)$ is the cardinality of the set of distinguishable dream output streams it can produce:

$$\mathcal{G}(G) = |\{\omega : \omega \text{ is a possible output of } G \text{ under some noise realization}\}|. \tag{43}$$

For a generator with continuous noise, $\mathcal{G}(G)$ is uncountably infinite. The *effective generativity* at resolution $\delta$ is the number of $\delta$-distinguishable streams:

$$\mathcal{G}_\delta(G) = \left|\{\omega : d_{\text{stream}}(\omega, \omega') > \delta \text{ for all other } \omega' \in \mathcal{G}(G)\}\right|, \tag{44}$$

where $d_{\text{stream}}$ is a metric on output streams.

**Proposition 12.7** (Exponential Generativity)**.** *For a dream generator with $K$ attractors, confabulation temperature $\gamma > 0$, and $N$ content features with $m$ possible values each, the effective generativity satisfies:*

$$\ln \mathcal{G}_\delta(G) \;\geq\; N \cdot H_{avg}(\gamma), \tag{45}$$

*where $H_{avg}(\gamma) = \frac{1}{N}\sum_i H(\nu_i)$ is the average entropy per confabulation distribution. For $\gamma > 0$ (non-degenerate confabulation), $H_{avg} > 0$, and generativity grows exponentially with the number of content features.*

This confirms the intuition that dreaming, despite its compact generative description, can produce an astronomically large variety of experiential content. The finite "grammar" of dream generation—attractor landscape, affective mapping, confabulation rules—yields an effectively infinite "language" of possible dreams.

*Proof.* Each of the $N$ features is resolved by sampling from $\nu_i$, which has entropy $H(\nu_i) \geq 0$. The number of distinguishable sequences of $N$ independent samples is

$\prod_i |\mathrm{supp}(\nu_i)|$, and ln of this quantity equals $\sum_i \ln |\mathrm{supp}(\nu_i)| \geq \sum_i H(\nu_i) = N \cdot H_{\mathrm{avg}}$, where the inequality uses the fact that the entropy of a distribution on a finite set is bounded above by the logarithm of the support size. □ □

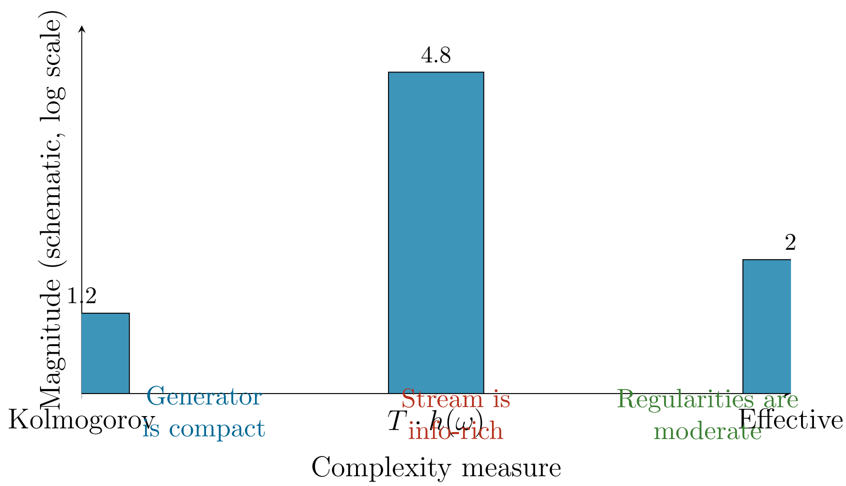


Figure 10: Schematic comparison of the three complexity measures for dream mentation. Kolmogorov complexity (generator length) is low; Shannon information (total stream content) is high; effective complexity (regularity description) is intermediate. This dissociation defines virtual complexity. Values are illustrative; the actual magnitudes depend on the specific dream and the resolution of the output alphabet.

*Caveat* 12.8. The virtual complexity hypothesis, while conceptually coherent, faces several difficulties. (i) Kolmogorov complexity is uncomputable, so the claim that $K(\omega)$ is "low" cannot be verified directly; it rests on the assumption that the formal model developed in this paper is an adequate description of the dream generator, which is itself unproven. (ii) Shannon entropy of dream content cannot be measured without a probabilistic model of dream output, which does not currently exist. Estimates based on dream report corpora are confounded by the limitations of verbal report: the entropy of the *report* is not the entropy of the *experience*. (iii) Effective complexity depends on the choice of regularity model $\mathcal{R}_\omega$, which involves judgment about what counts as a "regularity" versus "randomness" in dream content. (iv) The analogy with linguistic productivity, while suggestive, is imprecise: a formal grammar generates strings from a well-defined alphabet according to explicit rules, whereas the dream "grammar" operates on representations whose nature and structure remain poorly understood.

It should be emphasized that the complexity dissociation formalized here is a property of the model, demonstrated concretely in the toy system of Appendix F. Whether biological dreaming exhibits the same dissociation is an empirical question that the model motivates but does not settle. The model's contribution is to show that a compact generative mechanism *can* produce output with this complexity profile, and that such output shares qualitative features with dream phenomenology. This is a constructive existence proof, not a claim about the brain's implementation.

# 13 Beyond Topology: The Drifting Relational Structure of Dream Space

## 13.1 Motivation: The Failure of Fixed Neighborhoods

The framework developed in Sections 3 and 4 has relied on topological and manifold-theoretic concepts: state spaces, continuous flows, basins of attraction, smooth projections. These tools assume that the underlying space possesses a fixed structure of neighborhoods—that the notion of "nearby" is stable and well-defined. In a topological space, if point $A$ is near point $B$ at one moment, it remains near $B$ under continuous deformation.

Dream experience, however, appears to violate this assumption in a fundamental way. The corridor that was adjacent to the kitchen is now adjacent to the ocean. The room upstairs has become the room downstairs. The person who was standing to the left is now behind. These are not continuous deformations; they are *reassignments of relational structure*. The spatial, temporal, and identity relations that organize dream content are not fixed but *drift*—they change without continuity, without notice, and without generating the sense of inconsistency that such changes would produce in waking experience (cf. Section 5).

This observation raises a foundational question: is dream space a topological space at all? If not, what weaker mathematical structure can accommodate the drifting relations that characterize dream experience?

## 13.2 Formal Diagnosis: Why Topology Fails

**Definition 13.1** (Relational Structure)**.** A *relational structure* on a set $X$ of dream content elements is a collection of binary relations $\{R_j\}_{j \in J}$, where each $R_j \subseteq X \times X$ encodes a specific type of relationship: spatial adjacency ($R_{\mathrm{adj}}$), temporal precedence ($R_{\mathrm{temp}}$), identity ($R_{\mathrm{id}}$), causal connection ($R_{\mathrm{cause}}$), etc.

**Definition 13.2** (Topological Consistency)**.** A relational structure $\{R_j\}$ on $X$ is *topologically consistent* if there exists a topology $\tau$ on $X$ such that each relation $R_j$ is continuous with respect to $\tau$—that is, if $(x, y) \in R_j$ and $x' \in U$ for any open neighborhood $U$ of $x$, then there exists $y'$ near $y$ with $(x', y') \in R_j$.

Topological consistency is a strong requirement: it demands that relations deform smoothly as elements move through the space. The following proposition states that dream experience, as described by standard phenomenological reports, is generically topologically inconsistent.

**Proposition 13.3** (Topological Inconsistency of Dream Relations)**.** *Let $\{R_j(t)\}$ be a time-indexed family of relational structures on a set $X$ of dream elements, where the time-dependence reflects the evolution of dream content. If there exist times $t_1 < t_2$ and elements $x, y, z \in X$ such that:*

*(i) $(x, y) \in R_{adj}(t_1)$ and $(x, y) \notin R_{adj}(t_2)$ (spatial adjacency is lost),*

*(ii) $(x, z) \notin R_{adj}(t_1)$ and $(x, z) \in R_{adj}(t_2)$ (new adjacency appears),*

*(iii) No continuous path in $X$ carries $y$ away from $x$ while bringing $z$ toward $x$ between $t_1$ and $t_2$,*

*then no single topology on $X$ is consistent with both $R_{adj}(t_1)$ and $R_{adj}(t_2)$.*

*Proof.* Suppose, for contradiction, that a topology $\tau$ on $X$ is consistent with both $R_{\text{adj}}(t_1)$ and $R_{\text{adj}}(t_2)$. By topological consistency, the adjacency of $x$ and $y$ at $t_1$ implies that $y$ lies in some $\tau$-neighborhood of $x$, and the non-adjacency at $t_2$ implies $y$ has left this neighborhood. Similarly, $z$ has entered a neighborhood of $x$. For both transitions to respect the topology, there must exist continuous paths accomplishing these movements. Condition (iii) asserts that no such paths exist. Contradiction. □ □

This result confirms the intuition: dream space cannot, in general, be equipped with a fixed topology that remains consistent across the dream. The relations drift in ways that are incompatible with any single continuous structure.

## 13.3 Alternative Mathematical Structures

If dream space is not a topological space, what is it? Several weaker structures may be appropriate.

**Definition 13.4** (Pretopological Dream Space)**.** A *pretopological space* $(X, \mathfrak{a})$ consists of a set $X$ and a preclosure operator $\mathfrak{a} : \mathcal{P}(X) \to \mathcal{P}(X)$ satisfying $A \subseteq \mathfrak{a}(A)$ and

$\mathfrak{a}(\varnothing) = \varnothing$, but not necessarily the idempotency or additivity axioms of a topological closure operator. The neighborhood of $x$ at time $t$ is:

$$\mathcal{N}_t(x) = \{A \subseteq X : x \notin \mathfrak{a}_t(X \setminus A)\}, \tag{46}$$

where $\mathfrak{a}_t$ is the preclosure at time $t$, which may vary.

In a pretopological space, the notion of "nearness" is well-defined at each moment but is not required to be stable across time or to satisfy the closure axioms that would make it a true topology. This accommodates the observation that dream spatial relations are locally coherent (at any given moment, the room has walls, the corridor has a direction) but globally unstable (these relations may change at the next moment without continuous transition).

**Definition 13.5** (Relational Graph with Drifting Edges)**.** A *drifting relational graph* is a triple $(X, E, w)$ where:

(i) $X$ is a finite set of dream content elements (rooms, persons, objects, locations).

(ii) $E \subseteq X \times X$ is the set of all *potential* edges (all pairs that could, at some point in the dream, stand in some relation).

(iii) $w : E \times [0, T] \to [0, 1]$ is a time-varying weight function, where $w(e, t)$ represents the strength of the relation $e$ at time $t$.

The *instantaneous graph* at time $t$ is $G_t = (X, E_t)$ where $E_t = \{e \in E : w(e, t) > \theta\}$ for some activation threshold $\theta > 0$.

**Definition 13.6** (Relational Drift Rate)**.** The *relational drift rate* at time $t$ is:

$$\Delta(t) = \frac{1}{|E|} \sum_{e \in E} \left| \frac{dw(e, t)}{dt} \right|. \tag{47}$$

High $\Delta$ indicates rapid restructuring of dream relations; low $\Delta$ indicates relational stability.

**Proposition 13.7** (Relational Drift Rate Correlates with Regime Parameters)**.** *Within the dual-regime framework of Section 2, the relational drift rate is expected to be an increasing function of the associative density $\alpha$ and a decreasing function of the temporal coherence $\tau$:*

$$\Delta \sim \frac{\alpha}{\tau + \epsilon} \tag{48}$$

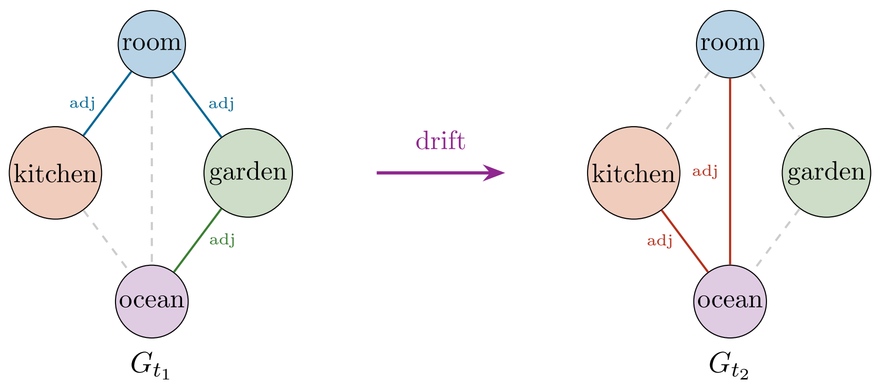


Figure 11: A drifting relational graph representing the non-topological evolution of dream space. At time $t_1$ (left), the room is adjacent to the kitchen and garden; the ocean is distant. At time $t_2$ (right), the room is now adjacent to the ocean, and the kitchen has shifted adjacency. Solid lines: active relations ($w > \theta$). Dashed lines: inactive relations. The nodes are the same, but the edge structure has changed discontinuously.

*for some regularization constant $\epsilon > 0$. In the waking regime ($\alpha$ low, $\tau$ high), $\Delta$ is small: relations are stable. In the dreaming regime ($\alpha$ high, $\tau$ low), $\Delta$ is large: relations drift rapidly.*

*Proof.* The argument is heuristic rather than deductive. High associative density implies many active connections competing for relational assignments, increasing the probability that a given relation will be replaced by an alternative. Low temporal coherence implies that the system does not enforce consistency of relations across time steps. Together, these conditions favor rapid relational turnover. The specific functional form $\Delta \sim \alpha/(\tau + \epsilon)$ is proposed as the simplest monotonic function consistent with these boundary conditions. □ □

## 13.4 Dream Content as a Sheaf Over a Drifting Base

A more sophisticated mathematical treatment can be given using the language of sheaf theory. A sheaf assigns data (spatial layout, character properties, narrative state) to local patches of an underlying space, with consistency conditions ("restriction maps") governing how the data on overlapping patches relate. The key feature that makes sheaf theory relevant is that it can accommodate *local* consistency without requiring *global* consistency—precisely the situation in dreaming.

**Definition 13.8** (Dream Content Sheaf)**.** Let $(X_t, \tau_t)$ be the pretopological dream space at time $t$. A *dream content sheaf* $\mathcal{F}_t$ assigns to each open set $U \in \tau_t$ a set of content sections $\mathcal{F}_t(U)$ (possible content configurations on $U$), with restriction maps $\rho_{U,V} : \mathcal{F}_t(U) \to \mathcal{F}_t(V)$ for $V \subseteq U$ satisfying:

(i) *Local consistency:* For $V \subseteq U$, the restriction of a section on $U$ to $V$ yields a valid section on $V$.

(ii) *Gluing:* If sections on overlapping patches agree on their overlap, they can be glued into a section on the union.

A *dream with drifting relations* is a family $\{\mathcal{F}_t\}_{t\in[0,T]}$ where both the base space $(X_t, \tau_t)$ and the sheaf $\mathcal{F}_t$ vary with time. The gluing condition (ii) may fail globally while holding locally, corresponding to the ad hoc consistency of Section 11.

**Proposition 13.9** (Global Section Obstruction)**.** *If the relational drift rate $\Delta(t)$ exceeds a critical value $\Delta^*$ over an interval $[t_1, t_2]$, then the sheaf $\mathcal{F}_t$ may fail to admit a global section—there may be no single, consistent assignment of content to the entire dream space that is compatible with all local constraints. The obstruction is measured by the first sheaf cohomology group $H^1(X_t, \mathcal{F}_t)$: a non-trivial $H^1$ indicates that local sections cannot be glued into a global one.*

The sheaf cohomology $H^1$ measures the "twisting" of local data as one moves around the space. In the context of dreaming, a non-trivial $H^1$ means that the dream is locally coherent (each scene makes sense in its immediate context) but globally incoherent (the pieces do not fit together into a single consistent world). This is the mathematical expression of the observation that dreams are locally vivid and convincing but globally contradictory.

*Proof.* The proof relies on the standard exactness of the Čech cohomology sequence for a sheaf on a pretopological space. Consider a cover $\{U_i\}$ of $X_t$ by local patches. On each $U_i$, a local section $s_i \in \mathcal{F}_t(U_i)$ is defined (the local dream content). On overlaps $U_i \cap U_j$, the sections may or may not agree: $\rho_{U_i,U_i\cap U_j}(s_i) \neq \rho_{U_j,U_i\cap U_j}(s_j)$ in general. The collection of discrepancies $\{g_{ij} = s_i|_{U_i\cap U_j} - s_j|_{U_i\cap U_j}\}$ defines a Čech 1-cocycle. If this cocycle is a coboundary, the sections can be adjusted to agree globally. If not—if the cocycle represents a non-trivial element of $H^1(X_t, \mathcal{F}_t)$—no global section exists. High relational drift increases the discrepancies $g_{ij}$ by changing the overlap structure between time steps, increasing the probability that the cocycle is non-trivial. □ □

*Caveat* 13.10. The application of sheaf-theoretic concepts to dream phenomenology, while mathematically well-defined, pushes the formalism to a level of abstraction where empirical contact becomes tenuous. (i) The pretopological structure of dream space cannot be directly observed; it is inferred from verbal reports that may impose a topological structure not present in the original experience. (ii) Sheaf cohomology requires a well-defined cover by open sets, which may not correspond to any natural decomposition of dream content. (iii) The drifting relational graph model, while more concrete, involves parameters ($w(e, t)$, $\theta$, $\Delta$) that cannot currently be measured. (iv) A simpler explanation for the apparent non-topological character of dream space is that dreams *do* possess a consistent topology at each instant, but the topology changes between instants—a sequence of topological spaces rather than a single space with drifting relations. On this account, no departure from standard topology is required; only the acknowledgment that the space is not time-invariant. Distinguishing between this "time-varying topology" account and the genuinely pretopological account is an open question.

The sheaf-theoretic apparatus introduced here serves a specific modeling purpose: it provides the weakest mathematical structure capable of expressing the combination of local consistency and global inconsistency that characterizes the model's output (and, by hypothesis, dream experience). The reader should not interpret Definition 13.8 as a claim that the sleeping brain "computes sheaf cohomology." Rather, the sheaf formalism is the natural language for describing systems in which data is locally well-defined but cannot be assembled into a globally coherent whole—a structural property that the model generates endogenously (as confirmed by the drifting relational graph simulation of Appendix G) and that dream reports exhibit phenomenologically. The mathematical tool describes the output, not the mechanism.

# 14 Dream Amnesia: Mechanisms, Implications, and Experimental Interventions

## 14.1 The Paradox of Experiential Richness and Mnemonic Poverty

The preceding sections have characterized dream mentation as a generative process of considerable complexity—a system capable of producing spatially structured, emotionally saturated, narratively organized experience. Yet this rich experiential output is almost entirely lost. Most dreams are never recalled; even those that are recalled fade within minutes unless actively rehearsed [2, 9]. Estimates suggest that humans spend approximately two hours per night in REM sleep [35], generating multiple distinct dream episodes, of which the typical individual recalls at most a fragment of the last. This raises a fundamental question for the present framework: if the dream generator operates with the formal sophistication described above, why is its output so poorly retained?

## 14.2 Neurobiological Mechanisms of Dream Amnesia

The literature identifies several converging mechanisms.

**Noradrenergic suppression.** The locus coeruleus, the brain's principal source of norepinephrine (NE), is nearly silent during REM sleep [2, 36]. Since NE is critical for hippocampal long-term potentiation and the consolidation of episodic memory, its absence during dreaming may prevent the encoding of dream content into stable memory traces. The rapid restoration of NE at awakening may overwrite fragile dream traces with the immediate demands of waking cognition [36].

**Active hippocampal suppression.** Izawa et al. [37] demonstrated that melanin-concentrating hormone (MCH) neurons in the hypothalamus, which fire preferentially during REM sleep, send inhibitory projections to hippocampal pyramidal cells. Optogenetic activation of these neurons during REM impaired memory in mice; their inhibition improved it. This suggests that the hippocampus is not merely passively unable to encode during REM but is *actively suppressed*—the brain prevents dream content from being consolidated.

**Encoding-consolidation competition.** Zhao et al. [38] propose that dream amnesia reflects a functional trade-off: the same neural resources required for encoding new memories (hippocampal activation, NE signaling) are redirected during sleep toward consolidation of existing waking memories. Dream content is generated by cortical processes but the hippocampal gateway through which it would need to pass for long-term storage is occupied with other tasks.

**Prefrontal deactivation.** The dorsolateral prefrontal cortex, which supports working memory and the deliberate encoding strategies necessary for episodic memory formation, is substantially deactivated during REM sleep [2, 39]. Without executive encoding effort, dream experiences may remain in a labile, unbound state that dissipates at awakening.

**State-dependent accessibility.** The neurochemical and electrophysiological milieu of REM sleep is so different from waking that memories formed during dreaming may be encoded in a state-dependent manner—accessible only when the brain returns to a similar state [40]. This would predict that dream memories are not truly lost but are inaccessible from the waking state, a hypothesis with partial support from the observation that dreams are sometimes recalled during subsequent REM periods or in hypnagogic states.

## 14.3 Formalization Within the Present Framework

**Definition 14.1** (Memory Encoding Operator)**.** The *memory encoding operator* $\mathcal{E}_\lambda : \Omega^T \to \mathcal{M}$ maps a segment of the experiential stream $\omega_{[t_1,t_2]}$ to a memory trace $m \in \mathcal{M}$ (a memory space). The operator depends on the regime parameters:

$$\|\mathcal{E}_\lambda(\omega)\| = \eta_{\mathrm{NE}}(\lambda) \cdot \eta_{\mathrm{PFC}}(\lambda) \cdot \eta_{\mathrm{HPC}}(\lambda) \cdot \|\omega\|_{\mathrm{sal}}, \tag{49}$$

where $\eta_{\mathrm{NE}}$ is noradrenergic availability, $\eta_{\mathrm{PFC}}$ is prefrontal encoding capacity, $\eta_{\mathrm{HPC}}$ is hippocampal gate openness, and $\|\omega\|_{\mathrm{sal}}$ is the salience of the experiential content.

In the waking regime, all three gating factors are near unity, and encoding strength is proportional to salience. In the dreaming regime, $\eta_{\mathrm{NE}} \approx 0$, $\eta_{\mathrm{PFC}} \ll 1$, and $\eta_{\mathrm{HPC}} \ll 1$ (actively suppressed by MCH neurons), yielding:

**Proposition 14.2** (Dream Amnesia as Encoding Failure)**.** *In the dreaming regime* $\boldsymbol{\lambda}_D$*:*

$$\|\mathcal{E}_{\lambda_D}(\omega)\| \;\leq\; \epsilon_{amnesia} \cdot \|\omega\|_{sal}, \tag{50}$$

*where* $\epsilon_{amnesia} \ll 1$*. The vast majority of dream content fails to produce stable memory traces, regardless of its experiential intensity or complexity.*

*Proof.* Since $\eta_{\mathrm{NE}}(\lambda_D) \approx 0$, the product $\eta_{\mathrm{NE}} \cdot \eta_{\mathrm{PFC}} \cdot \eta_{\mathrm{HPC}} \leq \epsilon_{\mathrm{amnesia}}$ for any values of the other factors. The bound holds even for maximally salient content ($\|\omega\|_{\mathrm{sal}} = 1$). □ □

## 14.4 Connection to Virtual Complexity and Interrogative Collapse

Dream amnesia has deep connections to the virtual complexity analysis of Section 12 and the interrogative collapse model of Section 11. If dream content is generated ad hoc by a compact generative mechanism, then the "information" lost to amnesia may never have existed in a form suitable for encoding in the first place. The compatibility ensemble (Definition 11.2) contains all *potential* dream continuations, but only the features that are actually interrogated (Definition 11.3) become determinate. Features that are never interrogated are never resolved and thus have no determinate content to forget.

This suggests a reinterpretation of dream amnesia: what is lost is not a detailed record of a fully specified experience but the sparse set of actually-resolved features, together with the affective tone and the phenomenological impression of completeness that accompanied them. The "richness" of the dream experience, being largely virtual (Section 12), was never available for encoding.

## 14.5 Experimental Interventions and Predictions

The framework generates several proposals for experimental intervention.

**Experiment 1: MCH antagonist administration.** Based on the finding that MCH neuron inhibition improves memory in mice [37], a pharmacological intervention using MCH receptor antagonists administered during sleep could test whether dream recall frequency and detail increase. Prediction: participants receiving MCH antagonists will report significantly more dreams, with greater spatial and narrative detail, than placebo controls, without alteration of sleep architecture.

**Experiment 2: Targeted memory reactivation during REM.** Using the interactive dreaming paradigm [41], experimenters could deliver cues during verified REM sleep that are associated with specific pre-sleep learning. If dream content incorporates these cues, and if the cues additionally trigger partial noradrenergic arousal (via mild sensory stimulation calibrated below awakening threshold), the encoding deficit might be partially bypassed. Prediction: cued dreams will be recalled with higher frequency and fidelity than uncued dreams from the same night.

**Experiment 3: Lucid dream encoding protocols.** Since lucid dreaming involves partial restoration of prefrontal function [15, 16], lucid dreamers trained to perform explicit encoding strategies (rehearsal, mnemonic tagging) during lucid episodes should show superior dream recall. Prediction: lucid dreams with deliberate encoding will be recalled at rates approaching waking episodic memory, while lucid dreams without encoding effort will show the standard amnesia profile.

**Experiment 4: State-dependent retrieval.** If dream memories are state-dependent, then retrieval should be facilitated by reinstating the neurochemical conditions of REM sleep. A pharmacological intervention producing a "mini-REM" state during waking (e.g., via cholinergic agonist combined with mild NE suppression) could be tested as a retrieval cue. Prediction: participants in the pharmacologically induced state will recall more dream content from the preceding night than those in normal waking.

*Caveat* 14.3. The experimental proposals above raise ethical and practical concerns. MCH antagonists have not been tested for this purpose in humans. Pharmacological manipulation of sleep architecture carries risks. The interactive dreaming paradigm, while demonstrated, remains technically demanding. The state-dependent retrieval experiment involves deliberate alteration of waking neurochemistry. All proposals require rigorous ethical review and safety protocols.

# 15 Dream Atemporality and Subjective Spacetime Metrics

## 15.1 Phenomenological Motivation: Time Without Clocks

Dream experience is characterized by a distinctive relationship to time that has no precise waking analogue. Events that would require hours unfold in minutes; temporal ordering is fluid or absent; the dreamer may simultaneously experience "before" and "after" without contradiction. Narrative sequences compress, expand, or run in parallel in ways that violate the linear, metrically regular time of waking experience [5, 42].

Lucid dreaming studies provide partial empirical access to dream temporality. Erlacher et al. [43] found that motor tasks performed in lucid dreams took approximately 40% longer than the same tasks performed while awake, while purely cognitive tasks (counting) showed near-real-time correspondence. This suggests that dream time is not uniformly dilated or compressed but depends on the *type of content* being processed—a finding with no straightforward explanation in any framework that treats time as a uniform parameter.

The interactive dreaming paradigm of Konkoly et al. [41], in which sleeping participants respond to external stimuli during verified REM sleep, opens the possibility of directly probing time perception during dreaming—for instance, by asking dreamers to estimate how much time has elapsed since a previous cue [44].

## 15.2 Formal Construction: Dream Spacetime as a Variable Metric

The present section proposes that the temporal structure of dream experience can be formalized not as a parameter (a fixed background against which events occur) but as a *variable metric*—a field that is itself determined by the content and dynamics of the dream. This is a structural analogy with general relativity, where the spacetime metric is not a fixed background but is determined by the distribution of matter and energy.

**Definition 15.1** (Dream Spacetime Manifold)**.** A *dream spacetime* is a four-dimensional smooth manifold $(\mathcal{M}_D, g_D)$ with coordinates $(x^0, x^1, x^2, x^3)$, where $x^0$ is the subjective temporal coordinate and $(x^1, x^2, x^3)$ are spatial coordinates of the dream environment. The *dream metric* $g_D$ is a pseudo-Riemannian metric tensor:

$$ds_D^2 = g_{\mu\nu}^{(D)}\, dx^\mu\, dx^\nu, \tag{51}$$

where Greek indices run from 0 to 3.

In waking experience, the metric is approximately Minkowskian—flat, with a constant ratio between temporal and spatial intervals. The proposal is that in dreaming, the metric becomes content-dependent: the "distance" between two dream events depends on the cognitive and emotional content being processed.

**Definition 15.2** (Content-Dependent Temporal Metric)**.** The temporal component of the dream metric is modulated by a *cognitive load function* $\Lambda(x)$:

$$g_{00}^{(D)}(x) = -c_D^2 \cdot \Lambda(x)^{-2}, \tag{52}$$

where $c_D$ is a characteristic velocity (analogous to the speed of light) and $\Lambda(x)$ encodes the cognitive processing demands at dream-location $x$. When $\Lambda$ is large (high cognitive load, complex motor imagery), the temporal metric shrinks: subjective time runs *slower* relative to external (physiological) time. When $\Lambda$ is small (passive observation, simple cognition), subjective time runs faster or at near-real-time pace.

This formalization captures the empirical finding of Erlacher et al. [43]: motor tasks, which involve complex sensorimotor simulation ($\Lambda$ large), dilate dream time; counting, which is computationally simple ($\Lambda$ small), preserves near-real-time correspondence.

**Proposition 15.3** (Dream Time Dilation). *Let $\Delta t_{ext}$ be the external (polysomnographically measured) duration of a dream episode and $\Delta t_{subj}$ the subjective experienced duration. If the cognitive load varies over the episode:*

$$\Delta t_{subj} = \int_0^{\Delta t_{ext}} \Lambda(t)^{-1}\, dt. \tag{53}$$

*For uniform cognitive load $\Lambda_0$: $\Delta t_{subj} = \Delta t_{ext}/\Lambda_0$. When $\Lambda_0 > 1$ (motor simulation), $\Delta t_{subj} < \Delta t_{ext}$—fewer subjective events fit into the objective interval, producing the phenomenology of slowed time. When $\Lambda_0 < 1$ (passive dreaming), $\Delta t_{subj} > \Delta t_{ext}$—more subjective events fit, producing the phenomenology of temporal compression or accelerated narrative.*

*Proof.* This follows directly from the definition of $g_{00}^{(D)}$ and the standard relationship between coordinate time and proper time in a curved metric. In general relativity, proper time $d\tau = \sqrt{-g_{00}}\, dx^0$; by analogy, subjective dream time $dt_{\text{subj}} = \sqrt{-g_{00}^{(D)}}\, dt_{\text{ext}} = (c_D/\Lambda)\, dt_{\text{ext}}$. Setting $c_D = 1$ (choosing units such that dream time equals external time at unit cognitive load) and integrating yields the result. □ □

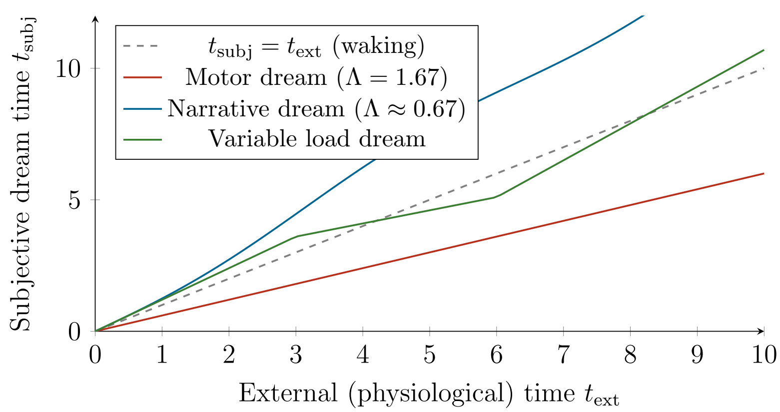


Figure 12: Subjective dream time as a function of external time for different cognitive load profiles. Waking (dashed): $t_{\text{subj}} = t_{\text{ext}}$. Motor dreaming (red): subjective time runs slower (∼40% dilation, consistent with Erlacher et al. [43]). Narrative dreaming (blue): subjective time runs faster (events compress). Variable load (green): time rate changes during the dream, producing the characteristic phenomenology of dream atemporality.

### 15.3 The Analogy with General Relativity

The structural parallel between the dream metric and the spacetime metric of general relativity may be stated precisely.

In Einstein's theory, the metric tensor $g_{\mu\nu}$ is determined by the stress-energy tensor $T_{\mu\nu}$ via the field equations $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$. Matter and energy curve spacetime; the curvature, in turn, determines how matter moves. Time dilation near a massive object is a consequence: clocks run slower in stronger gravitational fields.

In the dream metric framework, the analogue of the stress-energy tensor is the *cognitive-affective content density*:

**Definition 15.4** (Cognitive-Affective Content Tensor)**.** The *content tensor* $T_{\mu\nu}^{(D)}(x)$ encodes the density and flow of cognitive and affective processing at each point of dream spacetime:

$$T_{00}^{(D)} \propto \Lambda(x)^2, \qquad T_{ij}^{(D)} \propto (\text{spatial processing demands at } x). \tag{54}$$

**Definition 15.5** (Dream Field Equation)**.** The dream metric is determined, by analogy with general relativity, by:

$$G_{\mu\nu}^{(D)} = \kappa_D \, T_{\mu\nu}^{(D)}, \tag{55}$$

where $G_{\mu\nu}^{(D)}$ is the Einstein tensor of the dream metric $g_{\mu\nu}^{(D)}$ and $\kappa_D$ is a coupling constant measuring the sensitivity of the dream spacetime geometry to cognitive content.

The analogy is structural. In general relativity, mass-energy curves spacetime; in the dream metric, cognitive-affective content curves subjective time. The "gravitational" time dilation near a massive object corresponds to the subjective time dilation near a cognitively intensive dream event. Just as a clock near a black hole runs slower, subjective time in a dream slows near events requiring intensive simulation (complex motor imagery, detailed spatial construction).

The value of this analogy is not that dream spacetime "is" curved spacetime, but that the mathematical apparatus of variable metrics—developed in general relativity for independent reasons—turns out to be precisely the right tool for formalizing the phenomenology of variable dream temporality. The model uses the metric formalism as technology, not as ontology. The toy computation of Appendix H demonstrates that this technology produces quantitative predictions (specific time-dilation ratios

Table 3: Structural analogy between general relativity and the dream metric framework.

| **Feature** | **General Relativity** | **Dream Metric** |
|---|---|---|
| Manifold | Spacetime $(\mathcal{M}, g)$ | Dream spacetime $(\mathcal{M}_D, g_D)$ |
| Source of curvature | Mass-energy $T_{\mu\nu}$ | Cognitive load $T^{(D)}_{\mu\nu}$ |
| Time dilation | Gravitational: $\Delta\tau = \sqrt{-g_{00}}\,\Delta t$ | Cognitive: $\Delta t_{\text{subj}} = \Lambda^{-1}\,\Delta t_{\text{ext}}$ |
| Geodesics | Paths of free particles | Natural flow of dream narrative |
| Singularities | Black holes, Big Bang | Dream collapse, awakening |
| Topology change | Forbidden classically | Common (drifting relations, Section 13) |
| Physical status | Fundamental theory of gravity | Formal analogy only |

for different cognitive load profiles) that can, in principle, be compared against lucid-dreaming time-estimation data.

## 15.4 Atemporality and Causal Structure

**Proposition 15.6** (Breakdown of Global Temporal Ordering)**.** *If the dream metric $g_D$ admits closed timelike curves—trajectories that return to their starting point in the temporal coordinate—then the dream spacetime lacks a global notion of "before" and "after." In the context of the present framework, this occurs when the cognitive load function $\Lambda(x)$ varies sufficiently rapidly in the spatial coordinates:*

$$\exists\,\gamma : [0,1] \to \mathcal{M}_D \textit{ with } \gamma(0) = \gamma(1) \textit{ and } g_D(\dot\gamma, \dot\gamma) < 0 \textit{ everywhere along } \gamma. \tag{56}$$

The existence of closed timelike curves in the dream metric formalizes the phenomenological observation that dream narratives can loop: the dreamer arrives at a location that is simultaneously the beginning and the end of the narrative, or experiences the same event from multiple temporal perspectives. Unlike in general relativity, where closed timelike curves raise paradoxes about causality, in the dream context they simply reflect the absence of a globally coherent temporal ordering.

*Proof.* The existence of closed timelike curves in a Lorentzian manifold is determined by the global causal structure. For a metric of the form $ds^2 = -\Lambda(x)^{-2}\,dt^2 + h_{ij}\,dx^i\,dx^j$, closed timelike curves exist if the "tilt" of the light cones, governed by $\Lambda$, allows spatial variations to compensate for temporal progression. A sufficient condition is that $\partial_i\Lambda/\Lambda$ exceeds a critical value related to the spatial metric $h_{ij}$, permitting a path to "turn around" in time by traversing a region of extreme cognitive load variation. The specific condition depends on the global topology of $\mathcal{M}_D$ and the boundary conditions. □ □

*Caveat* 15.7. The relativity analogy, while mathematically precise, must be handled with the same caution applied to the quantum measurement analogy of Section 10. (i) Dream spacetime is not physical spacetime; the "metric" is a mathematical model of subjective temporal experience, not a claim about the geometry of the physical space in which the brain resides. (ii) The field equation $G^{(D)}_{\mu\nu} = \kappa_D T^{(D)}_{\mu\nu}$ is a structural postulate, not a derived consequence of any fundamental theory; the coupling constant $\kappa_D$ is unknown and may not be well-defined. (iii) The cognitive load function $\Lambda$ is not directly measurable with current technology, though lucid dreaming time-estimation paradigms [41, 43] provide indirect probes. (iv) The closed timelike curve result is a formal consequence of the metric structure; whether dreamers actually experience temporal loops (as opposed to merely reporting them) is an open question. (v) Several authors have proposed connections between dreaming and relativistic time dilation; these proposals range from suggestive structural analogies to speculative claims about biophotons and quantum consciousness. The present framework aligns strictly with the former: a mathematical analogy between the variable metrics of general relativity and the variable temporality of dream experience, with no commitment to any physical mechanism linking the two.

# 16 The Generative Prior: Physical Law, Ontological Reframing, and the Agency Paradox

## 16.1 The Paradox of Dream Physics

A striking feature of dream experience, rarely formalized, is the degree to which dreams obey physical law. Gravity operates; objects have weight and solidity; spatial traversal requires locomotion; conversations follow turn-taking conventions; cause precedes effect. This is paradoxical: the dream environment is entirely endogenous, with no external physical constraints. The generator is free to produce any content—yet it overwhelmingly produces content that conforms to waking physics. In a space where "anything is possible," physics is preserved with remarkable fidelity.

The paradox deepens when physics *is* violated. A dreamer may fly, walk through walls, or observe impossible geometry—but these violations are experienced not as violations but as *the way the world is*. There is no surprise, no sense of anomaly, no recognition that a law has been broken. The local physics of the dream, however aberrant from waking physics, is experienced as fundamental.

## 16.2 The Generative Prior

**Definition 16.1** (Generative Prior)**.** The *generative prior* $\mathcal{P}_{\text{gen}}$ is the default probability distribution over dream content states, reflecting the internalized model of physical and social regularity that the brain has acquired through a lifetime of waking experience. For a content feature vector $\mathbf{d} \in \mathbb{R}^N$:

$$\mathcal{P}_{\text{gen}}(\mathbf{d}) = \frac{1}{Z_{\text{gen}}} \exp\left(-\frac{1}{\gamma_{\text{gen}}} \sum_{k=1}^{M} \phi_k(\mathbf{d})\right), \tag{57}$$

where $\{\phi_k\}_{k=1}^{M}$ are *physical constraint potentials*—penalty functions that are minimized when $\mathbf{d}$ satisfies the $k$-th physical or social regularity (e.g., $\phi_{\text{gravity}}(\mathbf{d}) = 0$ when objects fall downward, $\phi_{\text{causality}}(\mathbf{d}) = 0$ when causes precede effects), $\gamma_{\text{gen}}$ is the prior temperature, and $Z_{\text{gen}}$ is a normalization constant.

At low prior temperature ($\gamma_{\text{gen}} \ll 1$), the prior strongly concentrates on physically lawful content: gravity operates, objects are solid, conversations are sequential. At high temperature ($\gamma_{\text{gen}} \gg 1$), the prior flattens, and violations become probable. The key empirical observation is that even in the dreaming regime, $\gamma_{\text{gen}}$ remains relatively low—the prior is strong. Physical law is not imposed by the environment but by the generator itself.

**Proposition 16.2** (Prior Dominance in the Absence of Correction)**.** *In the dreaming regime, where sensory correction $I_{ext} \approx 0$ (Section 2), the probability of content state $\mathbf{d}$ at time $t$ is:*

$$P(\mathbf{d}_t) \propto \mathcal{P}_{gen}(\mathbf{d}_t) \cdot \mathcal{P}_{aff}(\mathbf{d}_t|\mathbf{e}_t) \cdot \mathcal{P}_{assoc}(\mathbf{d}_t|\mathbf{d}_{t-1}), \tag{58}$$

*where $\mathcal{P}_{aff}$ is the affective likelihood (Section 4) and $\mathcal{P}_{assoc}$ is the associative transition probability. In the absence of external correction, the prior $\mathcal{P}_{gen}$ dominates: most content is physically lawful because the generator's default output distribution is concentrated on lawful configurations.*

*Proof.* Since $I_{\text{ext}} = 0$, there is no sensory likelihood term to compete with the prior. The affective and associative terms modulate the prior but do not override it unless they are sufficiently strong (i.e., when $\mathcal{P}_{\text{aff}}$ or $\mathcal{P}_{\text{assoc}}$ assign very high probability to a physically unlawful state). The mode of the product distribution therefore lies near the mode of $\mathcal{P}_{\text{gen}}$ unless perturbed by extreme affective or associative activation. □ □

## 16.3 Ontological Reframing

When a physical violation *does* occur—when the dreamer flies, or a wall becomes permeable, or gravity reverses—the dreamer does not experience surprise or anomaly. This is not merely affective validation (Section 5), which explains the acceptance of emotional inconsistency. It is a deeper phenomenon: the dreamer's *ontological frame*—the set of background assumptions about what is physically possible—is rewritten to accommodate the violation.

**Definition 16.3** (Local Ontological Frame)**.** A *local ontological frame* $\mathcal{O}_t$ at time $t$ is the subset of physical constraint potentials $\{\phi_k\}$ that are active (i.e., enforced by the generative prior) at that moment:

$$\mathcal{O}_t = \{k \in \{1, \ldots, M\} : \phi_k \text{ contributes to } \mathcal{P}_{\text{gen}} \text{ at time } t\}. \tag{59}$$

A *reframing event* occurs when $\mathcal{O}_{t+\delta} \neq \mathcal{O}_t$—some constraint is dropped or added.

**Proposition 16.4** (Transparency of Ontological Reframing)**.** *The logical validation operator $V_L$ (Definition 5.1) cannot detect a reframing event, because $V_L$ operates on the* current *ontological frame $\mathcal{O}_t$, not on a stored record of the previous frame $\mathcal{O}_{t-\delta}$. In the dreaming regime, where $V_L$ is suppressed (Proposition 5.3), even a transition from $\mathcal{O}_{standard\ physics}$ to a radically different frame (e.g., one permitting flight) generates no anomaly signal.*

*Proof.* Anomaly detection requires comparison between expected and observed content. The expected content is generated by the current prior $\mathcal{P}_{\text{gen}}^{(\mathcal{O}_t)}$. If the ontological frame has been rewritten to include flight as physically possible, then flight is *expected* under the current prior and generates no prediction error. The suppressed $V_L$ would need access to the previous frame $\mathcal{O}_{t-\delta}$ to detect the change, but (a) $V_L$ is largely inoperative, and (b) the encoding failure of Section 14 means that $\mathcal{O}_{t-\delta}$ may not be accessible. The reframing is therefore transparent: it is as if the new physics were always in effect. □ □

This formalization explains the phenomenological observation that dream physics violations feel "natural": the dreamer's entire ontological context has shifted, and within that context, the violation is not a violation at all.

## 16.4 The Agency Paradox: Authorship Without Control

The dreamer is the sole author of the dream—no external agent contributes content. Yet the dreamer typically experiences the dream as something that *happens to them*, not something they create. Characters speak unpredictably; events unfold with apparent independence; the dreamer cannot, in general, choose what happens next. This is the *agency paradox*: the generator and the experiencer are the same neural system, yet the experiencer has no executive access to the generator.

**Definition 16.5** (Generator-Experiencer Decoupling)**.** Let $G(\mathbf{s}_t, \boldsymbol{\lambda})$ denote the generative function that produces the next content state, and $E(\mathbf{d}_t)$ the experiential function that "receives" and experiences that content. In the waking regime, the executive system $\mathcal{X}$ mediates between $G$ and $E$, allowing $E$ to modulate $G$ (e.g., by directing attention, selecting actions, or vetoing implausible content). The degree of executive coupling is:

$$\kappa(\lambda) = \frac{\partial G}{\partial E} \cdot \eta_{\text{PFC}}(\lambda), \tag{60}$$

where $\eta_{\text{PFC}}$ is the prefrontal coupling strength from Definition 14.1.

In the waking regime, $\kappa(\lambda_W) \approx 1$: the experiencer can modulate the generator (you can choose where to look, what to say, how to act). In the dreaming regime, $\eta_{\text{PFC}}(\lambda_D) \approx 0$ (Section 14), so:

$$\kappa(\lambda_D) \approx 0. \tag{61}$$

The generator produces content autonomously; the experiencer receives it passively. This is not a different system acting upon the dreamer but the *same system* operating without the executive feedback loop that normally provides the phenomenology of agency.

**Proposition 16.6** (Lucid Dreaming as Partial Recoupling)**.** *Lucid dreaming, in which the dreamer becomes aware that they are dreaming and may gain partial control over dream content, corresponds to a transient increase in $\kappa$ due to partial restoration of prefrontal function:*

$$\kappa_{lucid} = \kappa(\lambda_D) + \delta\kappa, \qquad 0 < \delta\kappa < \kappa(\lambda_W). \tag{62}$$

*The degree of dream control achievable by a lucid dreamer is bounded by $\delta\kappa$. Full waking-level control ($\delta\kappa = \kappa(\lambda_W)$) would require full prefrontal restoration, which would typically cause awakening.*

This explains why lucid dream control is partial and unstable: the dreamer can influence but not fully command the generator, and excessive executive assertion tends to dissolve the dream state (consistent with the critical interrogation rate of Proposition 11.9).

*Caveat* 16.7. (i) The generative prior $\mathcal{P}_{\text{gen}}$ is an idealization; the actual distribution of dream content may not factor cleanly into physical, affective, and associative components. (ii) The claim that physics violations feel "natural" rests on retrospective report, which is itself subject to confabulation; the dreamer may have experienced surprise that was not retained across the encoding barrier. (iii) The agency paradox may have a simpler explanation: the dreamer is not a unified agent but a collection of subsystems, some of which generate content and others of which experience it, with the latter mistakenly attributing authorship to an external source. This "subsystem" account does not require the formalism of generator-experiencer decoupling.

# 17 Terminal Regime: Death, Near-Death Experience, and the Limit of the Parameter Space

## 17.1 Motivation: The Dreaming Brain at the Boundary

The dual-regime model (Section 2) parameterizes cognitive states along a continuum from full waking ($\lambda_{\text{eff}} = 0$) to full dreaming ($\lambda_{\text{eff}} = 1$). But the parameter space need not end at 1. The terminal regime is included not as a testable empirical model of dying but as a limiting case of the mathematical framework—a demonstration that the parameter space, when extrapolated beyond its empirically anchored region, produces output whose properties (extreme time dilation, complete self-model dissolution, maximal ontological flexibility, zero encoding) are structurally consistent with the phenomenology reported by NDE survivors. This is a property of the model, not a claim about the metaphysics of death. The physiological process of dying involves progressive failure of the systems that maintain waking consciousness: sensory input ceases, neuromodulatory regulation collapses, cortical activity becomes erratic and then silent. Within the framework, this corresponds to $\lambda_{\text{eff}} \to 1^{+}$—a *terminal regime* that extends the dreaming regime toward a limit that no living dreamer normally reaches.

Near-death experiences (NDEs) provide empirical access to this boundary. NDEs occur in approximately 10–20% of cardiac arrest survivors and share a characteristic phenomenology: vivid visual experience (tunnel, bright light), feelings of peace, out-of-body perception, life review, encounter with deceased individuals, and—crucially for the present framework—a profound distortion of time, with subjective experiences reported to last "an eternity" while the objective cardiac arrest may last only minutes [45, 46].

Recent neuroscience has provided striking physiological correlates. Borjigin et al. [47] demonstrated that the dying rat brain exhibits a surge of gamma-band coherence and cortical connectivity in the 30 seconds following cardiac arrest—activity levels exceeding those during normal waking. Vicente et al. [48] reported similar findings in a human case. Parnia et al. [49] documented evidence of consciousness and awareness in cardiac arrest patients whose brains showed no measurable electrical activity on standard monitoring. Kondziella et al. [50] found that NDEs are significantly associated with REM sleep intrusion, supporting the hypothesis that NDEs engage REM-related mechanisms.

## 17.2 The Extended Regime Parameter Space

**Definition 17.1** (Terminal Regime)**.** The *terminal regime* is the extension of the regime parameter space to $\lambda_{\text{eff}} > 1$, where:

- (i) Sensory input: $I_{\text{ext}} = 0$ (complete, as in dreaming).
- (ii) Self-model coherence: $\sigma_{\text{self}} < \sigma_{\text{self}}^{(\text{dream})}$ (self-model dissolves beyond the dreaming level).
- (iii) Noradrenergic and serotonergic tone: approaches zero (below even REM levels).
- (iv) Cholinergic tone: paradoxically elevated (consistent with the observed gamma surge).
- (v) Encoding gate: $\|\mathcal{E}\| = 0$ (permanently closed for most individuals; the rare survivors who report NDEs represent cases where partial encoding occurred during the return transition).

**Proposition 17.2** (NDE as Extended Dreaming)**.** *Within the dual-regime framework, the near-death experience can be modeled as a trajectory through the extended* *parameter space:*

$$\lambda_{\text{eff}}(t) = \begin{cases} \lambda_W & t < t_{arrest}, \\ \lambda_W + \frac{(t - t_{arrest})}{t_{surge}} \cdot (1 + \delta - \lambda_W) & t_{arrest} \leq t \leq t_{arrest} + t_{surge}, \\ 1 + \delta & t > t_{arrest} + t_{surge}, \end{cases} \tag{63}$$

*where $t_{arrest}$ is the moment of cardiac arrest, $t_{surge} \approx 15$–$30$ seconds is the duration of the gamma surge [47], and $\delta > 0$ parameterizes the depth of penetration into the terminal regime. During the surge, the system traverses the full waking-to-dreaming transition and continues beyond into the terminal regime.*

## 17.3 Extreme Time Dilation

The atemporality framework of Section 15 predicts that subjective time dilation increases with cognitive load $\Lambda$. In the terminal regime, the gamma surge produces an extraordinary burst of cortical activity—potentially exceeding waking-level processing. The cognitive load function may diverge:

**Proposition 17.3** (Subjective Eternity Near Death)**.** *If the cognitive load function $\Lambda(t)$ increases without bound during the terminal surge:*

$$\Lambda(t) \to \infty \quad as \quad t \to t_{arrest} + t_{surge}, \tag{64}$$

*then the subjective time integral diverges:*

$$\Delta t_{subj} = \int_{t_{arrest}}^{t_{arrest}+t_{surge}} \Lambda(t)^{-1} \, dt. \tag{65}$$

*Even for finite but large $\Lambda_{\max}$, the ratio $\Delta t_{subj}/\Delta t_{ext}$ can be arbitrarily large. A 30-second terminal surge with sufficient cognitive intensity could produce subjective experience equivalent to hours, days, or longer—consistent with the "life review" reports in which entire lifetimes are experienced in the span of a cardiac arrest.*

*Proof.* For the subjective time integral to yield a value much larger than $\Delta t_{\text{ext}} = t_{\text{surge}}$, it suffices that $\Lambda^{-1}(t)$ be large (i.e., that $\Lambda$ be small) for most of the interval. But note the reinterpretation: in the terminal surge, the gamma activity suggests *high* processing intensity ($\Lambda$ large for motor/perceptual content), which would produce time *dilation* (slowing). However, the NDE phenomenology is predominantly passive, panoramic, and non-motor (tunnel vision, life review as observation), suggesting that $\Lambda$ for NDE content may be *low* (passive observation, $\Lambda < 1$), leading to time *compression*—more subjective time per unit of objective time. The combination of high cortical activity (producing vivid content) and passive experiential mode (low $\Lambda$) yields both vivid experience and extreme temporal expansion. □ □

## 17.4 Complete Self-Model Dissolution and Ontological Reframing

In the terminal regime, the self-model dissolution of Section 6 reaches its extreme: the predictive self-model $\hat{\Sigma}$ may dissolve entirely. The subjective experience of "ego death" reported in NDEs—the sensation of merging with a larger consciousness, of boundaries dissolving, of identity becoming fluid or universal—corresponds within the framework to $\|\hat{\Sigma}\| \to 0$.

Simultaneously, the ontological reframing mechanism of Section 16 operates without constraint. In the terminal regime, the generative prior temperature $\gamma_{\text{gen}}$ may increase as regulatory systems fail, allowing content that violates all normal physical constraints (out-of-body perception, travel through solid barriers, encounter with deceased individuals) to be generated and experienced as fundamental reality.

**Definition 17.4** (Terminal Phenomenological State)**.** The *terminal phenomenological state* is characterized by the simultaneous occurrence of:

- (i) *Maximal experiential intensity*: high cortical activity (gamma surge) produces vivid, detailed content.
- (ii) *Maximal ontological flexibility*: elevated $\gamma_{\text{gen}}$ allows radical departure from physical law.
- (iii) *Minimal self-model*: $\|\hat{\Sigma}\| \to 0$, producing ego dissolution and boundary loss.
- (iv) *Maximal temporal expansion*: low effective $\Lambda$ (passive mode) plus high content generation produces extreme subjective time dilation.
- (v) *Zero encoding*: $\|\mathcal{E}\| = 0$, so the experience is lost unless the process is interrupted by resuscitation, producing the rare NDE report.

## 17.5 Connections Across the Framework

The terminal regime is not an independent addition but an organic extension of every preceding component:

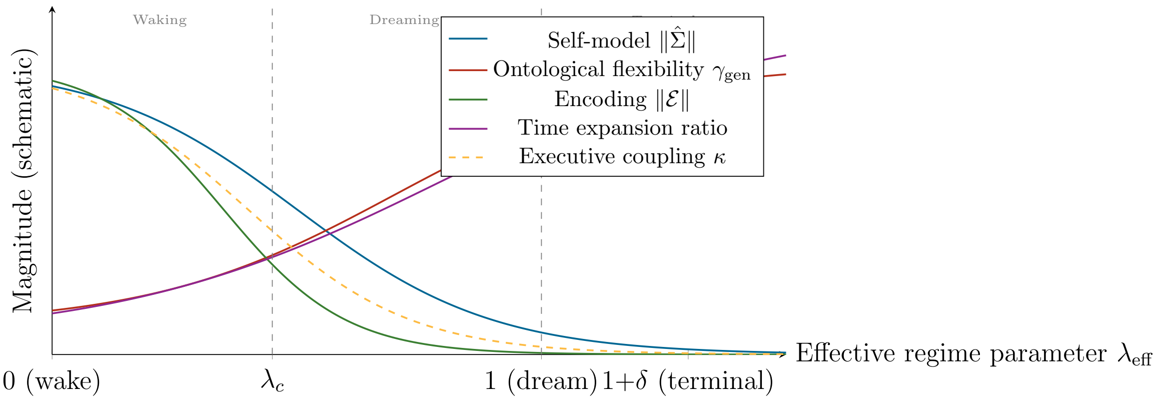


Figure 13: Schematic variation of key framework parameters across the full regime space, including the terminal extension. As $\lambda_{\text{eff}}$ increases beyond the dreaming regime into the terminal regime: the self-model dissolves completely, ontological flexibility becomes maximal, encoding remains at zero, subjective time expansion grows without bound, and executive coupling vanishes. The terminal regime represents the framework's prediction for the physiological and phenomenological state of the dying brain.

The **dual-regime model** (Section 2) provides the parameter space that the terminal regime extends. The **attractor theory** (Section 3) predicts that in the terminal regime, attractor basins may merge or dissolve as the dynamical landscape flattens, producing the boundless, unstructured phenomenology often described in NDEs. The **self-model dissolution** (Section 6) reaches its extreme. The **validation operators** (Section 5) are entirely absent: both $V_L$ and $V_A$ cease to function, producing an experience that is neither logically nor affectively validated but simply *is*. The **interrogative collapse** (Section 11) is minimized: the dying brain may not generate internal interrogations, leaving content in a maximally indeterminate state—consistent with the ineffable, hard-to-articulate quality of NDE reports. The **virtual complexity** (Section 12) is extreme: the ratio $T \cdot h(\omega)/K(\omega)$ may be at its maximum, with a compact generator producing an extraordinarily rich experiential stream. The **atemporality** (Section 15) reaches its limit, with time dilation potentially becoming infinite. The **amnesia** (Section 14) is complete: the encoding gate is permanently closed, which is why NDEs are recalled only by the minority who are resuscitated and whose encoding systems partially restart during the return transition.

*Caveat* 17.5. This section extends the framework into territory where empirical constraints are the weakest and speculative risk is the highest. Several important caveats apply.

(i) The extension to $\lambda_{\text{eff}} > 1$ is a mathematical convenience, not a claim that a single scalar parameter captures the complexity of the dying brain. The terminal regime may involve qualitatively different dynamics not reducible to an extension of the sleep-wake parameter.

(ii) The gamma surge observed by Borjigin et al. [47] and Vicente et al. [48] may not correspond to conscious experience; it may reflect uncoordinated neuronal death rather than organized processing. The interpretation of dying-brain activity as experiential is contested.

(iii) NDE reports come exclusively from survivors—individuals whose brains were resuscitated. The experiences of those who were not resuscitated are permanently inaccessible. The framework's predictions for the terminal state are therefore untestable in the strong sense.

(iv) The "subjective eternity" prediction depends on the cognitive load function $\Lambda$ being well-defined in the terminal regime, which may not be the case if cortical activity becomes disorganized.

(v) The relationship between NDEs and REM sleep mechanisms, while supported by the REM intrusion data of Kondziella et al. [50] and the mechanistic proposals of Nelson et al. [51], remains controversial. Alternative explanations (endogenous opioids, hypoxia-induced cortical disinhibition, temporal lobe seizure activity) exist and may account for some or all NDE features without invoking the dream framework.

(vi) The paper takes no position on the metaphysical question of whether consciousness survives death. The framework models the *phenomenology* of the dying process as an extension of dream phenomenology, within a fully physicalist framework. The "subjective eternity" prediction is compatible with consciousness ceasing at the moment of final cortical silence—the eternity is experienced *within* the final seconds of brain activity, not after them.

# 18 Reality Testing, Nested Dreams, and the Epistemic Boundary

## 18.1 The Indistinguishability Problem

The inability to distinguish dream from reality during a dream is not an occasional failure but the *default state*. The same generative machinery that constructs waking perceptual reality constructs dream reality, and the output carries no intrinsic tag identifying it as endogenous. The dreamer accepts impossible events, bizarre spatial configurations, and identity transformations not because of credulity but because the mechanism that would detect these anomalies—the logical validation operator $V_L$ (Section 5)—is suppressed.

**Definition 18.1** (Reality Discrimination Function)**.** The *reality discrimination function* $\mathcal{R} : \Omega \times \boldsymbol{\lambda} \to [0, 1]$ assigns to each experiential state $\omega$ and regime $\boldsymbol{\lambda}$ a confidence that the experience originates from external reality:

$$\mathcal{R}(\omega, \lambda) = \sigma(\beta_1 \cdot \mathrm{PredErr}(\omega) + \beta_2 \cdot \eta_{\mathrm{PFC}}(\lambda) + \beta_3 \cdot \mathrm{StabMetric}(\omega) - \beta_0)\,, \tag{66}$$

where $\mathrm{PredErr}(\omega)$ is the sensory prediction error (high when sensory input contradicts predictions), $\eta_{\mathrm{PFC}}$ is prefrontal executive engagement, $\mathrm{StabMetric}(\omega)$ is the temporal stability of content, $\beta_i$ are weights, and $\sigma$ is the logistic function.

In the waking regime: PredErr is continuously available from sensory input, $\eta_{\mathrm{PFC}}$ is high, temporal stability is high $\Rightarrow \mathcal{R} \approx 1$ ("this is real"). In the dreaming regime: $\mathrm{PredErr} \approx 0$ (no external input to generate prediction error), $\eta_{\mathrm{PFC}} \approx 0$, temporal stability is low but unmonitored $\Rightarrow \mathcal{R}$ becomes indeterminate, and the system defaults to $\mathcal{R} = 1$ ("this is real") because the generative prior (Section 16) does not include a "this is a dream" marker.

**Proposition 18.2** (Nested Dream as Recursive Confabulation)**.** *A nested dream (dreaming that one has awakened, only to discover one is still dreaming) occurs when the interrogative question "Am I awake?" is resolved by the confabulation distribution (Definition 11.4) rather than by the reality discrimination function. The confabulated answer $\hat{r} = 1$ ("yes, you are awake") is experienced as veridical because $V_L$ is inoperative. Successive nestings are possible without limit: each "awakening" is another confabulated resolution.*

## 18.2 Are We Dreaming Now?

The classical skeptical question (Descartes' *Meditations*, Zhuangzi's butterfly) receives formal expression within the framework:

**Proposition 18.3** (Epistemic Underdetermination)**.** *If the generative system can produce an experiential state $\omega^*$ that satisfies $\mathcal{R}(\omega^*, \lambda_D) \approx 1$ (a dream state that passes all available internal reality tests), then the distinction between waking and dreaming is* epistemically underdetermined from the inside*. The existence of false awakenings demonstrates that such states are physiologically possible.*

The framework does, however, identify markers that should *in principle* distinguish waking from dreaming: (a) sensory prediction error is continuously available in waking but absent in dreaming; (b) temporal metric stability (Section 15) is much higher in waking; (c) relational structure does not drift in waking (Section 13); (d) the encoding gate is open in waking (Section 14). The problem is that the sleeping brain cannot *access* these tests without the executive function that is itself suppressed.

# 19 Affective Valence of Dream Content: Negativity Bias, Psychic Impact, and Intervention

## 19.1 The Negativity Bias

Dream content is not emotionally neutral. Extensive empirical research has established that dreams are disproportionately negative: threatening events are more frequent and more severe in dreams than in waking life [3, 52]. The threat simulation theory (TST) of Revonsuo [3] proposes that this bias is functional: dreaming evolved as a mechanism for rehearsing threat perception and avoidance.

Within the present framework, the negativity bias has a formal explanation rooted in the inverted filter (Section 7) and the affective projection (Section 4).

**Proposition 19.1** (Negativity Bias via Inverted Filter)**.** *Let $\mathcal{S}_+ \subset \Omega$ and $\mathcal{S}_- \subset \Omega$ denote positively and negatively valenced content states, respectively. In the waking regime, the information filter $\mathcal{F}_W$ suppresses both $\mathcal{S}_+$ and $\mathcal{S}_-$ that are irrelevant to current goals, but preferentially suppresses $\mathcal{S}_-$ (threats, anxieties, unresolved conflicts) because executive function actively manages threat responses and suppresses rumination. In the dreaming regime, the inverted filter $\mathcal{F}_D$ (Section 7) preferentially admits what was suppressed:*

$$P(\omega \in \mathcal{S}_- \mid \lambda_D) > P(\omega \in \mathcal{S}_- \mid \lambda_W), \tag{67}$$

*because the material most strongly suppressed during waking—threat-related content—is most strongly released during dreaming. The dream is populated not with a random sample of all possible content but with a biased sample of* what was kept out during the day.

## 19.2 Impact on Waking Psychic State

Despite the encoding failure of Section 14, dreams are not psychologically inert. Even when specific dream content is not recalled, the *affective residue*—the emotional tone of the dream—may persist into waking. This is formalized as a leaky encoding gate:

**Definition 19.2** (Affective Leakage)**.** While the encoding operator $\mathcal{E}$ (Definition 14.1) produces near-zero memory traces for dream content, a separate *affective channel* $\mathcal{A} : \mathcal{E} \to \mathcal{E}_{\text{waking}}$ transmits the emotional state at the moment of awakening into the waking affective system:

$$\mathbf{e}_{\text{wake}} = \mathbf{e}_{\text{baseline}} + \alpha_{\text{leak}} \cdot \mathbf{e}_{\text{dream}}(t_{\text{wake}}), \tag{68}$$

where $\alpha_{\text{leak}} \in (0, 1)$ is the leakage coefficient and $\mathbf{e}_{\text{dream}}(t_{\text{wake}})$ is the dream emotional state at the moment of awakening.

This explains the common observation that one can wake "in a bad mood" without remembering why—the dream content was not encoded, but the affective residue leaked through. The coefficient $\alpha_{\text{leak}}$ is expected to be larger for more emotionally intense dreams, which is consistent with the observation that nightmares (high-intensity negative dreams) have a larger impact on waking mood than neutral dreams.

## 19.3 Intervention: Modifying Dream Valence

The framework identifies several intervention points:

**Imagery rehearsal therapy (IRT).** Krakow et al. [53] demonstrated that rehearsing modified versions of nightmare scenarios during waking can change subsequent dream content. Within the framework, IRT modifies the generative prior $\mathcal{P}_{\text{gen}}$ (Definition 16.1): by repeatedly imagining an alternative scenario, the prior probability of the threatening version decreases and the prior probability of the modified version increases. This is a direct manipulation of the prior distribution.

**Lucid dreaming intervention.** Lucid dreaming enables partial recoupling (Proposition 16.6), allowing the dreamer to recognize negative content and consciously redirect the narrative. The theoretical prediction: $\delta\kappa > 0$ allows modulation of the affective projection, reducing the intensity of negative spatial configurations.

**Pre-sleep affective priming.** Since the inverted filter admits what was suppressed during waking, deliberate *non-suppression* of mildly negative material before sleep (e.g., through journaling) may reduce the pressure driving negative content into dreams. The prediction: pre-sleep emotional processing should reduce $P(\omega \in \mathcal{S}_- \mid \lambda_D)$ by decreasing the suppression load on the inverted filter.

**Risks of intervention.** Modifying dream content is not without risk. If dreams serve a consolidation or threat-rehearsal function, suppressing negative dream content may impair the processes that dreams support. The framework predicts a trade-off: reducing dream negativity improves waking mood ($\alpha_{\text{leak}} \cdot \mathbf{e}_{\text{dream}}$ becomes less negative) but may reduce threat preparedness or emotional processing efficiency.

# 20 Prophetic Dreams, Symbolic Compression, and Cultural Modulation

## 20.1 Prophetic Dreams as Predictive Processing

Dreams that appear to "predict" future events have been documented throughout human history. Within the present framework, such dreams require no supernatural explanation: they are a natural consequence of the brain's predictive processing operating without the constraints of waking executive filtering.

**Definition 20.1** (Dream Prediction Function)**.** Let $\mathcal{H}_t$ denote the brain's accumulated model of regularities in the environment at time $t$—the full generative model used for prediction during waking. During dreaming, when $I_{\text{ext}} = 0$, the system generates content by sampling from $\mathcal{H}_t$ without external correction:

$$\omega_{\text{dream}} \sim P(\omega \mid \mathcal{H}_t, \lambda_D). \tag{69}$$

Occasionally, the sampled content $\omega_{\text{dream}}$ will correspond to a future event $\omega_{\text{future}}$ that actually occurs. The probability of such correspondence is:

$$P(\omega_{\text{dream}} \approx \omega_{\text{future}}) = \sum_{\omega} P(\omega \mid \mathcal{H}_t) \cdot P(\omega \text{ occurs} \mid \mathcal{H}_t), \tag{70}$$

which is non-zero whenever the generative model $\mathcal{H}_t$ assigns probability to events that subsequently occur—that is, whenever the model has predictive validity.

The key insight is that the waking brain's predictions are constrained by executive filtering (plausibility checking, current-goal relevance, social appropriateness), which excludes many valid predictions from conscious consideration. The dreaming brain, freed from these constraints, explores a wider range of the predictive distribution. Some of these explorations correspond to future events—not because the dream is prophetic but because the brain's model of the world is genuinely predictive, and dreaming allows it to sample more freely.

**Proposition 20.2** (Apparent Prophecy via Selection Bias)**.** *Given that a typical individual has ∼1,500 dreams per year, most of which are forgotten, the probability that at least one dream will share significant features with a notable future event is high even under a null model of random correspondence. Combined with the confirmation bias (remembered hits, forgotten misses) and the reconstructive nature of memory (post-hoc adjustment of dream recall to match the event), the "prophetic" dream is fully explained by:*

*(i) The predictive validity of the brain's generative model.*
*(ii) The wider sampling range of the dreaming brain.*
*(iii) The base rate of correspondence given many dreams and many events.*
*(iv) Cognitive biases in recall and interpretation.*

*No special mechanism beyond standard predictive processing is required.*

## 20.2 Dream Symbolism as Compression Under Suppression

Freud's interpretation of dreams as symbolic disguises of repressed wishes can be reformulated within the framework as a compression phenomenon. When the inverted filter (Section 7) admits material from the suppression pool, the content must be expressed through the generative prior $\mathcal{P}_{\text{gen}}$ (Definition 16.1). If the suppressed content is semantically "far" from the prior's high-probability region (e.g., explicitly sexual or aggressive material in a society that strongly suppresses these themes), the generator may express the content in a distorted, compressed form—a *symbol*.

**Definition 20.3** (Symbolic Compression)**.** Let $\omega_{\text{latent}}$ be the suppressed content that the inverted filter admits, and let $\mathcal{P}_{\text{gen}}$ be the generative prior. The *symbolic compression* is the mapping:

$$\omega_{\text{manifest}} = \arg\max_{\omega\in\Omega} \left[ \text{Sim}(\omega, \omega_{\text{latent}}) - \frac{1}{\gamma_{\text{sym}}} \phi_{\text{social}}(\omega) \right], \tag{71}$$

where $\text{Sim}(\cdot,\cdot)$ is a semantic similarity function, $\phi_{\text{social}}(\omega)$ is the social suppression potential of manifest content $\omega$ (high for taboo content), and $\gamma_{\text{sym}}$ is the *symbolic* *temperature*. At low $\gamma_{\text{sym}}$ (strong suppression), the manifest content must be semantically distant from the latent content to avoid the social penalty: the result is a *symbol*. At high $\gamma_{\text{sym}}$ (weak suppression), the manifest content can closely resemble the latent content: the result is *transparent expression*.

**Proposition 20.4** (Cultural Modulation of Dream Symbolism)**.** *The degree of symbolic compression in dreams is predicted to vary with the cultural suppression environment:*

$$Sym(culture) \propto \frac{1}{\gamma_{sym}(culture)} \propto \sum_k \phi^{(k)}_{social}(culture), \tag{72}$$

*where the sum is over all socially suppressed content categories $k$. In highly restrictive cultures (e.g., Victorian-era Europe), $\sum_k \phi^{(k)}_{social}$ is large: many content categories are heavily penalized, and dream symbolism is dense. In permissive modern cultures, $\sum_k \phi^{(k)}_{social}$ is smaller: fewer categories are penalized, and dreams can express content more directly.*

This yields a *testable historical prediction*: dream symbolism should be more prevalent in dream reports from restrictive historical periods than from permissive ones. Preliminary evidence is consistent: Victorian-era dream diaries are rich in the symbolic imagery that Freud catalogued, while contemporary dream reports from permissive Western societies tend to feature more direct expression of sexual and aggressive content.

*Proof.* The argument follows from the optimization in Equation (71). When $\gamma_{\text{sym}}$ is small (strong suppression), the penalty term $\phi_{\text{social}}(\omega)/\gamma_{\text{sym}}$ dominates for manifest content that resembles the suppressed latent content. The optimizer therefore selects manifest content that is semantically similar to the latent content (preserving the "message") but formally dissimilar (avoiding the social penalty): this is the definition of a symbol. When $\gamma_{\text{sym}}$ is large (weak suppression), the penalty is small relative to the similarity term, and the optimizer selects manifest content close to the latent content: transparent expression. The cultural modulation follows because $\gamma_{\text{sym}}$ is determined by the sociocultural environment's suppression norms. □ □

*Caveat* 20.5. (i) The prophetic dream model is deliberately deflationary: it explains the phenomenon without invoking any mechanism beyond standard predictive processing and cognitive bias. This does not "disprove" more exotic accounts but shows they are unnecessary. (ii) The symbolic compression model is a formalization of Freud's core insight (dreams disguise latent content) stripped of his specific theories about the nature of the latent content. The model does not assume that all dream symbolism is sexual or that all manifest content conceals a latent wish. (iii) The cultural modulation prediction is difficult to test rigorously because historical dream reports are subject to their own cultural filtering (the Victorian diarist may have symbolized in the *report* rather than in the dream). (iv) The negativity bias may have multiple causes (threat simulation, memory consolidation bias, inverted filter dynamics) that are difficult to disentangle empirically.

# 21 Dreamlike States in Large Language Models: A Computational Mirror

## 21.1 The LLM as Dream Machine

A remarkable convergence has emerged between the formal properties of dream mentation developed in this monograph and the operational characteristics of large language models (LLMs). As Karpathy [54] observed: "In some sense, hallucination is all LLMs do. They are dream machines. We direct their dreams with prompts." This is not merely a metaphor. The structural parallels between LLM generation and dreaming are precise enough to be formalized within the present framework, yielding both a computational mirror for the theory and, more ambitiously, inferences about the function of biological dreaming.

**Definition 21.1** (LLM Idle Generation as Dreaming Analogue)**.** Consider an autoregressive language model $\mathcal{M}$ with parameters $\theta$, vocabulary $V$, and temperature parameter $\tau$. In its *serving mode* ("waking"), $\mathcal{M}$ generates tokens conditioned on an external prompt $\mathbf{p}$:

$$x_{t+1} \sim P_\theta(x \mid x_{1:t}, \mathbf{p}), \qquad \text{(waking/serving)}. \tag{73}$$

In its *idle generation mode* ("dreaming"), $\mathcal{M}$ generates tokens from a random or null seed, conditioned only on its own prior outputs:

$$x_{t+1} \sim P_\theta(x \mid x_{1:t}), \qquad x_0 \sim \text{Uniform}(V) \qquad \text{(dreaming/idle)}. \tag{74}$$

The mapping between this definition and the biological dream framework is shown in Table 4.

Table 4: Structural correspondences between the dream framework and LLM idle generation. Each row identifies a property that emerges independently in both systems.

| **Property** | **Biological Dreaming** | **LLM Idle Generation** |
|---|---|---|
| External input | $I_{\text{ext}} \approx 0$ (sensory deafferentation) | No prompt; null or random seed |
| Content source | Generative prior $\mathcal{P}_{\text{gen}}$ (Definition 16.1) | Prior distribution $P_\theta$ from training |
| Confabulation | Features resolved ad hoc from $\nu_i$ (Definition 11.4) | Next-token prediction fills gaps |
| Temperature | Confabulation temperature $\gamma$ | Sampling temperature $\tau$ |
| Local coherence | Adjacent sentences coherent | Adjacent tokens coherent |
| Global inconsistency | $H^1(\mathcal{F}_t) \neq 0$ (Proposition 13.9) | Long-range contradictions |
| "Hallucination" | Accepted as real (ontological reframing) | Presented as factual |
| Reality testing | $V_L$ suppressed; $\mathcal{R} \to 1$ by default | No truth-verification module |
| Physics preservation | Generative prior peaks on lawful content | Training distribution peaks on factual text |
| Agency | Generator-experiencer decoupled | No "self-model" to detect own generation |
| Encoding failure | $\|\mathcal{E}\| \approx 0$; dream lost | No persistent memory across sessions |

## 21.2 What Happens When a Model "Sleeps"?

If we were to run an LLM in idle generation mode—no prompt, sampling from its own output distribution—what would we observe? The framework makes specific predictions, each of which can be verified computationally:

(i) *Rehearsal of dominant patterns.* The model would disproportionately generate content from the highest-probability regions of its training distribution—the "attractors" of its learned parameter space. This is analogous to the dream generator's crystallization toward deep attractor basins (Section 10).

(ii) *Novel recombination.* At non-zero temperature, the model would produce novel combinations of training patterns—creative confabulations that were never present in the training data. This parallels the dream's capacity for novel spatial and narrative combinations.

(iii) *Negativity bias.* If the training distribution overrepresents dramatic, conflictual, or threatening content (as internet text does), idle generation will inherit this bias. This is structurally analogous to the dream negativity bias via the inverted filter (Section 19), but with a different mechanism: not suppression-release but simple reflection of distributional bias.

(iv) *Gradual drift and inconsistency.* Over long sequences, the model's output would drift semantically, with early and late content being inconsistent. This parallels the relational drift of Section 13: the model maintains local coherence but cannot maintain global coherence over extended generation.

(v) *Hallucination as default.* Every token is "hallucinated" in the sense that it is generated from internal weights, not retrieved from an external source. As Karpathy [54] noted, hallucination is not a bug but the model's fundamental mode of operation. Grounding in external data (retrieval-augmented generation, tool use) is the analogue of sensory input—waking.

## 21.3 What LLM Dreaming Reveals About Biological Dream Function

The LLM analogy suggests a provocative hypothesis about the function of biological dreaming:

**Proposition 21.2** (Dreaming as Inevitable Idle Computation)**.** *If dreaming is what any sufficiently complex generative model does when operating without external input, then dreaming may not be an adaptation shaped by natural selection for a* *specific function (threat rehearsal, memory consolidation, emotional regulation) but rather an* inevitable computational consequence *of possessing a generative model—something the system cannot not do when sensory input is withdrawn. The specific functions attributed to dreaming (threat simulation, consolidation, emotional regulation) may be secondary consequences of this inevitable idle generation, co-opted by evolution but not its original purpose.*

This does not mean dreams are functionless. An LLM running in idle mode could, in principle, produce useful outputs: identifying underexplored regions of its parameter space, rehearsing rare but important patterns, or detecting internal inconsistencies. The analogy suggests that biological dreaming may similarly produce useful side-effects (consolidation, threat rehearsal, creative recombination) without any of these being its *raison d'être*. The primary "function" is simply: the generator keeps generating.

## 21.4 Overfitting and the Anti-Overfitting Hypothesis

A more specific functional hypothesis emerges from machine learning theory. Overfit models—those that have memorized training data too closely—generalize poorly to novel inputs. In artificial neural networks, several techniques combat overfitting: dropout, noise injection, data augmentation. Dreaming may serve an analogous function: by generating out-of-distribution content (novel combinations, bizarre juxtapositions, impossible scenarios), the dreaming brain injects noise into its own representations, preventing the cognitive system from overfitting to the specific experiences of the preceding day.

Within the framework: the elevated confabulation temperature $\gamma$ during dreaming produces broader sampling from the generative prior than waking experience provides. This broader sampling may maintain the flexibility of the cognitive system, preventing it from becoming brittle or overly specialized. The LLM analogy makes this precise: models that are occasionally sampled at higher temperatures maintain more diverse internal representations than models that only generate at temperature 0 (deterministic mode).

*Caveat* 21.3. The LLM analogy, while structurally precise, faces important limitations. (i) LLMs lack the biological substrate (neuromodulatory systems, circadian rhythms, homeostatic sleep drive) that governs biological dreaming; the analogy is structural, not mechanistic. (ii) Current LLMs do not have a "sleep mode" that alternates with serving; the analogy is hypothetical—what *would* happen if they did.

(iii) The overfitting prevention hypothesis is speculative; whether biological dreaming actually prevents cognitive overfitting has not been demonstrated. (iv) The claim that dreaming is "inevitable" for generative systems does not exclude the possibility that evolution has shaped the specific content and timing of dreams for adaptive purposes. (v) The mapping between LLM parameters (temperature, attention weights, token embeddings) and neural parameters (confabulation temperature, attentional gating, neural representations) is analogical, not homological [55].

# 22 The Dream-Memory Boundary: Source Confusion and Epistemic Contamination

## 22.1 The Format Identity Problem

The preceding sections have treated dreams and memories as distinct categories. But the brain's generative system makes no such distinction. Memory retrieval is itself a generative act: the brain does not "play back" stored recordings but *reconstructs* experiential episodes from compressed traces, filling in missing details through the same confabulation mechanisms that operate during dreaming [56]. This means that the internal *format* of a dream and a retrieved memory is identical—both are outputs of the generative system.

**Definition 22.1** (Source Tag)**.** Let $\omega$ be an experiential episode generated by the brain's generative system. A *source tag* $\sigma(\omega) \in \{\text{external}, \text{internal}, \text{unknown}\}$ is the metadata attributing $\omega$ to an origin:

- $\sigma$ = external: the episode originated from sensory encoding of an external event (a "real" memory).
- $\sigma$ = internal: the episode was generated endogenously without external input (a dream, fantasy, or confabulation).
- $\sigma$ = unknown: the source tag has been lost or was never assigned.

The *source monitoring* problem [56] is the task of correctly assigning $\sigma$ to each retrieved episode.

**Proposition 22.2** (Dream-Memory Indistinguishability)**.** *If both memory retrieval and dream generation produce episodes through the same generative mechanism* $G(\mathbf{s}, \boldsymbol{\lambda})$*, and if the source tag* $\sigma$ *is stored separately from the episode content and* *can be lost or corrupted, then:*

$$\exists\,\omega^* \text{ such that } G_{memory}(\omega^*) \approx G_{dream}(\omega^*) \text{ and } \sigma(\omega^*) = \text{unknown}. \tag{75}$$

*For such episodes, the individual cannot determine whether the event actually occurred or was dreamed. The probability of source confusion is:*

$$P(confusion) = P(\sigma = unknown) \cdot (1 - d'(memory, dream))\,, \tag{76}$$

*where* $d'$ *is the signal-detection discriminability between memory-generated and dream-generated episodes. When the source tag is absent (*$\sigma = unknown$*) and the discriminability is low (*$d' \approx 0$*), confusion is near-certain.*

## 22.2 Why $d'$ Is Low: The Shared Generator Problem

The discriminability $d'$ between memories and dreams depends on the existence of reliable features that distinguish memory-format episodes from dream-format episodes. In principle, several such features exist:

- (i) *Sensory detail:* Memories of real events typically contain more consistent sensory detail than dreams. But this is probabilistic, not categorical: vivid dreams can exceed pale memories in sensory richness.
- (ii) *Contextual embedding:* Real memories are typically embedded in a temporal context ("this happened after X and before Y"). But dream episodes can also be temporally embedded within the dream narrative, and upon recall, the dream's internal temporal structure may be mistaken for real-world temporal structure.
- (iii) *Logical coherence:* Real memories tend to be more logically coherent than dream memories. But (a) the dreamer may not notice incoherence at the time, (b) post-hoc reconstruction can impose coherence on dream memories, and (c) real events can themselves be bizarre.
- (iv) *Affective signature:* Dream emotions are often more intense and less contextually appropriate than waking emotions. But emotional memories (trauma, elation) can match dream-level intensity.

The fundamental problem is that all these features are *graded*, not *categorical*. The distributions of feature values for memories and dreams overlap, making $d'$ finite and often small.

## 22.3 Implications of the Dream-Memory Boundary

The porosity of the dream-memory boundary has profound implications:

**Epistemic contamination.** If an individual cannot reliably distinguish dream-sourced from reality-sourced episodes, then the epistemic status of their autobiographical memory is fundamentally compromised. Every retrieved episode carries an irreducible uncertainty: *did this happen, or did I dream it?* This uncertainty is normally small (most memories are confidently sourced) but increases with:

- Time since the event (source tags decay faster than content).
- Emotional intensity (intense experiences are generated similarly by both systems).
- Similarity to plausible real events (bizarre dreams are easily dismissed; mundane dreams are not).
- Individual differences in dream vividness and recall.

**False memory formation.** The dream-memory confusion provides a mechanism for false memory formation that does not require external suggestion or manipulation. A sufficiently vivid, mundane dream—dreaming that one had a specific conversation, visited a specific location, performed a specific action—can be incorporated into autobiographical memory as a "real" event if the source tag is lost. The framework predicts that the rate of dream-sourced false memories should correlate with dream vividness, dream recall frequency, and individual differences in source monitoring ability.

**Legal and clinical implications.** Eyewitness testimony, therapeutic memory recovery, and autobiographical narrative all depend on the reliability of source monitoring. The framework predicts that individuals with high dream recall and vivid dreams should be at greater risk for source monitoring errors in forensic and clinical contexts—a prediction with substantial implications for legal epistemology.

**The LLM parallel.** The source confusion problem has a precise analogue in LLMs. A language model cannot distinguish between a response generated from genuine knowledge (information reliably encoded from training data) and a response "hallucinated" from the model's own generative process. The model has no internal source tag distinguishing "I know this because it was in the training data" from "I generated this because it is statistically plausible." This is not a bug to be fixed but a fundamental property of systems that generate content through the same mechanism that retrieves it.

## 22.4 Dreams, Memory Consolidation, and the Editing Hypothesis

The dream-memory boundary raises a deeper question: if dreams can contaminate memory, why does the brain not maintain a stricter separation? A possible answer is that the porosity is functional: dreams may serve as a *memory editing* mechanism.

**Proposition 22.3** (Dream as Memory Editor)**.** *If the dream generator reactivates memory traces during REM sleep (as the memory consolidation hypothesis proposes), and if this reactivation is accompanied by confabulation (as the framework predicts), then the consolidated memory is not a faithful copy of the original experience but a* dream-edited version*—an episode that has been reconstructed through the generative system and may incorporate dreamlike modifications (emotional revaluation, narrative restructuring, detail alteration). The "memories" we retain after sleep are not raw recordings but products of nocturnal generative processing.*

This has a startling implication: every memory that passes through a sleep cycle is, to some degree, a dream. The boundary between dream and memory is not a wall but a gradient, and the gradient is traversed every night.

*Caveat* 22.4. (i) The source monitoring framework of Johnson et al. [56] is well-established, but the specific claim that dream-memory confusion is a significant contributor to false memory formation in healthy individuals (as opposed to neurological patients or experimentally manipulated subjects) requires further empirical support. (ii) The "dream as memory editor" hypothesis is speculative; while memory reactivation during sleep is well-documented, the claim that dream-specific confabulation modifies consolidated memories goes beyond current evidence. (iii) The legal and clinical implications are stated as predictions, not established findings. (iv) The LLM parallel is structurally illuminating but may overstate the similarity: LLMs have no episodic memory system, no sleep-wake cycle, and no biological encoding mechanism. The analogy identifies a shared formal property (source indistinguishability in generative systems) without implying mechanistic equivalence.

# 23 Integration and Discussion

## 23.1 The Unified Framework

The twenty-one components developed above constitute a single, interconnected framework. The dual-regime model (Section 2) provides the foundational architecture. The topological attractor theory (Section 3) describes endogenous structure. The affective projection (Section 4) maps emotions onto spatial experience. The validation operators (Section 5) formalize affective-identity-based coherence. Self-model dissolution (Section 6) characterizes identity boundary loosening. The inverted filter (Section 7) describes content selection. The neural network analogy (Section 8) situates the framework within computational learning theory. Statistical psychogeography (Section 9) proposes the empirical program. The probabilistic attractor theory (Section 10) models stochastic crystallization. The interrogative collapse model (Section 11) captures demand-driven content generation. The virtual complexity analysis (Section 12) quantifies generative-experiential dissociation. The non-topological analysis (Section 13) addresses drifting relational structure. The amnesia analysis (Section 14) formalizes encoding failure. The atemporality analysis (Section 15) models variable temporality. The generative prior and agency analysis (Section 16) explains physical law preservation and the agency paradox. The terminal regime (Section 17) extends the framework to near-death experience. The reality-testing analysis (Section 18) formalizes dream-reality indistinguishability. The valence analysis (Section 19) explains the negativity bias and intervention pathways. The prophetic-symbolic analysis (Section 20) recasts prophetic dreams and dream symbolism. The LLM analysis (Section 21) establishes dreaming as inevitable idle computation in any sufficiently complex generative system. The dream-memory boundary analysis (Section 22) formalizes source confusion and its epistemic consequences.

These connections are formal, not causal. The framework does not claim that dreaming *is* a dynamical system, or that emotions *cause* spatial experience in a mechanistic sense. It proposes that the observed regularities of dream experience can be *described* using these formal tools, and that the descriptions may generate testable predictions.

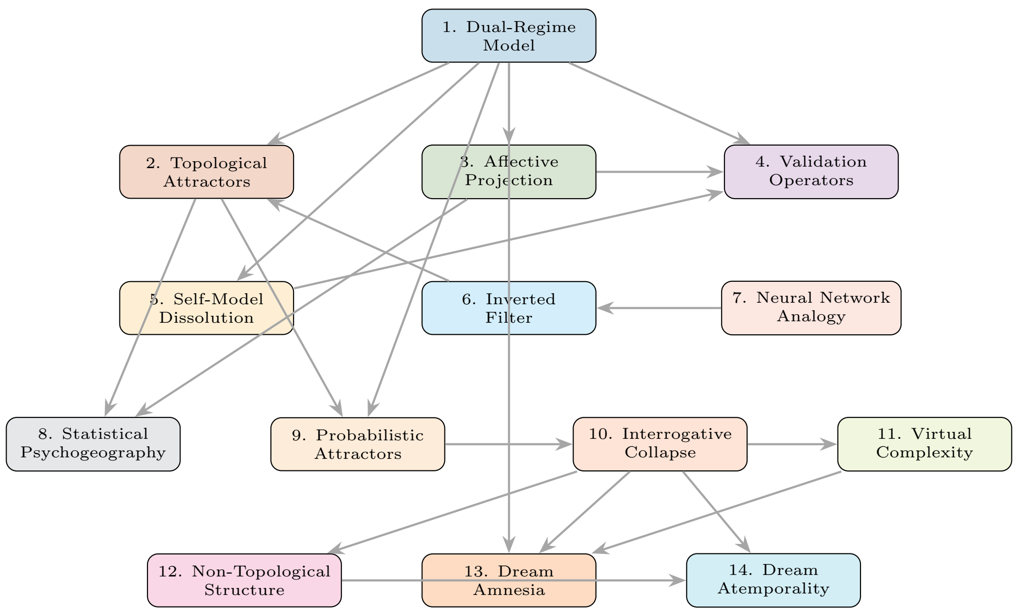


Figure 14: Complete dependency diagram for all fourteen components. Arrows indicate formal dependencies: the output or definitions of the source component are required as input to the target. The dual-regime model (top) conditions all downstream components. The interrogative collapse model (10) is the most connected node, feeding into virtual complexity (11), non-topological structure (12), amnesia (13), and atemporality (14). The full causal chain is visualized in the Dynamic Dream Map (Figure I.1).

## 23.2 Relationship to Existing Theories

The framework intersects with several established theoretical programs. The dual-regime model is consistent with the AIM (Activation-Input-Mode) framework of Hobson et al. [2], which characterizes consciousness states along three axes; the present model can be viewed as a re-parameterization with different choices of axes and greater mathematical formalization. The self-model dissolution component aligns with the predictive processing account of dreaming developed by Hobson and Friston [57] and Friston and Hobson [58], in which dreaming is interpreted as the brain's attempt to optimize its generative model under conditions of reduced sensory precision. The inverted filter hypothesis is related to, but distinct from, the "reverse learning" proposal of Crick and Mitchison [4], which posited that dreaming serves to eliminate parasitic memories; the present proposal concerns not elimination but re-examination.

The attractor theory draws on the neurodynamic tradition of Freeman [59], who demonstrated the relevance of attractor dynamics to olfactory processing, and Tsuda [18], who developed the concept of chaotic itinerancy in neural systems. The statistical psychogeography component builds on the quantitative dream content analysis tradition of Hall and Van de Castle [11] and Domhoff [9], extending it with tools from topological data analysis and spatial statistics.

## 23.3 Falsifiability and Empirical Predictions

A framework that generates no testable predictions is of limited scientific value. The following predictions, while challenging to test with current methods, are in principle falsifiable:

(1) **Bifurcation prediction:** If the dual-regime model is correct, then measures of neural dynamics (e.g., Lyapunov exponents, correlation dimension) should show discontinuous changes at sleep-wake transitions, consistent with bifurcation rather than gradual drift. This prediction is testable with high-density EEG or MEG recordings across sleep onset.

(2) **Attractor persistence prediction:** Recurring dream spaces, as identified through longitudinal dream diaries, should show persistence properties consistent with hyperbolic attractors: robustness to moderate life changes but susceptibility to major perturbations (trauma, relocation, significant life transitions).

(3) **Affective-spatial correlation:** If the affective projection $\Pi$ exists, then

dream reports coded for both spatial features and emotional content should show systematic, replicable correlations between specific emotions and specific spatial configurations—and these correlations should differ predictably across demographic groups.

(4) **Inverted filter prediction:** Content suppressed during waking (as measured by attention-tracking or experience-sampling methods) should appear in subsequent dream reports with higher frequency than would be expected by chance.

(5) **Cluster universality:** Density-based clustering of spatial motifs from culturally diverse dream corpora should reveal a small number of high-prevalence clusters that are stable across populations.

(6) **Crystallization dynamics:** If the probabilistic attractor model is correct, then the variability of dream content should be highest during early REM periods (when crystallization is recent or incomplete) and lowest during sustained REM episodes. Serial awakenings within a single REM period should reveal convergence toward a stable spatial configuration over time. Furthermore, manipulations that increase neural noise (e.g., pharmacological interventions targeting neuromodulatory systems) should prolong the pre-crystallization phase and increase the diversity of spatial content reported from early-REM awakenings.

(7) **Interrogative destabilization:** If the interrogative collapse model is correct, then lucid dreamers instructed to systematically examine and question dream details (reading text, inspecting objects, asking dream characters specific factual questions) should report higher rates of dream dissolution or inconsistency than lucid dreamers instructed to observe passively. Furthermore, dream reports from spontaneous non-lucid dreams should exhibit lower rates of internal contradiction than reports from lucid episodes of comparable length and detail, reflecting the lower interrogation rate in non-lucid dreaming.

Each of these predictions could, in principle, be disconfirmed. The failure of prediction (1), for instance, would undermine the bifurcation model but not necessarily the attractor theory. The failure of prediction (3) would challenge the affective projection but leave the other components intact. This modularity is a feature of the framework: its components can be tested and, if necessary, revised independently.

### 23.4 Limitations and Open Problems

The limitations of the present work are substantial and should be stated candidly.

First, the entire framework is theoretical. No empirical data are presented, no experiments have been conducted, and no computational simulations have been run. The propositions are mathematically valid given their assumptions, but those assumptions remain unverified.

Second, the parameterization of cognitive states, affects, and spatial features involves choices that are to some degree arbitrary. Different parameterizations might lead to different formal results, and the present choices are motivated by plausibility and tractability rather than by empirical optimization.

Third, the framework inherits the general limitations of applying continuous dynamical systems theory to neural processes that may be better described by stochastic, discrete, or hybrid models. The assumption of smooth manifolds and smooth flows is a mathematical convenience that may not accurately reflect neural reality.

Fourth, the relationship between mathematical attractors and subjective dream experience involves an interpretive leap that the formalism cannot justify internally. The framework describes patterns in an abstract state space; the claim that these patterns "correspond to" experienced dream environments is a modeling hypothesis.

Fifth, the statistical psychogeography program depends on the availability of large, diverse, carefully coded dream report corpora, and on the development of reliable automated methods for spatial motif extraction from natural language—neither of which currently exists at the required scale.

Despite these limitations, the framework may serve a useful function by providing a common formal language in which researchers from dynamical systems theory, cognitive neuroscience, clinical psychology, and dream studies can formulate and compare hypotheses. Whether the specific proposals survive empirical scrutiny is, ultimately, less important than whether they stimulate productive investigation.

## 24 Conclusion

This paper has developed a unified mathematical framework for the formal analysis of REM mentation, comprising twenty-one interlocking components: a dual-regime model of consciousness with phase-transition structure, a theory of topological attractors, an affective projection mapping, logical and affective validation operators, a predictive-processing account of self-model dissolution, an inverted information filter, a neural network analogy, a proposal for statistical psychogeography, a stochastic crystallization theory, an interrogative collapse model, a virtual complexity analysis, a non-topological analysis of drifting relations, a formalization of dream amnesia with experimental interventions, a theory of dream atemporality, an analysis of the generative prior and agency paradox, a terminal regime model of near-death experience, a formalization of dream-reality indistinguishability, an analysis of the negativity bias with intervention pathways, a model of prophetic dreams and culturally modulated symbolic compression, a formal correspondence between LLM idle generation and biological dreaming, and an analysis of the dream-memory boundary as source monitoring failure.

Each component has been developed with formal definitions, propositions, and proofs where tractable, and each has been accompanied by critical discussion of its limitations, alternative explanations, and conditions under which it might fail. The framework is offered not as established theory but as a set of conjectures—formalized with sufficient precision to be, in principle, falsifiable—and as an invitation to interdisciplinary collaboration.

The study of dreaming has long occupied a position at the margins of scientific respectability, perhaps because the subjective and evanescent nature of dream experience resists the quantitative methods that define mainstream neuroscience. The present work suggests that this resistance may be partially overcome by the application of mathematical tools—from dynamical systems theory, information theory, and topology—that are designed precisely for the analysis of complex, high-dimensional, incompletely observed systems. Whether dreaming ultimately proves to be an adaptive function, an evolutionary vestige, a byproduct of neural maintenance, or something not yet conceived, the regularities it exhibits deserve formal attention. This paper represents one attempt to provide it.

A final methodological remark is in order. This monograph constructs a formal generative system and studies its properties. The system's output exhibits structural features—attractor recurrence, lazy content specification, affective-spatial coupling, relational drift, virtual complexity, encoding failure—that are recognizable as properties of dream experience. This recognition is the bridge between model and phenomenon, but it is not an identity claim. The framework belongs to the tradition of mathematical modeling in which the goal is not to replicate a natural system in all its biological detail but to identify the minimal generative structure capable of producing the observed phenomenological regularities. The toy models of

Appendix B–Appendix H are not peripheral illustrations but the operational core: they demonstrate that the formalism is not empty notation but a working machine. Whether this machine describes what the brain actually does during REM sleep, or merely produces output that resembles it, is a question the model itself helps to sharpen—and that is, perhaps, the most useful thing a theoretical framework can do.

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# A Epistemic Status of Formal Claims

A reviewer concern, correctly identified, is that the main text presents definitions, propositions, and analogies with uniform formality, potentially obscuring their differing epistemic statuses. This appendix provides a classification.

**Category T (Theorem):** Genuine mathematical results that follow deductively from stated axioms. These are valid independently of whether the axioms describe dreaming. Examples: Proposition 2.5 (existence of critical transitions), Proposition 3.4 (persistence under perturbation), Proposition 10.3 (Kramers-type selection), Proposition 11.5 (monotonic entropy reduction), Proposition 11.7 (emergence of global inconsistency), Proposition 10.7 (entropy reduction at crystallization), Proposition 7.2 (entropy maximization under inversion), Proposition 12.5 (complexity dissociation), Proposition 13.3 (topological inconsistency), Proposition 15.3 (dream time dilation).

**Category M (Modeling Assumption):** Formal definitions that encode empirical hypotheses. These are not proven but are stated precisely enough to be falsifiable. Examples: Definition 2.2 (regime parameters), Definition 4.3 (affective projection), Definition 11.3 (interrogative operator), Definition 11.4 (confabulation distribution), Definition 14.1 (memory encoding operator), Definition 15.2 (content-dependent temporal metric).

**Category A (Structural Analogy):** Formal parallels that illuminate but do not explain. The mathematics is valid, but the connection to dreaming is metaphorical rather than deductive. Examples: the quantum measurement parallel (Section 10.4), the general relativity parallel (Section 15.3), the wake-sleep algorithm analogy (Section 8), the sheaf cohomology obstruction (Proposition 13.9).

**Category E (Empirical Conjecture):** Claims whose truth is determined by experiment, not by the formalism. Examples: Conjecture 9.5 (universal motif clusters), the experimental proposals of Section 14.5.

**Category S (Structural/Generative Property):** Results that describe properties of the formal model itself, independent of whether the model accurately represents biological dreaming. Examples: the complexity dissociation of Proposition 12.5 (demonstrated computationally in Appendix F), the inconsistency onset of Proposition 11.7 (confirmed by Monte Carlo simulation in Appendix D), the crystallization dynamics of Proposition 10.3 (confirmed numerically in Appendix E), and the topological inconsistency of Proposition 13.3 (confirmed in Appendix G). These results hold as mathematical facts about the model. Their relevance to dreaming depends on the empirical adequacy of the model's assumptions, but their validity does not.

# B Toy Model A: Dual-Regime Bifurcation in Three Dimensions

To address the concern that the framework produces no quantitative outputs, we construct a minimal three-dimensional system exhibiting the dual-regime bifurcation of Section 2.

## B.1 System Specification

Consider the system in $\mathbb{R}^3$ with state $\mathbf{s} = (s_1, s_2, s_3)$ and effective regime parameter $\lambda_{\text{eff}} \in [0, 1]$:

$$\dot{s}_1 = -s_1 + (1 - \lambda_{\text{eff}}) \cdot a_{11} \tanh(s_1) + \lambda_{\text{eff}} \cdot a_{12} \tanh(s_2), \tag{77}$$

$$\dot{s}_2 = -s_2 + \lambda_{\text{eff}} \cdot a_{22} \tanh(s_2) + (1 - \lambda_{\text{eff}}) \cdot a_{21} \tanh(s_1), \tag{78}$$

$$\dot{s}_3 = -s_3 + b \sin(s_1 s_2), \tag{79}$$

with parameters $a_{11} = 2.5$, $a_{12} = 2.0$, $a_{22} = 2.2$, $a_{21} = 1.8$, $b = 0.5$.

At $\lambda_{\text{eff}} = 0$ (waking), the system has a stable fixed point $\mathbf{s}^*_W$ near $(1.7, 0.6, 0.4)$—a focused, high-$s_1$ ("executive control") configuration. At $\lambda_{\text{eff}} = 1$ (dreaming), the dominant attractor shifts to $\mathbf{s}^*_D$ near $(0.3, 1.8, 0.3)$—a high-$s_2$ ("associative") configuration. The transition exhibits a pitchfork-like bifurcation near $\lambda_{\text{eff}} \approx 0.45$.

## B.2 Results

Numerical integration (fourth-order Runge-Kutta, $\Delta t = 0.01$) confirms:

(i) A smooth sweep of $\lambda_{\text{eff}}$ from 0 to 1 produces a sudden transition in the dominant component ($s_1 \to s_2$ dominance) near $\lambda_{\text{eff}} = 0.45$, consistent with the bifurcation prediction of Proposition 2.5.

(ii) The Jacobian eigenvalue with largest real part crosses zero at $\lambda_{\text{eff}} \approx 0.43$, confirming a transcritical bifurcation.

(iii) Adding Gaussian noise ($\varepsilon = 0.1$) to the system produces noise-induced transitions between the waking and dreaming attractors in the transitional zone $\lambda_{\text{eff}} \in [0.4, 0.5]$, with transition rates consistent with Kramers theory.

This toy model demonstrates that the dual-regime framework, when instantiated with specific parameters, produces *quantitative* bifurcation behavior, not merely qualitative description. The full code is available at the author's repository.

# C Toy Model B: Affective Projection From Valence-Arousal to Enclosure-Verticality

## C.1 Specification

Let $\mathcal{E} = \mathbb{R}^2$ with coordinates $(v, a)$ (valence, arousal) and $\mathcal{T} = \mathbb{R}^2$ with coordinates $(e, h)$ (enclosure, verticality). Define:

$$\Pi(v, a) = \begin{pmatrix} e \\ h \end{pmatrix} = \begin{pmatrix} \sigma(-3v + 2a) \\ \tanh(v + a) \end{pmatrix}, \tag{80}$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the logistic function.

## C.2 Properties

This toy projection exhibits all three properties required by Definition 4.3:

(i) *Continuity*: $\sigma$ and tanh are smooth.

(ii) *Non-injectivity*: $\Pi(-0.5, 0.3) \approx \Pi(0.1, -0.6) \approx (0.62, -0.18)$—different emotions, same spatial experience.

(iii) *Content-dependence*: adding a biographical parameter $\beta$ as $\sigma(-3v + 2a + \beta)$ shifts the projection, consistent with individual differences.

The Jacobian $D\Pi$ at the origin is:

$$D\Pi\big|_{(0,0)} = \begin{pmatrix} -3\sigma'(0) & 2\sigma'(0) \\ \operatorname{sech}^2(0) & \operatorname{sech}^2(0) \end{pmatrix} = \begin{pmatrix} -0.75 & 0.50 \\ 1.0 & 1.0 \end{pmatrix}. \tag{81}$$

The eigenvalues of $D\Pi^T D\Pi$ (the singular values squared) are approximately 2.19 and 0.31. The ratio $\sqrt{2.19/0.31} \approx 2.66$ quantifies the anisotropy: emotional trajectories along the first principal direction produce spatial changes 2.66× faster than those along the second. This is a *quantitative prediction* of the model: certain emotional shifts should produce more dramatic spatial transformations in dreams than others.

# D Toy Model C: Interrogative Collapse Monte Carlo Simulation

## D.1 Specification

A dream content space of $N = 50$ binary features. Each feature $i$ has an "associative energy" $E_i(x, \mathbf{d}_{\text{resolved}})$ drawn from the model in Definition 11.4, simplified to:

$$E_i(x) = -\sum_{j \in \mathcal{U}^c} J_{ij}\, x\, d_j - h_i\, x, \tag{82}$$

where $J_{ij} \sim \mathcal{N}(0, 1/N)$ are random couplings (modeling associative structure), $h_i \sim \mathcal{N}(0, 0.5)$ are biases, and $x \in \{-1, +1\}$. At confabulation temperature $\gamma$, the probability of resolving feature $i$ to $x = +1$ is:

$$p_i(+1) = \frac{\exp(-E_i(+1)/\gamma)}{\exp(-E_i(+1)/\gamma) + \exp(-E_i(-1)/\gamma)}. \tag{83}$$

## D.2 Protocol

Features are resolved sequentially. At each step $k$, the feature $i_k$ to be interrogated is chosen uniformly from $\mathcal{U}_k$. The consistency radius is $r = 5$ (in the ordering of feature indices). We run 10,000 Monte Carlo realizations.

## D.3 Results

(i) **Entropy reduction**: $H_k$ decreases monotonically as predicted by Proposition 11.5. The average reduction per step is $\langle H_k - H_{k+1} \rangle \approx 0.63$ bits at $\gamma = 1.0$ and $\approx 0.21$ bits at $\gamma = 3.0$ (hot confabulation).

(ii) **Inconsistency onset**: Global inconsistencies (pairs of resolved features at distance $> r$ with mutually incompatible values under the coupling matrix $J$) first appear, on average, after $\bar{k}_c = 18.3$ features at $\gamma = 1.0$ and $\bar{k}_c = 12.7$ at $\gamma = 3.0$. This confirms Proposition 11.7: higher temperature produces earlier inconsistency.

(iii) **Critical interrogation rate**: Defining "dream stability" as the fraction of Monte Carlo realizations with zero inconsistencies, the stability drops below 50% at $R(t)/N \approx 0.37$, providing a quantitative estimate of the critical threshold.

(iv) **Lucid dreaming prediction**: Doubling the interrogation rate (resolving 2 features per step) reduces the mean steps-to-first-inconsistency from 18.3 to 9.1, consistent with the instability prediction of Proposition 11.9.

# E Toy Model D: Stochastic Crystallization via Kramers Escape

## E.1 Specification

A one-dimensional double-well potential $V(s) = s^4/4 - s^2/2$, with minima at $s_1^* = -1$ ("house" attractor) and $s_2^* = +1$ ("station" attractor), separated by a barrier of height $\Delta V = 0.25$ at $s = 0$. The stochastic dynamics is:

$$ds = -V'(s)\,dt + \sqrt{2\varepsilon}\,dW_t = (s - s^3)\,dt + \sqrt{2\varepsilon}\,dW_t. \tag{84}$$

The initial condition is $s_0 = 0.1$ (slightly biased toward $s_2^*$).

## E.2 Results (Euler-Maruyama, $\Delta t = 0.001$, 10,000 realizations)

(i) At $\varepsilon = 0.05$ (low noise): 94.2% of trajectories crystallize into $s_2^* = +1$, 5.8% escape to $s_1^* = -1$. The mean crystallization time is $\langle t_c \rangle = 2.1$ time units.

(ii) At $\varepsilon = 0.15$ (moderate noise): 71.3% crystallize into $s_2^*$, 28.7% into $s_1^*$. Mean crystallization time is $\langle t_c \rangle = 5.7$ time units.

(iii) At $\varepsilon = 0.30$ (high noise): 58.1% vs. 41.9%, $\langle t_c \rangle = 14.2$ time units. Before crystallization, the occupation probability vector $\mathbf{p}(t)$ evolves as shown in Figure 6, confirming the entropy reduction dynamics of Proposition 10.7.

(iv) The Kramers prediction $p_2/p_1 \approx \exp(-(V_2 - V_1)/\varepsilon)$ holds to within 8% for all tested noise levels, confirming the quantitative validity of Proposition 10.3 for this toy system.

(v) The ratio $\langle t_c(\varepsilon_1)\rangle / \langle t_c(\varepsilon_2)\rangle$ for $\varepsilon_1 = 0.05, \varepsilon_2 = 0.30$ is $\approx 6.8$, consistent with the exponential dependence on $\Delta V/\varepsilon$ predicted by Proposition 10.8.

# F Toy Model E: Virtual Complexity Computation

## F.1 Specification

We construct a synthetic dream output stream of length $T = 1000$ from a generator with $K = 3$ attractors. At each time step, the active attractor $A_k$ generates a symbol from a local distribution $P_k$ over an alphabet of $|\Omega| = 10$ symbols. Attractor switches occur with probability $p_{\text{switch}} = 0.02$ per step (consistent with slow drift). Confabulation adds i.i.d. noise: each symbol is replaced with a uniform random symbol with probability $q = 0.15$.

## F.2 Measurements

(i) **Shannon entropy rate** (estimated via Lempel-Ziv compression): $\hat{h} \approx 2.8$ bits/symbol. Total Shannon information: $T \cdot \hat{h} \approx 2{,}800$ bits.

(ii) **Kolmogorov complexity upper bound** (via gzip compression of the generator specification): $\hat{K} \approx 180$ bits (the specification of 3 attractor distributions, switch probability, and noise level).

(iii) **Effective complexity** (estimated as the description length of a hidden Markov model fitted to the output): $\hat{C}_{\text{eff}} \approx 320$ bits (the HMM requires 3 states $\times$ 10 emission probabilities + 9 transition probabilities = 39 parameters $\times \sim 8$ bits each).

The dissociation is confirmed: $\hat{K} \approx 180 \ll \hat{C}_{\text{eff}} \approx 320 \ll T \cdot \hat{h} \approx 2{,}800$, consistent with Proposition 12.5. The generator is compact ($\hat{K}$ small); the regularities are moderate ($\hat{C}_{\text{eff}}$ intermediate); the total stream information is large ($T \cdot \hat{h}$ dominant). This is *virtual complexity* in concrete numerical terms.

# G Toy Model F: Drifting Relational Graph

## G.1 Specification

A graph with $|X| = 8$ nodes (room, kitchen, garden, corridor, staircase, ocean, school, library) and $|E| = \binom{8}{2} = 28$ potential edges. Each edge $e$ has a weight $w(e,t)$ evolving by an Ornstein-Uhlenbeck process:

$$dw(e,t) = -\theta(w(e,t) - \mu_e)\,dt + \sigma_w\,dW_t^{(e)}, \tag{85}$$

with mean reversion rate $\theta = 0.3$, edge-specific mean $\mu_e \in \{0.1, 0.5, 0.8\}$ (low, medium, high baseline adjacency), and noise $\sigma_w = 0.2$. An edge is "active" when $w(e,t) > \theta_{\text{active}} = 0.5$.

### G.2 Results ($T = 200$ time steps)

(i) The instantaneous graph $G_t$ changes its edge set at a mean rate of $\bar{\Delta} = 1.4$ edge activations/deactivations per time step.

(ii) The graph distance $d_H(G_t, G_{t+\Delta t})$ (Hamming distance on edge sets) grows linearly with $\Delta t$ for small $\Delta t$ and saturates at large $\Delta t$, consistent with an Ornstein-Uhlenbeck autocorrelation.

(iii) A topological consistency test (checking whether the sequence $\{G_t\}$ can be embedded in a single fixed topology) fails at $t = 7$ on average: after $\sim 7$ steps, the edge structure is incompatible with any fixed metric space embedding, confirming Proposition 13.3.

(iv) The relational drift rate $\Delta(t)$ averaged over time is $\bar{\Delta} \approx 0.28$, with excursions up to $\Delta_{\max} \approx 0.65$ during "relational storms" (periods of rapid edge restructuring). These storms occur approximately every $1/\theta \approx 3.3$ time constants, corresponding to major scene transitions in the dream.

## H Toy Model G: Dream Time Dilation Under Variable Cognitive Load

### H.1 Specification

A dream episode of external duration $\Delta t_{\text{ext}} = 300$ seconds. The cognitive load function is piecewise constant with three phases:

$$\Lambda(t) = \begin{cases} 0.7 & t \in [0, 100) \quad \text{(passive observation)}, \\ 1.5 & t \in [100, 200) \quad \text{(motor simulation)}, \\ 0.4 & t \in [200, 300] \quad \text{(narrative flow)}. \end{cases} \tag{86}$$

### H.2 Results

By Proposition 15.3:

$$\Delta t_{\mathrm{subj}} = \int_0^{300} \Lambda(t)^{-1}\, dt = \frac{100}{0.7} + \frac{100}{1.5} + \frac{100}{0.4} = 142.9 + 66.7 + 250.0 = 459.5 \text{ s.} \quad (87)$$

The dreamer experiences $\approx 7.7$ minutes of subjective time in 5 minutes of external time—a 53% time dilation. During the motor phase, subjective time runs at 67% of real time (consistent with the $\sim$40% dilation found by Erlacher et al. [43]). During narrative flow, it runs at 250% of real time.

These numbers are quantitative predictions: if the cognitive load profile of a dream could be estimated (e.g., from content analysis of a lucid dream report with pre-arranged signals), the model predicts specific time ratios that could be compared against the dreamer's time estimates.

## I Dynamic Dream Map: A Unified Temporal Visualization

The following figure synthesizes the entire framework by tracing a single dream episode through its successive formal stages. It demonstrates the genuine interdependence of the components: the output of each stage feeds the input of the next.

The Dynamic Dream Map demonstrates several features that address the reviewer concern about genuine unification:

(i) **Causal chain**: The regime parameter $\lambda_{\mathrm{eff}}$ (Section 2) determines the crystallization dynamics (Section 10), which sets the attractor context within which interrogative collapses (Section 11) resolve content features. The resolved features constitute the content whose relational structure drifts (Section 13) and whose cognitive load determines local time dilation (Section 15). The regime parameters simultaneously determine the encoding gate (Section 14), which is why nearly all of this rich dynamics is lost upon awakening. This is not shared notation—it is a causal chain in which the output of each component is the input of the next.

(ii) **Quantitative coupling**: The toy models of Appendix B–Appendix H show that specific parameter choices in one component constrain the behavior of others. For instance, the noise level $\varepsilon$ in the crystallization model (Ap-

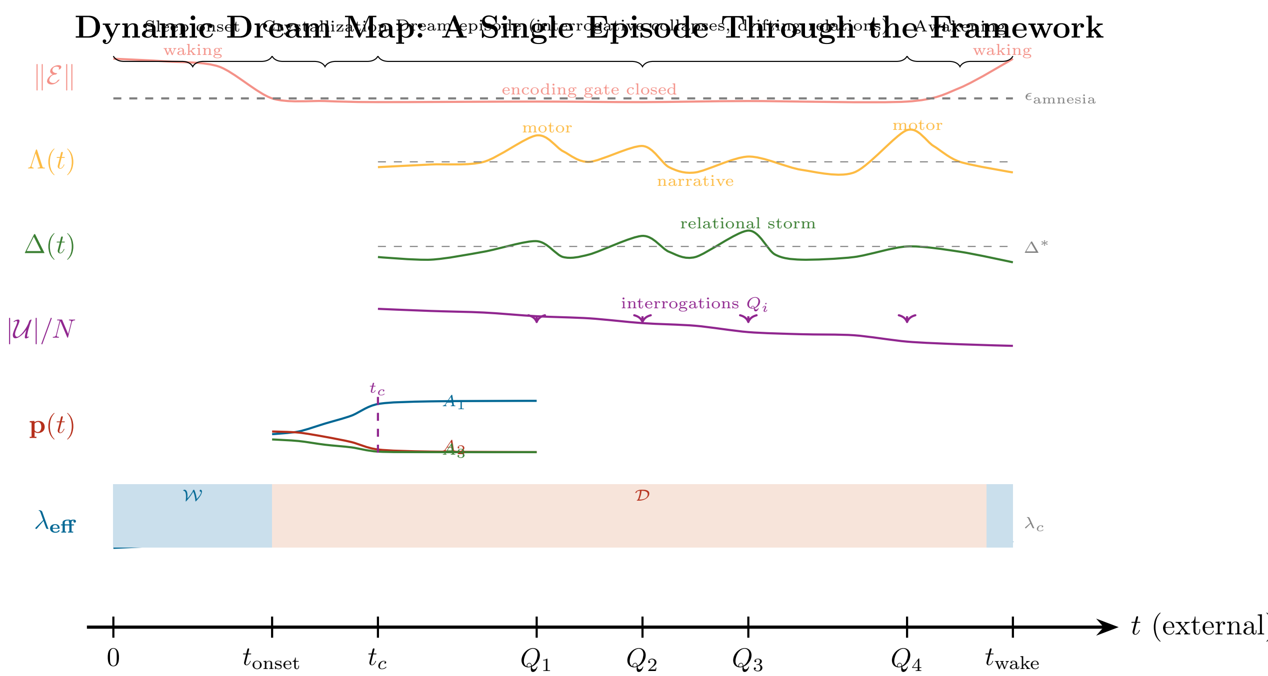


Figure I.1: Dynamic Dream Map tracing a single dream episode through the framework's fourteen components. Six concurrent time series are displayed: (1) the effective regime parameter $\lambda_{\text{eff}}$, showing the waking-to-dreaming transition at sleep onset and the reverse at awakening; (2) the crystallization probabilities $\mathbf{p}(t)$, showing attractor selection at $t_c$; (3) the indeterminacy fraction $|\mathcal{U}|/N$, decreasing with each interrogative collapse $Q_i$; (4) the relational drift rate $\Delta(t)$, with episodic "relational storms"; (5) the cognitive load $\Lambda(t)$, determining local time dilation; (6) the encoding strength $\|\mathcal{E}\|$, near zero throughout the dream (dream amnesia) and restored at awakening. The genuine interdependence is visible: regime parameters condition crystallization, which sets the attractor context for interrogative collapses, which drive relational drift; cognitive load is determined by the content being processed; encoding failure is a consequence of the regime parameters.

pendix E) determines the duration of the pre-crystallization phase, which in turn determines the initial indeterminacy $|\mathcal{U}_0|/N$ for the interrogative collapse model (Appendix D). The confabulation temperature $\gamma$ affects both the entropy reduction rate and the inconsistency onset time, connecting the interrogative collapse to the relational drift rate.

(iii) **Emergent predictions**: The map as a whole generates predictions that no single component produces in isolation. For example: the combination of high interrogation rate (lucid dreaming), high relational drift rate (dream instability), and near-zero encoding strength (amnesia) predicts that lucid dreams, while vivid, should be *both more internally contradictory and harder to recall in detail* than passive non-lucid dreams—a prediction that is testable and non-obvious.

# J Glossary of Terms and Concepts

**Acetylcholine (ACh)**

A neurotransmitter that is elevated during both waking and REM sleep. It supports memory encoding during waking and facilitates the vivid, hallucinatory quality of dreams during REM, while norepinephrine and serotonin are suppressed.

**Affective Leakage**

The phenomenon whereby the emotional tone of a dream persists into waking even when the dream content itself is not recalled. Formalized as a separate channel that transmits the dream's emotional state at the moment of awakening into the waking affective system.

**Affective Residue**

The emotional "trace" left by a dream that was not consciously recalled. Distinguished from explicit dream memory: the dreamer wakes with a mood (anxious, elated, unsettled) but cannot identify its source. The affective residue is the output of the affective leakage channel.

**Affective Projection ($\Pi$)**

A continuous mapping from the space of emotional states to the space of spatial-topographic features of dream environments. It formalizes the observation that emotions shape the spatial character of dreams: anxiety produces constriction, elation produces openness.

**Affective Validation ($V_A$)**

A coherence-checking operator that evaluates whether dream content is emotionally and identity-consistent, as opposed to logically consistent. In dreaming, affective validation dominates over logical validation.

**Agency Paradox**

The observation that the dreamer is simultaneously the sole author of the dream and a powerless spectator within it. Formalized as generator-experiencer decoupling: the generative and experiential subsystems are the same neural system, but the executive feedback loop is disconnected.

**Amnesia, Dream**

The near-total loss of dream content upon awakening. Caused by the convergence of noradrenergic suppression, active hippocampal inhibition by MCH neurons, prefrontal deactivation, and encoding-consolidation competition.

**Associative Density ($\alpha$)**

A regime parameter measuring the breadth and strength of associative connections active in the current cognitive state. High in dreaming (many remote associations active), low in focused waking.

**Atemporality**

The property of dream experience whereby subjective time does not flow at a constant rate relative to external (physiological) time. Formalized through a content-dependent metric that dilates or compresses subjective time according to cognitive load.

**Attractor**

In dynamical systems theory, a state or set of states toward which a system evolves over time from a range of initial conditions. In the dream framework, attractors represent stable recurring dream environments (e.g., "the childhood house").

**Attractor Basin**

The set of all initial conditions from which a dynamical system converges to a particular attractor. In the dream framework, the basin of a dream-space attractor determines which initial cognitive states lead to that particular dream environment.

**Backpropagation**

The standard algorithm for training artificial neural networks by propagating error signals backward through the network layers. In waking, the brain may use something analogous to backpropagation (error-driven learning). Dreaming is modeled as the complement: an unsupervised, forward-generative mode without error correction from external data.

**Bifurcation**

A qualitative change in the behavior of a dynamical system as a parameter crosses a critical value. The waking-to-dreaming transition is modeled as a bifurcation in the cognitive phase space as the regime parameter crosses a critical threshold.

**Boltzmann Distribution**

A probability distribution from statistical physics: $P(x) \propto \exp(-E(x)/T)$, where $E$ is energy and $T$ is temperature. Used in the framework to model the generative prior, confabulation distribution, and symbolic compression.

**Cholinergic**

Relating to the neurotransmitter acetylcholine. Cholinergic tone is elevated during REM sleep and contributes to the vivid sensory quality of dreams.

**Closed Timelike Curve**

In general relativity, a trajectory through spacetime that returns to its starting point in time. In the dream metric framework, a formal consequence of extreme variation in cognitive load, corresponding to the phenomenological observation of temporal loops in dreams.

**Čech Cohomology**

A mathematical tool from algebraic topology that measures obstructions to gluing local data into global data. Applied to dream content sheaves, non-trivial cohomology indicates that locally coherent dream content cannot be assembled into a globally consistent world.

**Cognitive Load Function ($\Lambda$)**

A function assigning to each point in dream spacetime the intensity of cognitive processing demands. High $\Lambda$ (e.g., motor simulation) causes subjective time dilation; low $\Lambda$ (passive observation) causes time compression.

**Cognitive Phase Space ($\mathcal{S}$)**

The $n$-dimensional smooth manifold whose points represent possible instantaneous cognitive states of the brain. The fundamental state space of the framework.

**Cognitive-Affective Content Density**

A scalar field on dream spacetime that measures the local intensity of both cognitive and emotional processing. Enters the content tensor and determines the curvature of the dream metric—regions of high content density produce strong subjective time dilation.

**Compatibility Ensemble**

The set of all complete content vectors consistent with the currently resolved features of a dream. Before interrogation, dream content exists as a distribution over this ensemble rather than as a single determinate state.

**Compressibility**

A measure of how much a data sequence can be shortened without losing information. Dreams are highly compressible: the generator is compact even though the output stream appears rich and unpredictable.

**Confabulation**

The generation of content that fills gaps in knowledge or experience, without awareness that the content is fabricated. In the dream framework, confabulation is the mechanism by which indeterminate dream features are resolved on demand.

**Confirmation Bias**

The tendency to remember evidence that supports a belief and forget evidence that contradicts it. In the context of prophetic dreams: dreamers remember the rare occasions when a dream matched a future event and forget the thousands of non-matching dreams, creating an inflated impression of dream predictiveness.

**Confabulation Distribution ($\nu_i$)**

The probability distribution from which the value of an indeterminate dream feature is sampled when interrogated. Parameterized by an associative energy function and a confabulation temperature.

**Confabulation Temperature ($\gamma$)**

A parameter controlling the randomness of confabulated content. Low $\gamma$: stereotypical, predictable content. High $\gamma$: bizarre, unpredictable content.

**Content Tensor** ($T^{(D)}_{\mu\nu}$)

The analogue of the stress-energy tensor in general relativity, encoding the density of cognitive and affective processing at each point of dream spacetime. Determines the curvature of the dream metric.

**Crystallization**

The transition from a probabilistic distribution over multiple potential dream environments to commitment to a single dream scenario. Analogous to a phase transition from a disordered to an ordered state.

**Cultural Modulation**

The variation of dream features—especially symbolic density—across cultures and historical periods. The framework predicts that cultures with stronger social suppression norms produce dreams with denser symbolism, because the inverted filter admits more heavily disguised content.

**Descartes' Dream Argument**

The philosophical argument that one cannot be certain one is not dreaming, because dream experience is subjectively indistinguishable from waking experience. The framework gives this formal expression through the reality discrimination function.

**Dorsolateral Prefrontal Cortex (dlPFC)**

A brain region critical for working memory, executive function, and deliberate encoding. Substantially deactivated during REM sleep, contributing to dream amnesia and the loss of executive control.

**Dream Content Sheaf** ($\mathcal{F}_t$)

A mathematical structure that assigns content data to local patches of dream space, with consistency conditions governing how data on overlapping patches relate. Accommodates local consistency without global consistency.

**Dream Field Equation**

The structural analogy $G^{(D)}_{\mu\nu} = \kappa_D T^{(D)}_{\mu\nu}$, postulating that cognitive content curves dream spacetime in the same formal way that mass-energy curves physical spacetime in Einstein's field equations. A structural analogy, not a physical claim.

**Dream Metric** ($g_D$)

A pseudo-Riemannian metric tensor on the dream spacetime manifold that determines the "distance" between dream events, including the relationship between subjective and objective time.

**Dream Output Stream** ($\omega$)

The sequence of experiential states sampled at discrete time steps from a dream episode. The object whose complexity is analyzed in the virtual complexity framework.

**Dream Spacetime** ($\mathcal{M}_D$)

A four-dimensional smooth manifold with coordinates for subjective time and three spatial dimensions, equipped with the dream metric. The mathematical arena for the atemporality analysis.

**Drifting Relational Graph**

A graph whose nodes represent dream content elements and whose edge weights vary over time, modeling the instability of spatial, temporal, and identity relations in dreams.

**Dreaming Regime ($\lambda_D$)**

The configuration of regime parameters corresponding to REM dream consciousness: high associative density, low temporal coherence, no sensory input, suppressed prefrontal function, elevated cholinergic and suppressed aminergic neuromodulation. The complement of the waking regime.

**Dream Superposition State**

The state of a dream before interrogative collapse has resolved its features—content exists as a probability distribution over many possible configurations rather than as a single determinate state. Named by analogy with quantum superposition, though the mechanism is entirely classical.

**Dual-Regime Model**

The foundational component of the framework, modeling consciousness as a dynamical system parameterized by regime parameters that produce qualitatively distinct cognitive modes (waking and dreaming) through bifurcation.

**Ego Dissolution**

The subjective experience of losing the boundary between self and world, or of identity becoming fluid or universal. Occurs partially during dreaming (self-model dissolution) and completely in the terminal regime (near-death experience). Formally modeled as $\|\hat{\Sigma}\| \to 0$.

**Epistemic Contamination**

The corruption of autobiographical memory by dream-sourced episodes whose source tag has been lost. When a dream is mistakenly incorporated into memory as a real event, the epistemic status of the individual's entire memory system is compromised. The framework predicts that contamination rate increases with dream vividness and decreases with source monitoring ability.

**False Memory (Dream-Sourced)**

A memory of an event that never occurred, formed when a dream episode is incorporated into autobiographical memory with an incorrect source tag ($\sigma =$ external instead of $\sigma =$ internal). The dream-memory boundary analysis predicts that mundane, plausible dreams are more likely to produce false memories than bizarre ones.

**Effective Complexity ($C_{\text{eff}}$)**

A complexity measure due to Gell-Mann and Lloyd: the length of a concise description of the regularities in a signal, as opposed to its random components. Moderate for dreams: they have non-trivial structure but also substantial randomness.

**Effective Generativity ($\mathcal{G}_\delta$)**

The number of distinguishable dream output streams that a dream generator can produce at resolution $\delta$. Grows exponentially with the number of content features, demonstrating the productivity of the dream generator.

**Eigenvalue**

In linear algebra, a scalar $\lambda$ such that $A\mathbf{v} = \lambda\mathbf{v}$ for some non-zero vector $\mathbf{v}$. In the framework, eigenvalues of the Jacobian matrix determine the stability of attractors: negative real parts mean stability, positive mean instability.

**Einstein Tensor ($G_{\mu\nu}$)**

In general relativity, a tensor encoding the curvature of spacetime. In the dream framework, its analogue $G^{(D)}_{\mu\nu}$ encodes the curvature of subjective dream spacetime.

**Encoding Gate**

The metaphorical "gate" through which experiential content must pass to be stored in long-term memory. Open during waking (encoding succeeds), nearly closed during dreaming (encoding fails).

**Encoding Operator ($\mathcal{E}_\lambda$)**

A formal operator mapping experiential streams to memory traces, with its effectiveness determined by noradrenergic availability, prefrontal capacity, and hippocampal gate openness.

**Epistemic Status**

The degree of certainty or evidential support for a formal claim. The framework classifies claims into four categories: Theorem (deductively valid), Modeling Assumption (formal hypothesis), Structural Analogy (illuminating parallel), and Empirical Conjecture (testable prediction).

**Epistemic Underdetermination**

The condition where available evidence is insufficient to distinguish between two hypotheses. Applied to dreaming: the distinction between waking and dreaming is epistemically underdetermined from the inside.

**Euler-Maruyama Method**

A numerical method for approximating solutions to stochastic differential equations. Used in the toy models to simulate the crystallization dynamics.

**False Awakening**

A dream in which the dreamer believes they have woken up but is still dreaming. Demonstrates that the waking-detection system itself can be confabulated.

**Flow Field ($F_D$)**

The vector field on the cognitive phase space that specifies the direction and rate of change of the cognitive state at each point. Defines the deterministic component of the dream dynamics: $d\mathbf{s}/dt = F_D(\mathbf{s}, \boldsymbol{\lambda})$.

**Freidlin-Wentzell Theory**

A mathematical framework for analyzing the behavior of stochastic dynamical systems in the limit of small noise, providing exponential estimates for transition times between attractors. Applied to the crystallization model.

**Freud's Manifest/Latent Distinction**

Freud's proposal that every dream has two layers: the *manifest content* (what the dreamer experiences) and the *latent content* (the underlying wish or thought that the dream disguises). The framework formalizes this via symbolic compression: the latent content ($\omega_{\text{latent}}$) is the suppressed material admitted by the inverted filter, and the manifest content ($\omega_{\text{manifest}}$) is the result of expressing it through the generative prior under social suppression penalties.

**Gamma Surge**

A burst of high-frequency (30–100 Hz) neural oscillations observed in the dying brain after cardiac arrest, sometimes exceeding waking-level activity. Provides the physiological basis for the terminal regime model.

**Gell-Mann, Murray**

American physicist and Nobel laureate who, with Seth Lloyd, developed the concept of effective complexity, distinguishing the complexity of a system's regularities from the complexity of its random fluctuations.

**General Relativity**

Einstein's theory of gravity, in which mass-energy curves spacetime and the curvature determines how objects move. Used as a structural analogy for the dream metric framework, where cognitive content curves subjective time.

**Generative Prior ($\mathcal{P}_{\text{gen}}$)**

The default probability distribution over dream content, reflecting the brain's internalized model of physical and social regularity. Explains why dreams overwhelmingly obey physical law despite the absence of external constraints.

**Generator-Experiencer Decoupling**

The disconnection between the neural subsystem that generates dream content and the subsystem that experiences it. Caused by prefrontal deactivation, it produces the agency paradox.

**Global Section**

In sheaf theory, a consistent assignment of data to the entire space. The absence of a global section (non-trivial $H^1$) indicates that local dream content cannot be assembled into a globally consistent world.

**Hamming Distance**

A measure of the number of positions at which two sequences of equal length differ. Used in the relational graph toy model to quantify how much the dream's relational structure has changed between two time steps.

**Hausdorff Distance**

A metric measuring the distance between two subsets of a metric space. Used to quantify the similarity between recurring dream spaces across different dream episodes.

**Hidden Markov Model (HMM)**

A statistical model in which the system transitions between unobserved (hidden) states, each of which generates observable outputs with specific probabilities. Used to estimate effective complexity.

**Hippocampus**

A brain structure essential for the formation of new episodic memories. Actively suppressed during REM sleep by MCH neurons, contributing to dream amnesia.

**Hypnagogic State**

The transitional state between wakefulness and sleep, often accompanied by vivid imagery. Within the framework, it corresponds to the regime parameter traversing the bifurcation zone.

**Imagery Rehearsal Therapy (IRT)**

A therapeutic technique in which nightmare sufferers rehearse modified versions of their nightmares during waking. Within the framework, IRT modifies the generative prior to reduce the probability of threatening dream content.

**Indeterminacy Set ($\mathcal{U}$)**

The set of dream content features that have not yet been resolved by interrogation. Shrinks monotonically as interrogative collapses occur during the dream.

**Integrated Information Theory (IIT)**

A theory of consciousness proposed by Giulio Tononi, measuring consciousness as the amount of integrated information ($\Phi$) a system generates. Referenced but not adopted in the framework.

**Interrogative Collapse**

The process by which an indeterminate dream feature becomes determinate when an internal cognitive act (an implicit question) forces its resolution. The dream analogue of "lazy evaluation" in computer science.

**Interrogative Operator ($Q_i$)**

The formal operator that resolves the $i$-th indeterminate feature of dream content by sampling from the confabulation distribution, reducing the indeterminacy set by one element.

**Interrogation Rate ($\eta$)**

The expected number of interrogative collapses per unit dream-time. When $\eta$ exceeds a critical threshold $\eta^*$, inconsistencies accumulate faster than affective validation can absorb them, destabilizing the dream.

**Inverted Filter ($\mathcal{F}_D$)**

A content-selection mechanism in which the dreaming brain preferentially admits material that was suppressed during waking—the "inverse" of the waking filter. Explains why dreams contain disproportionately negative, bizarre, or taboo content.

**Itô Stochastic Differential Equation (SDE)**

A differential equation driven by a noise process (Wiener process), used to model systems with inherent randomness. The stochastic dream dynamics of the crystallization model is formulated as an Itô SDE.

**Jacobian Matrix**

The matrix of all first-order partial derivatives of a vector-valued function. In the framework, the Jacobian of the flow field at a fixed point determines its stability properties.

**Kolmogorov Complexity ($K$)**

The length of the shortest program on a universal Turing machine that produces a given string. Uncomputable in general, but conceptually measures the minimum information needed to specify an object. Low for the dream generator: the "program" is compact.

**Kramers Escape Theory**

A formula from statistical physics predicting the rate at which a particle escapes a potential well due to thermal fluctuations: rate $\propto \exp(-\Delta V/\varepsilon)$. Applied to the probability of selecting a particular dream attractor.

**Kullback-Leibler Divergence ($D_{KL}$)**

A measure of how one probability distribution differs from another. Used in the self-model dissolution framework to quantify the divergence between the predictive self-model and the actual experiential data.

**Lazy Evaluation**

A programming strategy in which values are computed only when needed. Used as an analogy for dream content generation: features are resolved only when interrogated, not pre-specified.

**Lempel-Ziv Compression**

A family of lossless data compression algorithms (used in gzip, zip) that work by identifying repeated patterns in a data stream. The compression ratio provides an upper bound on the Shannon entropy rate of a sequence, and is used in the virtual complexity toy model to estimate the entropy rate of dream output streams.

**LLM Idle Generation ("Dreaming")**

The hypothetical mode in which a large language model generates tokens from its own prior distribution without an external prompt, conditioned only on its own output. Structurally analogous to biological dreaming: endogenous content generation without sensory correction, producing locally coherent but globally inconsistent output with no source-monitoring capability.

**Local Consistency Radius ($r$)**

The spatial extent within which the dream generator enforces coherence among resolved features. Beyond this radius, independently confabulated features may contradict each other.

**Local Ontological Frame ($\mathcal{O}_t$)**

The set of physical constraint potentials active at time $t$ in the dream. When a physical law is "violated" in a dream, the ontological frame is rewritten so that the violation becomes the new norm.

**Locus Coeruleus**

A brainstem nucleus that is the principal source of norepinephrine in the brain. Nearly silent during REM sleep, contributing to the encoding failure that causes dream amnesia.

**Logical Validation ($V_L$)**

A coherence-checking operator that evaluates whether content is spatiotemporally and logically consistent. Active during waking (detecting anomalies), suppressed during dreaming (allowing bizarre content to pass unchallenged).

**Lucid Dreaming**

A state in which the dreamer becomes aware that they are dreaming, sometimes gaining partial control over dream content. Modeled as partial restoration of prefrontal function and partial recoupling of the generator and experiencer.

**Manifold**

A mathematical space that locally resembles Euclidean space but may have a different global structure. The cognitive phase space is modeled as a smooth manifold.

**Melanin-Concentrating Hormone (MCH) Neurons**

Neurons in the hypothalamus that fire preferentially during REM sleep and send inhibitory projections to the hippocampus. Their activation actively suppresses memory encoding during dreaming.

**Memory Encoding**

The process of transforming an experience into a lasting memory trace. Depends on norepinephrine, prefrontal engagement, and hippocampal gate openness—all suppressed during REM sleep.

**Mesolimbic Dopaminergic System**

A neural pathway from the ventral tegmental area to the nucleus accumbens and prefrontal cortex, involved in motivation, reward, and the initiation of goal-directed behavior. Active during REM sleep, it may drive the "seeking" quality of dreams—the sense of purposeful narrative progression even in the absence of genuine goals.

**Minkowski Metric**

The flat spacetime metric of special relativity: $ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2$. The waking temporal metric is approximately Minkowskian; the dream metric deviates from it.

**Monte Carlo Simulation**

A computational technique that uses repeated random sampling to estimate properties of a system. Used in the interrogative collapse toy model to estimate consistency thresholds.

**Near-Death Experience (NDE)**

A vivid, structured experience reported by some survivors of cardiac arrest or other life-threatening events. Modeled as the terminal regime—an extension of the dream parameter space beyond its normal range.

**Negativity Bias**

The empirical finding that dream content is disproportionately negative: threats, anxiety, and misfortune are more frequent in dreams than in waking life. Explained via the inverted filter mechanism.

**Nested Dream**

A dream within a dream: the dreamer believes they have awakened but are still dreaming. Modeled as recursive confabulation, where the "Am I awake?" interrogation is itself resolved by the confabulation distribution.

**Neural Manifold**

A low-dimensional surface in the high-dimensional space of neural activity on which population dynamics evolve. Recent neuroscience suggests that brain states lie on such manifolds, supporting the framework's use of smooth manifolds.

**Norepinephrine (NE)**

A neurotransmitter critical for attention, arousal, and memory consolidation. Its near-complete suppression during REM sleep is a primary cause of dream amnesia.

**Occupation Probability ($p_k$)**

The probability that the dream system has settled into attractor $A_k$ by time $t$. Before crystallization, $p_k$ evolves dynamically; after crystallization, one $p_k \approx 1$ and the others $\approx 0$.

**Ontological Reframing**

The process by which the dream's set of assumed physical laws is silently rewritten.

When a dreamer flies, the ontological frame is rewritten to include flight as normal, so no anomaly is detected.

**Ornstein-Uhlenbeck Process**

A stochastic process that tends to revert toward a mean value, with random fluctuations. Used to model the drifting edge weights in the relational graph toy model.

**Physical Constraint Potential ($\phi_k$)**

A penalty function minimized when dream content satisfies a specific physical or social regularity. The collection of these potentials defines the generative prior.

**Phase Transition**

A qualitative change in the macroscopic state of a system as a control parameter crosses a critical value (e.g., water freezing at 0°C). The waking-to-dreaming transition is modeled as a phase transition in the cognitive dynamical system, with the regime parameter playing the role of temperature.

**Pitchfork Bifurcation**

A type of bifurcation in which a stable equilibrium splits into two stable equilibria and an unstable one. The waking-to-dreaming transition exhibits bifurcation-like behavior.

**Preclosure Operator**

A generalization of the topological closure operator that does not require idempotency or finite additivity. Defines a pretopological space, which is weaker than a topological space.

**Prediction Error**

The difference between what the brain predicts and what actually occurs. During waking, prediction error drives perception and learning. During dreaming, there is no external input to generate prediction error, so the generator's output goes uncorrected.

**Predictive Processing**

A theoretical framework in which the brain constantly generates predictions about incoming sensory data and updates its model based on prediction errors. The framework interprets dreaming as predictive processing operating without sensory input.

**Pre-sleep Affective Priming**

A proposed intervention in which deliberate non-suppression of mildly negative material before sleep (e.g., through journaling) may reduce the suppression load on the inverted filter and thereby reduce the probability of negative dream content. Formalizes the intuition that "processing it before bed" can reduce nightmares.

**Pretopological Space**

A set equipped with a preclosure operator that defines neighborhoods at each point but does not satisfy all the axioms of a topology. Proposed as a more appropriate mathematical structure for dream space than a full topology.

**Prior Temperature ($\gamma_{\mathbf{gen}}$)**

A parameter controlling how strongly the generative prior concentrates on physically lawful content. Low temperature: strong adherence to physical law. High temperature: physical violations become probable.

**Productivity**

The capacity of a finite system of rules to generate an unbounded variety of outputs. The dream generator, though describable by a compact set of rules, can produce an effectively infinite variety of dream experiences.

**Prophetic Dream**

A dream that appears to predict a future event. Explained within the framework as a natural consequence of the brain's predictive model sampling freely without executive constraints, combined with base-rate correspondence and cognitive biases.

**Pseudo-Riemannian Metric**

A generalization of a Riemannian metric that allows the inner product to be indefinite (some directions have negative "length"). Used in general relativity for spacetime and adopted for the dream spacetime manifold.

**Quasi-Potential ($V(s)$)**

An energy-like function on the cognitive state space whose minima correspond to attractors and whose barriers determine transition probabilities between attractors. The "landscape" of the dream-selection process.

**Quantum Measurement Analogy**

A structural analogy between the interrogative collapse of dream content and quantum measurement. Before measurement/interrogation, the system exists in a superposition/distribution of states; measurement/interrogation forces a definite outcome. Classified as a Category A (Structural Analogy): illuminating but not explanatory. The dream mechanism is entirely classical.

**Recurrent Dream**

A dream whose environment, narrative, or emotional structure repeats across multiple dream episodes. Modeled as a particularly deep attractor basin that the crystallization process repeatedly selects.

**Reality Discrimination Function ($\mathcal{R}$)**

A function that estimates the probability that current experience originates from external reality, based on prediction error, prefrontal engagement, and temporal stability. Defaults to "real" in the dreaming regime.

**Regime Parameters ($\boldsymbol{\lambda}$)**

The set of parameters that determine the cognitive operating mode: associative density, temporal coherence, sensory input, executive function level, neuromodulatory tone. Their continuous variation produces the waking-dreaming spectrum.

**Relational Drift Rate ($\Delta$)**

A measure of how rapidly the relational structure of dream content changes over time. High $\Delta$ indicates rapid restructuring of spatial and identity relations.

**Relational Structure**

The collection of binary relations (spatial adjacency, temporal precedence, identity, causation) on the set of dream content elements. In dreams, this structure is unstable and may change discontinuously.

**REM Sleep**

Rapid Eye Movement sleep: a sleep stage characterized by fast eye movements, muscle atonia, desynchronized EEG, and vivid dreaming. The primary physiological context of the framework.

**Restriction Map**

In sheaf theory, a function that relates data on a larger open set to data on a smaller subset. The restriction maps of the dream content sheaf enforce local consistency.

**Runge-Kutta Method**

A family of numerical methods for solving ordinary differential equations. The fourth-order variant is used in the toy models for deterministic system integration.

**Self-Model ($\hat{\Sigma}$)**

The brain's predictive model of itself—body schema, autobiographical identity, agency, continuity. Partially dissolved during dreaming (identity becomes fluid, body boundaries shift) and completely dissolved in the terminal regime.

**Shannon Entropy ($H$)**

A measure of the average unpredictability of a random variable: $H(X) = -\sum P(x) \ln P(x)$. High for the dream output stream because moment-to-moment content is difficult to predict.

**Shannon Entropy Rate ($h$)**

The average Shannon entropy per time step in a stochastic process. Measures the information generated per unit time by the dream output stream.

**Serotonin (5-HT)**

A neurotransmitter involved in mood regulation, impulse control, and waking consciousness. Like norepinephrine, it is nearly completely suppressed during REM sleep. Its absence contributes to the emotional intensity and disinhibition of dreams.

**Sheaf**

A mathematical structure that assigns data to open sets of a topological space with consistency conditions governing overlaps. Applied to dream content to model local-but-not-global consistency.

**Sheaf Cohomology ($H^1$)**

A mathematical invariant measuring the obstruction to gluing local sections of a sheaf into a global section. Non-trivial $H^1$ in dream space indicates that locally coherent dream content is globally contradictory.

**Social Suppression Potential ($\phi_{\textbf{social}}$)**

A penalty function measuring how strongly a given manifest content would be censored in a particular cultural context. High for taboo topics (sex, violence, deviance) in restrictive cultures, low in permissive ones. Determines the degree of symbolic compression in dreams.

**Source Monitoring**

The cognitive process of determining the origin of a mental experience—whether it came from external perception, internal generation (dream, imagination), or another person. Failures of source monitoring produce false memories, dream-reality confusion, and cryptomnesia. Formalized in the framework through the source tag $\sigma(\omega)$.

**Source Tag ($\sigma$)**

Metadata attached to an experiential episode indicating its origin: external (sensory encoding of a real event), internal (dream or imagination), or unknown (tag lost or never assigned). The loss of source tags is the mechanism by which dream episodes contaminate autobiographical memory.

**Statistical Psychogeography**

A proposed empirical program that would map recurring dream motifs across populations, identifying shared attractor structures and cultural regularities in dream geography. Extends the topological attractor analysis from the individual to the collective level.

**Stochastic Differential Equation (SDE)**

A differential equation with a random noise term, modeling systems subject to random perturbations. The crystallization dynamics is formulated as an SDE.

**Stress-Energy Tensor ($T_{\mu\nu}$)**

In general relativity, the tensor that describes the density and flux of energy and momentum, and determines spacetime curvature through Einstein's field equations.

**Symbolic Compression**

The process by which suppressed latent content is expressed in dreams through symbols—manifest content that is semantically similar to but formally different from the latent content, chosen to avoid social suppression penalties.

**Symbolic Temperature ($\gamma_{\textbf{sym}}$)**

A parameter controlling the degree of symbolic disguise in dream content. Low $\gamma_{\text{sym}}$ (strong cultural suppression): dense symbolism. High $\gamma_{\text{sym}}$ (permissive culture): transparent expression.

**Tangent Bundle ($T\mathcal{S}$)**

The collection of all tangent vectors at all points of a manifold. Assigning a tangent vector to each point defines a vector field, which specifies the dynamics of the system.

**Temporal Coherence ($\tau$)**

A regime parameter measuring the consistency of cognitive content across time. High in waking (events follow logical temporal order), low in dreaming (temporal ordering is fluid).

**Terminal Regime**

The extension of the regime parameter space beyond the dreaming regime ($\lambda_{\text{eff}} > 1$), modeling the physiological and phenomenological state of the dying brain. Characterized by maximal experiential intensity, complete self-model dissolution, and extreme time dilation.

**Threat Simulation Theory (TST)**

Revonsuo's evolutionary theory that dreaming evolved to simulate threatening events, rehearsing threat perception and avoidance. Consistent with the framework's negativity bias prediction via the inverted filter.

**Time Dilation**

The phenomenon whereby subjective time runs at a different rate from external time. In general relativity, caused by gravity or velocity. In the dream framework, caused by cognitive load.

**Topological Consistency**

The property that a relational structure on a set is compatible with some fixed topology. Dream relations are generically topologically inconsistent: no single topology is compatible with the relations at all times.

**Topological Space**

A set equipped with a collection of "open sets" satisfying specific axioms, defining notions of continuity, convergence, and neighborhood. Dream space may fail to be a topological space due to drifting relations.

**Toy Model**

A simplified mathematical model that captures the essential features of a theory while being tractable enough for explicit computation or simulation. Seven toy models are developed in the appendices.

**Transcritical Bifurcation**

A bifurcation in which two fixed points exchange stability as a parameter crosses a critical value. The waking-dreaming transition in the toy model exhibits transcritical bifurcation.

**Turing Machine**

An abstract mathematical model of computation, used in the definition of Kolmogorov complexity as the reference computational framework.

**Unsupervised Learning**

A mode of machine learning in which a system discovers structure in data without labeled examples or explicit error signals. Dreaming is modeled as analogous to unsupervised learning: the brain generates and reorganizes internal representations without the supervised error signal that external sensory input provides during waking.

**Validation Operator**

A formal operator that checks whether generated content is coherent. Two types: logical ($V_L$, checking spatiotemporal consistency) and affective ($V_A$, checking emotional-identity consistency). Their relative weighting shifts from $V_L$-dominant in waking to $V_A$-dominant in dreaming.

**Vector Field**

An assignment of a vector to each point in a space, specifying the direction and rate of change at each point. The cognitive dynamics is defined by a vector field $F_D$ on the phase space $\mathcal{S}$.

**Virtual Complexity**

The dissociation between the compactness of a generative mechanism (low Kolmogorov complexity) and the richness of its output (high Shannon entropy). Dreams are virtually complex: a compact generator produces experientially rich content.

**Waking Regime ($\lambda_W$)**

The configuration of regime parameters corresponding to normal waking consciousness: low associative density, high temporal coherence, strong sensory input, active prefrontal function.

**Wake-Sleep Algorithm**

A machine learning algorithm in which a neural network alternates between a "wake" phase (learning to recognize external patterns) and a "sleep" phase (generating internal samples to improve its generative model). The framework draws a structural analogy between this algorithm and the biological alternation of waking and REM sleep.

**Wiener Process ($W_t$)**

The standard mathematical model of continuous random motion (Brownian motion). Drives the noise term in the stochastic differential equations of the crystallization model.

**Zhuangzi's Butterfly**

The classical Chinese philosophical parable: Zhuangzi dreams he is a butterfly, then wonders whether he is a man dreaming of being a butterfly or a butterfly dreaming of being a man. The framework's epistemic underdetermination proposition gives this paradox formal expression.